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Identifying Non-striatal MRI-derived Biomarkers for Huntington's Disease for Use in Gene Therapy Trials

Data driven project

Candidate number: 24218073

Submitted as partial fulfilment of the requirements for the MSc in Clinical Neuroscience

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Statement of Contribution

Study design: HD-YAS investigators
Experimental procedures: HD-YAS investigators
Data Collection: HD-YAS investigators
Dataset preparation (combining HD-YAS data in excel): Candidate 24218073 with support of Dr Nicola Hobbs
Image segmentation: Dr Nicola Hobbs
Quality control: Candidate 24218073 with support of Dr Nicola Hobbs and Dr Rachael Scahill
Volume extraction: Dr Nicola Hobbs
Atrophy rate and effect size analysis: Candidate 24218073 with support of Dr Nicola Hobbs
Statistical analysis and graphing (STATA): Candidate 24218073
Overall Data analysis: Candidate 24218073 with support of Dr Nicola Hobbs and Dr Rachael Scahill, and advice from Dr Graziella Favarato (Statistician)
Writing and preparation of manuscript: Candidate 24218073, with feedback from Dr Nicola Hobbs and Dr Rachael Scahill

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Abbreviations

- ALN-HTT02 – RNA interference therapy targeting *HTT* (Alnylam Pharmaceuticals)
- AMT-130 – Gene therapy candidate for HD (uniQure, AAV5-miHTT)
- CAG – Cytosine-Adenine-Guanine trinucleotide repeat
- CAP100 – Normalised CAG-Age Product (standardised to 100 at expected age of onset)
- CED – Convection-Enhanced Delivery
- CI – Confidence Interval
  CSF – Cerebrospinal Fluid
- DBS – Disease Burden Score
- DNA – Deoxyribonucleic Acid
- ES – Effect Size (Cohen's d)
- FDR – False Discovery Rate
- FLAIR – Fluid-Attenuated Inversion Recovery (MRI sequence)
- FOV – Field of View
- FP – Fully Penetrant (CAG repeat category)
- GABA – Gamma-Aminobutyric Acid
- GM – Grey Matter
- GM-CSF – Grey Matter / Cerebrospinal Fluid boundary
- GM-WM – Grey Matter / White Matter boundary
- HD – Huntington's Disease
- HD-ISS – Huntington's Disease Integrated Staging System
- HD-YAS – Huntington's Disease Young Adult Study
- HDGE – Huntington's Disease Gene-Expanded
- HDYO – Huntington's Disease Youth Organisation
- HTT – Huntingtin gene
- ICBM152 – International Consortium for Brain Mapping 152 template
- ICV – Intracranial Volume
- IFG – Inferior Frontal Gyrus
- INT – Intermediate allele (CAG 27–35 repeats)
- IQR – Interquartile Range
- MAGNETOM – Siemens 3T Prisma MRI scanner model
- MALP-EM – Multi-Atlas Label Propagation with Expectation-Maximisation
- MRI – Magnetic Resonance Imaging
- N3 – Nonparametric Nonuniform intensity Normalisation (bias field correction method)
- PVS – Perivascular Spaces
- QC – Quality Control
- ROI – Region of Interest
- RP – Reduced Penetrance allele (CAG 36–39 repeats)
- SD – Standard Deviation

- SOP – Standard Operating Procedure
- STATA – Statistical software package (StataCorp)
- TFC – Total Functional Capacity
- TR/TE/TI – Repetition Time / Echo Time / Inversion Time (MRI parameters)
- WM – White Matter
- WT – Wild-Type allele (CAG ≤ 26 repeats)

Abstract

Huntington's disease (HD) is a genetic neurodegenerative disorder characterised by early pathological changes in the caudate and putamen, deep grey-matter structures collectively known as the striatum. MRI has become a pivotal tool for tracking HD progression, with striatal measures established as sensitive biomarkers and widely adopted as outcome measures in clinical trials. However, in gene therapy studies, where therapeutics are delivered directly into the caudate and putamen via neurosurgical procedures, post-operative MRI signals may distort structural boundaries, reducing the reliability of striatal markers. This study therefore aimed to identify alternative MRI-derived biomarkers by quantifying whole-brain atrophy rates in HD gene-expanded (HDGE) participants compared to healthy controls, ranking affected regions by effect size, and examining associations with disease burden (CAP100) and HD-ISS stage. Eighty-eight participants (54 HDGE, 34 controls) from the HD-YAS cohort underwent 3T MRI at baseline and 4.5-year follow-up, with T1-weighted scans segmented using MALP-EM to extract 132 regions of interest (ROIs). Following structured quality control, 69 bilateral ROIs were retained, and annualised atrophy rates were computed, adjusted for age and sex. Linear regression with FDR correction identified sixteen ROIs showing significant between-group differences. As expected, the caudate (d = 1.305) and putamen (d = 1.269) showed the largest effect sizes, while non-striatal regions including the pallidum, accumbens, ventral diencephalon, and cortical regions within the frontal and parietal lobes also exhibited significant atrophy, albeit with smaller effect sizes. Atrophy rates increased with higher CAP100 scores in selected regions, whereas associations with HD-ISS stage could not be clearly determined due to group size imbalance and limited statistical power. Together, these findings confirm the striatum as a key site of vulnerability while highlighting non-striatal regions as promising complementary biomarkers. Conclusively, this work provides preliminary evidence supporting the development of alternative MRI outcome measures for gene therapy trials where direct striatal targeting may confound interpretation, but further validation in larger, longitudinal cohorts will be essential before such markers can be reliably applied in clinical settings.

Chapter 1. Introduction

1.1. Huntington's disease: an overview

Huntington's disease (HD) is a neurodegenerative disorder and a major inherited cause of dementia, characterized by chorea, an involuntary, dance-like movement typically seen in later stages. Although the underlying mutation is present from birth, clinical diagnosis occurs between ages 30-50, based on the emergence of motor impairments, accompanied by cognitive decline and neuropsychiatric deterioration. These symptoms progressively worsen over time and ultimately lead to death (McColgan and Tabrizi, 2018).

In the United Kingdom, HD affects approximately 12.3 per 100,000 individuals, that is roughly 1 in 8,100 people, with an incidence of 7.2 new cases per million individuals annually (Wexler *et al.*, 2016). While HD follows a Mendelian inheritance pattern, 10% of cases occur without a known family history, often due to undiagnosed ancestry or rare de novo mutations (Penn Huntington's Disease Center). Globally, prevalence varies by region and ancestry, with the highest rates observed in populations of European descent and substantially lower rates reported in Asia (Baig, Strong and Quarrell, 2016). Recent increases in HD prevalence are primarily driven by improved genetic testing and earlier clinical recognition, which enable more accurate diagnosis. These improvements, combined with longer survival due to enhanced clinical care and disease management, as well as growing awareness and support, account for the rise in recorded cases, rather than an actual increase in incidence (Wexler *et al.*, 2016).

Despite major advances in HD research, no disease-modifying treatment currently exists. As George Huntington wrote, "if the thread is broken then the grandchildren of the original shakers may rest assured that they are free from the disease" (Huntington, 1872). Breaking this thread requires early, preventive interventions before symptoms emerge. Such strategies rely on identifying sensitive and reproducible biomarkers capable of detecting subtle preclinical changes and tracking progression in HD gene-expanded (HDGE) individuals, yet these biomarkers remain limited. Neuroimaging

offers a promising, non-invasive tool for biomarker development, and recent advances have intensified efforts to identify neuroimaging biomarkers that reflect HD pathogenesis and reliably monitor disease course. These biomarkers are essential for evaluating therapeutic efficacy in clinical trials and ultimately achieving a cure (van de Zande *et al.*, 2022).

1.2. Symptoms of Huntington's Disease

HD is a relentlessly progressive neurodegenerative disorder characterized by a core triad of motor, cognitive, and psychiatric symptoms, each contributing to functional decline and reduced quality of life (Ross *et al.*, 2014) (Figure 1). Motor symptoms typically begin with involuntary movements such as chorea and may progress to dystonia, bradykinesia, rigidity, and impaired coordination. As HD advances, voluntary motor control deteriorates, affecting gait, speech, and swallowing, and increasing aspiration risk (Reilmann, Leavitt and Ross, 2014). Cognitive symptoms include deficits in memory, attention, executive function, and processing speed. Difficulties with planning, multitasking, and emotion recognition are common, resembling subcortical dementias and often emerging more than a decade before clinical motor diagnosis (Vaccarino *et al.*, 2011). Psychiatric symptoms are common and can be as disabling as motor or cognitive features. Depression, anxiety, irritability, apathy, and obsessive-compulsive behaviours are frequently reported. Apathy and mood disturbances often appear early and intensify with disease progression, significantly impacting social functioning and caregiver burden (van Duijn *et al.*, 2014).

Cognitive and psychiatric symptoms in HD are often more detrimental to quality of life than motor impairments. This disparity underscores the need for broader clinical frameworks that reflect the full spectrum of disease burden. Together, these symptoms evolve over time, with variable onset and severity across individuals. Their combined impact underscores the need for multidimensional clinical assessment and integrative therapeutic strategies (Gil-Salcedo *et al.*, 2025).

1.3. Pathophysiology of HD

1.3.1. Genetics basis

HD is inherited in an autosomal dominant pattern, meaning a single copy of the mutated gene is sufficient to cause disease. It is caused by an abnormal expansion of a cytosine-adenine-guanine (CAG) repeat, which directly drives HD pathogenesis (Table 1). The hallmark of HD is the accumulation of mutant HTT protein in the brain, primarily composed of truncated fragments with expanded polyQ tracts, generated via proteolytic cleavage of aberrant splicing of exon 1 (Neueder and Bates, 2018). The production of this toxic protein increases with CAG repeat length in exon 1, which underpins disease onset and progression (Romo, Mohn and Aronin, 2018). In unaffected individuals, CAG repeats remain below a pathological threshold, whereas in HD, excessive repeats disrupt neuronal function and drive progressive degeneration. Repeat length is inversely correlated with age of onset and death (Table 1).

In addition to inherited repeat length, HD pathogenesis is further influenced by somatic expansion, an age-dependent increase in CAG repeat length within the HTT gene across tissues. Longer repeats are associated with earlier onset, in a phenomenon known as anticipation leads to increasing expansion across generations, typically through the paternal line (Black *et al.*, 2024). Somatic expansion is most pronounced in the striatum and cortex, regions highly vulnerable in HD, and especially evident in individuals with unexpectedly early clinical motor diagnosis (Swami *et al.*, 2009). Once striatal neurons exceed a critical threshold of ~150 CAG repeats, the transcriptome shifts, neuronal identity is lost, and cell death occurs within months (Handsaker *et al.*, 2025a).

1.3.2. Neuropathological basis

The primary site of degeneration in HD is the striatum, comprising the caudate and putamen, which undergo progressive atrophy and astrogliosis, a reactive increase in astrocyte density in response to neuronal injury (Waldvogel *et al.*, 2014; Hedreen, Berretta and White III, 2024). This region shows marked loss of medium spiny GABAergic neurons, which constitute ~95% of striatal neurons and are essential for inhibitory signalling within basal ganglia circuits. Their degeneration leads to reduced inhibitory output and contributes to motor dysfunction (Blumenstock and Dudanova, 2020).

Degeneration typically begins in the caudal striatum and progresses rostrally, accompanied by loss of key neurotransmitters, GABA, dopamine, and glutamate, which further destabilize basal ganglia output (André, Cepeda and Levine, 2010). Dopamine signalling becomes increasingly dysregulated; early HD shows elevated dopamine levels and reduced receptor expression, exacerbate chorea, while late-stage HD is marked by dopamine depletion, resembling Parkinsonism and contributing to bradykinesia and rigidity. This biphasic shift in dopamine tone parallels changes in glutamatergic transmission, disrupts corticostriatal communication, and amplifies the impact of GABAergic medium spiny neuron degeneration (Chen *et al.*, 2013) (Figure 1).

Table 1. CAG repeat length categories and their clinical significance in Huntington's disease (HD).

CAG Repeat Length	Category	Association with HD	Typical age of onset
26 or fewer	Wild type	Not associated with HD	
27 – 35	Intermediate	Generally considered healthy, but some may develop HD in later life	50s or older
36 – 39	Reduced penetrance	Some may develop HD but not all	Late 30s to early 50s
40 or more	Fully penetrant	Will develop HD within a typical lifespan under current diagnostic criteria	Typically, 30 to 50 years old

The number of CAG repeats in the HTT gene determines the risk and onset of Huntington's disease. Wild-type alleles (≤26 repeats) are not associated with the disease. Intermediate alleles (27-35 repeats) typically do not cause HD but may expand in future generations. Reduced penetrance alleles (36-39 repeats) may or may not lead to HD symptoms. Fully penetrant alleles (≥40 repeats) are invariably associated with disease onset, typically between the ages of 30 and 50 (Black *et al*., 2024).

Abbreviations: HD, Huntington's disease; HTT, huntingtin gene; CAG, cytosine-adenine-guanine repeat; WT, wild-type; INT, intermediate; RP, reduced penetrance; FP, fully penetrant; AO, age of onset.

While the caudate and putamen are well-established as primary sites of degeneration in HD, additional basal ganglia structures, including the nucleus accumbens and pallidum, also show HD-related pathology, though the evidence base is comparatively smaller (van den Bogaard *et al.*, 2011). The nucleus accumbens, integral to reward processing and motivated behaviour, exhibits functional and structural disruption in HD, contributing to apathy and impaired reward sensitivity (Xu *et al.*, 2024). The pallidum, central to motor output pathways, undergoes early atrophy, further destabilizing basal ganglia signalling and movement regulation (Blumenstock and Dudanova, 2020). The thalamus, positioned within cortico-basal ganglia-thalamo-cortical loops, plays a critical role in relaying sensorimotor and cognitive signals; its disruption contributes to circuit breakdowns and amplifys motor and motivational deficits (Mehrabi *et al.*, 2016).

Although striatal degeneration defines early HD, non-striatal regions, including cortical areas, are progressively affected. Cortical thinning, especially in frontal and parietal regions, tends to emerge near motor clinical diagnosis and is linked to executive and motor dysfunction (Rosas *et al.*, 2002, 2008). Unlike other cortical regions that typically show structural changes later in the disease course, the occipital cortex has been consistently identified as an early site of atrophy in HD, preceding clinical motor diagnosis (Johnson *et al.*, 2015a). Given its role in visual and visuospatial processing, early occipital involvement aligns with premanifest deficits in visual cognition. Thinning in lateral occipital and lingual regions correlates with impaired visual performance, underscoring early cortical vulnerability (Johnson *et al.*, 2015b; Coppen *et al.*, 2018). This overlap between structural vulnerability and cognitive relevance positions the occipital cortex as a valuable region for studying neurodegenerative changes. These alterations are detectable via neuroimaging and histology, including early white matter disruption and caudate atrophy years before motor onset (Johnson *et al.*, 2015b).

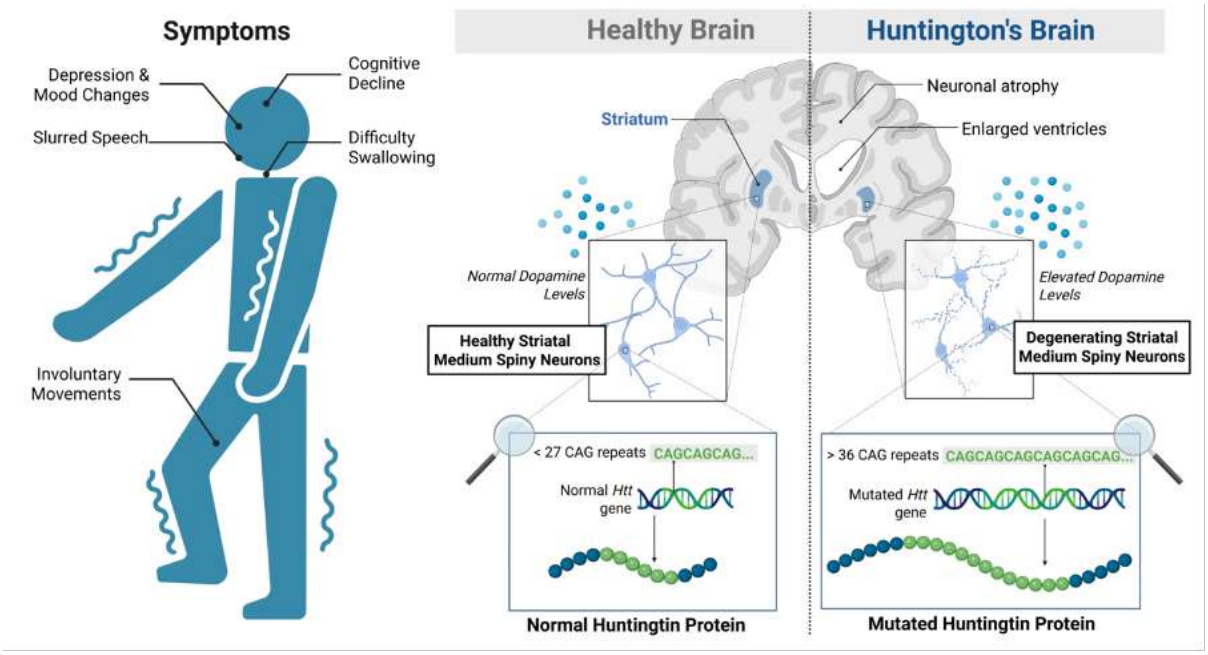

Figure 1. Clinical and neuropathological features of Huntington's disease (HD). *Left:* Overview of motor and non-motor symptoms commonly observed in HD, including cognitive decline, psychiatric disturbances, and movement abnormalities. *Right:* Schematic representation of HD-related brain pathology, highlighting striatal degeneration and disrupted dopamine signalling resulting from mutant huntingtin (mHTT) expression. Adapted from McArdle, C. (2025).

1.4. Stages of HD

HD has traditionally been classified into two stages: premanifest and manifest, based on the presence or absence of clear motor symptoms (Tabrizi *et al.*, 2009). The Huntington's Disease Integrated Staging System (HD-ISS) (Tabrizi *et al.*, 2022) (Figure 3) was developed to address limitations of the traditional model, which overlooks decades of biological change preceding clinical motor diagnosis. HD-ISS provides a standardized, evidence-based framework for research and trial design spanning the entire disease course, from genetic confirmation through advanced stages.

The HD-ISS defines four stages, stage 0 (*HTT* mutation without pathology), Stage 1 (biomarker evidence), Stage 2 (clinical symptoms - motor, cognitive, or psychiatric), and Stage 3 (functional decline), based on genetic status, pathology, and clinical impact. By replacing vague terms like "presymptomatic" and "prodromal" with threshold-based criteria, it enables consistent classification, supports early intervention, and improves trial design across the disease continuum (Tabrizi *et al.*, 2022).

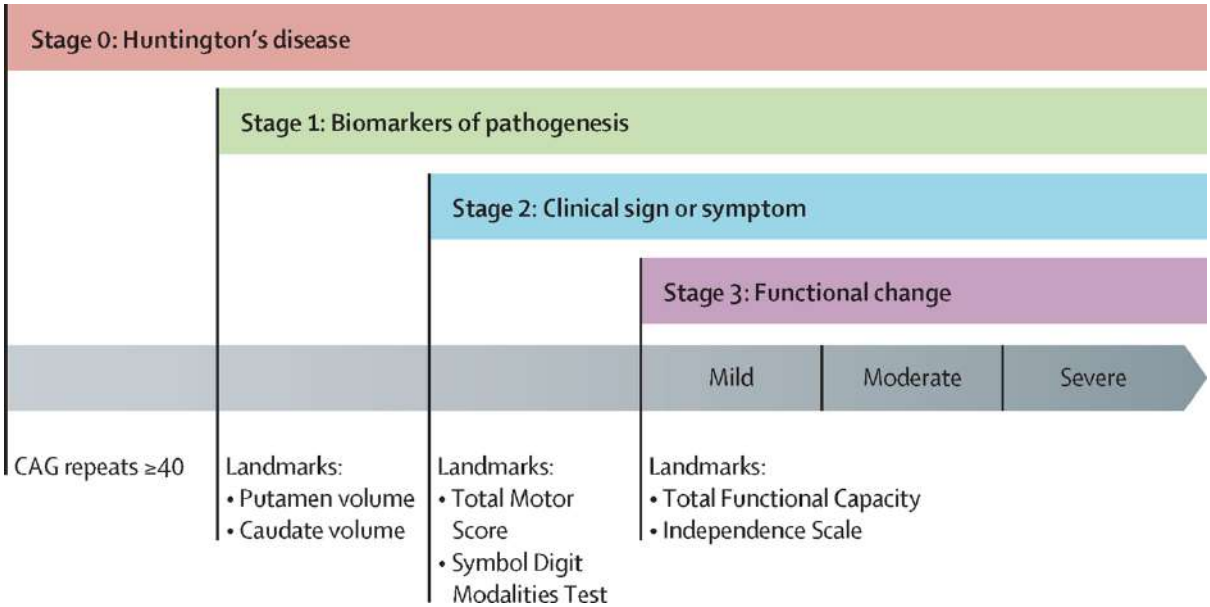

Figure 3. HD-ISS staging framework for Huntington's disease (HD). Illustration of the four-stage HD-ISS model, integrating genetic status, biomarker evidence, clinical signs, and functional decline. Stage 0 includes individuals with ≥40 CAG repeats and no detectable pathology. Stage 1 reflects neurodegeneration, typically indexed by striatal atrophy. Stage 2 captures the emergence of clinical symptoms, and Stage 3 denotes functional impairment based on Total Functional Capacity and Independence Scale scores. Adapted from Tabrizi et al., 2022 (HD-ISS Working Group).

Abbreviations: HD, Huntington's disease; HD-ISS, Huntington's Disease Integrated Staging System; TFC, Total Functional Capacity; CAG, cytosine-adenine-guanine repeat.

1.5. Quantifying CAG repeat exposure: CAP100

CAP100 is a normalized version of the CAG-Age Product (CAP) score, designed to quantify cumulative exposure to mutant huntingtin (*mHTT*) and predict disease progression in HD. It refines the earlier Disease Burden Score (DBS) by standardizing the calculation across studies, enabling consistent comparisons and trial eligibility criteria. It is calculated as:

$$CAP = Age \times (CAG - L) / K$$

Where L = 30 and K = 6.49, chosen so that CAP = 100 corresponds to the expected age of motor diagnosis. This standardization was proposed by Warner et al. (2022) and are widely adopted in HD trials to identify participants at similar stages of disease progression, especially in premanifest phases.

1.6. Treatments for HD

Currently, no disease-modifying treatment exists for HD, but several clinical trials are targeting its underlying biology. A key strategy involves lowering mutant huntingtin (mHTT) protein using antisense oligonucleotides (*Tominersen, Roche*) that block HTT mRNA translation. Oral splicing modulators (*PTC518, PTC Therapeutics*) introduce premature stop codons to suppress mHTT production. RNA interference therapies (*ALN-HTT02, Alnylam*) degrade HTT transcripts, while gene therapy platforms (*AMT-130, uniQure*) deliver silencing microRNA via adeno-associated virus directly into brain cells. Together, these approaches reflect a multifaceted strategy to slow HD progression by targeting the genetic driver (Tabrizi *et al.*, 2019).

AMT-130 is the first gene therapy for HD to enter human trials, involving direct neurosurgical infusion into the caudate and putamen, regions that undergo early and pronounced degeneration. In the Phase I/II trial, participants receive bilateral burr hole placement, followed by MRI-guided stereotactic infusion of AAV5 vectors into six striatal sites via convection-enhanced delivery (CED), a technique that distributes fluid evenly and minimise reflux (Reilmann *et al.*, 2020; Stimming *et al.*, 2022). Intraoperative MRI refines cannula placement and monitors infusion dynamics, ensuring accurate targeting of vulnerable striatal nuclei (*AMT-130, uniQure*). Post-

surgical MRI provides insight into delivery fidelity, with localized signal changes such as T2-weighted hyperintensities or gadolinium enhancement reflecting vector spread, tissue response, or procedural effects (Spronck *et al.*, 2021; *AMT-130, uniQure*). These features confirm caudate, and putamen targeting and provide non-invasive readout of therapeutic engagement in deep brain regions, enabling longitudinal monitoring in gene therapy trials.

Wild-type HTT is essential for transcriptional regulation, axonal transport, and embryonic development (Reiner *et al.*, 2003). While partial reduction of WT HTT maybe tolerable in animal models, preserving its function is critical (Leavitt, Kordasiewicz and Schobel, 2020). Continued refinement is needed to balance efficacy with safety in targeting this complex and fatal disease.

In parallel, a promising strategy is emerging that targets somatic expansion of the HTT CAG repeat, a process thought to initiate the neurodegenerative cascade and drive disease progression. Stabilising CAG repeats at the DNA level offers a way to delay or prevent disease onset by targeting the root cause rather than downstream effects. Unlike therapies such as tominersen, which lower mHTT protein, this approach directly addresses the genetic trigger, offering a mutation-specific and potentially durable solution (Handsaker *et al.*, 2025b).

A recent Nature Medicine study by Scahill et al. (2025), conducted within the Huntington's Disease Young Adult Study (HD-YAS) cohort, provides compelling evidence that neurodegeneration begins nearly two decades before clinical symptoms. By studying HD gene-expanded (HDGE) individuals with intact cognitive, motor, and behavioural function, the team identified subtle but consistent changes in brain imaging and biofluid biomarkers, defining a critical window for early intervention (Tabrizi et al., 2024). Most notably, they established a mechanistic link between somatic CAG expansion, the progressive lengthening of CAG repeats in specific brain regions and early neurodegenerative changes in living humans (Scahill *et al.*, 2025). As therapies aimed at suppressing this process advance, they lay the foundation for prevention trials that intervene before symptom onset offering a transformative opportunity to alter the trajectory of HD and delay or prevent disease progression.

1.7. MRI-derived Neuroimaging Biomarkers of HD

Magnetic Resonance Imaging (MRI) offers high-resolution, non-invasive brain images, with volumetric analysis serving as a robust tool for tracking anatomical change, disease progression, and therapeutic impact in HD (Kinnunen *et al.*, 2021). To support claims of disease modification, such measures must function as potential or validated surrogate endpoint biomarkers, demonstrating sensitivity and reliability across the disease course (Hobbs *et al.*, 2024).

Volumetric biomarkers, particularly caudate, putamen, whole brain, and CSF volumes, are widely used in HD trials due to their reproducibility and responsiveness to progression (Tabrizi *et al.*, 2009; Scahill *et al.*, 2020). Caudate and putamen atrophy emerge years before motor symptoms, while whole-brain atrophy and CSF expansion become more prominent closer to onset (Kinnunen et al., 2021). These patterns support their inclusion in HD-ISS Stage 1 criteria and enable early intervention and longitudinal monitoring (Tabrizi *et al.*, 2022).

Early work by Aylward et al. (2004) suggested a nonlinear trajectory of striatal atrophy. In contrast, Hobbs et al. (2010), using clinical diagnosis as a reference, identified a more linear decline, particularly in the caudate. Langbehn et al. (2019) confirmed that caudate and putamen volumes follow a steady, consistent trajectory throughout the disease course. This steady volume loss in clinically asymptomatic individuals supports striatal atrophy as a reliable progression marker, enabling early intervention and longitudinal monitoring of therapeutic efficacy.

Post-surgical MRI, while confirming accurate delivery in gene therapy trials, can introduce localized signal changes, especially in the caudate, that reflect procedural effects rather than underlying pathology. This complicates interpretation and underscores the need for complementary biomarkers that remain stable post-intervention and reliably capture therapeutic impact (*AMT-130, uniQure*).

1.7.1. Challenges in Applying Volumetric MRI to Gene Therapy Trials

While volumetric MRI is a cornerstone for monitoring HD progression, its interpretive reliability may be compromised in trials involving direct intracerebral delivery. In gene therapy studies such as AMT-130, the caudate and putamen, regions typically used as biomarkers, are also the sites of intervention. This overlap introduces confounds localized signal changes and morphological alterations may reflect procedural effects rather than disease-related atrophy (Hobbs *et al.*, 2024).

Evidence from other neurodegenerative conditions suggests that catheter placement and vector infusion can induce MRI abnormalities at target sites, including T2 hyperintensities and structural distortion. Such changes may obscure natural history trajectories and weaken associations between volumetric measures and clinical outcomes (Sondhi *et al.*, 2020). As HD gene therapy trials advance, complementary biomarkers that remain unaffected by surgical intervention will be essential for accurately assessing therapeutic impact.

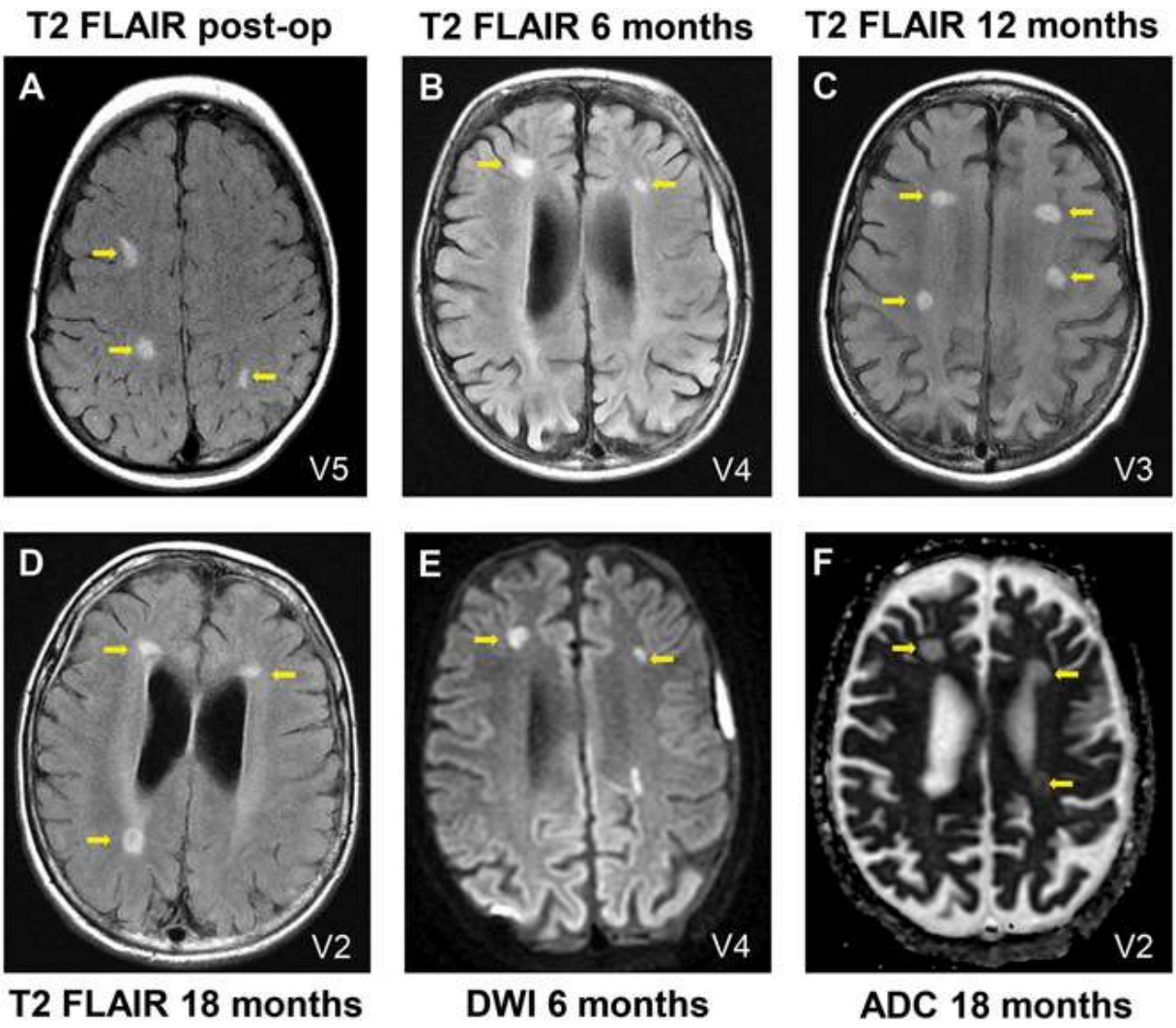

Figure 4. MRI signal changes post-gene therapy delivery. FLAIR images showing persistent hyperintensities at catheter sites following intraparenchymal vector infusion in a non-HD disorder. Similar artifacts may confound volumetric MRI interpretation in Huntington's disease (HD) trials involving direct striatal delivery. Adapted from Sondhi et al., 2020

Abbreviations: HD, Huntington's disease; MRI, magnetic resonance imaging; FLAIR, fluid-attenuated inversion recovery.

1.8. Observational studies in HD

Observational studies in HD increasingly target the preclinical phase to identify early biomarkers that enable preventative trial design. HD-YAS and its extension, HD-YAS 2.0, characterise early biological changes nearly two decades before predicted onset, using multimodal imaging, biofluid biomarkers, and cognitive profiling. HD-YAS revelated, that neurodegeneration begins well before clinical diagnosis (Scahill *et al.*, 2020), with *somatic CAG expansion* in blood correlating with in vivo changes, highlighting its therapeutic relevance (Scahill *et al.*, 2025). HD-YAS 2.0 longitudinally tracks biomarker trajectories to refine progression models and intervention timing (Mena *et al.*, 2023). Complementary initiatives like PREVENT-HD (University of Wisconsin, Madison, 2025), and the HD-ISS framework (Tabrizi *et al.*, 2022), now enable biologically anchored trial inclusion, supporting early-intervention designs.

1.9. Study Objectives and Aims

This study is divided into three parts. First, I will use annualized atrophy rates to screen the whole brain for regions affected in HDGE participants as compared to controls. Second, I will rank the most impacted areas by effect size. Third, I will examine how these atrophy rates relate to CAP100 and how they vary across HD-ISS stages.

Together, these steps aim to identify candidate non-striatal volumetric MRI-derived biomarkers of disease progression for use as outcome measures in gene therapy trials involving neurosurgical delivery directly to the caudate and putamen.

***Objective 1:* Between-Group Differences in Atrophy Rates**

To quantify longitudinal volumetric differences between HDGE participants and healthy controls across the whole brain using a spatially unbiased, whole-brain ROI analysis.

Hypothesis 1: a) HDGE participants will show significant longitudinal volumetric atrophy in the caudate and putamen compared to controls, consistent with previous findings from this dataset.

b) Deep grey matter regions structurally and functionally connected to the caudate and putamen, such as the nucleus accumbens, pallidum, and thalamus, will show significant longitudinal atrophy in HDGE participants, though to a lesser extent than the caudate and putamen, when compared to healthy controls.

c) Selective cortical regions, such as the occipital cortex, will exhibit measurable longitudinal atrophy that may serve as early biomarkers of disease progression, with HDGE participants showing greater atrophy compared to healthy controls.

Objective 2: **Longitudinal Effect sizes**

To calculate longitudinal effect sizes for each brain region and generate a ranked table based on significant between-group differences in atrophy rates, adjusted for age and sex, and corrected for multiple comparisons. Regions will be prioritized by the magnitude of their effect size as indicators of sensitivity to disease progression.

Hypothesis 2: The caudate and putamen will show the largest longitudinal effect sizes. In comparison, non-striatal regions like the nucleus accumbens, pallidum, and cortical regions such as the occipital cortex, are expected to show smaller, yet meaningful, effect sizes that needs further investigation as potential alternative outcome measures for clinical trials.

Objective 3: **Association of Regional Atrophy Rates with HD-ISS Stage and CAP100**

Within the HDGE group, to determine whether longitudinal atrophy rates are associated with HD-ISS stage or disease burden, measured by the CAP100 score. This analysis was restricted to regions of interest (ROIs) that showed significant differences between controls and HDGE participants.

Hypothesis 3: Longitudinal atrophy rates increase progressively:a) across HD-ISS stages 0-2, and b) with increasing CAP100 scores, across striatal regions (caudate, putamen), non-striatal deep grey matter (nucleus accumbens, pallidum, thalamus), and selected cortical regions (e.g., occipital cortex), with the strength of these associations varying by region.

Chapter 2. Materials and Method

2.1. Study design & Participants

This study analysed a subset of participants from the HD-YAS, a prospective, observational, case-control study (Scahill *et al.*, 2020a). Only individuals with longitudinal imaging at both baseline and follow-up (visit 2) were included, comprising 54 HDGE participants and 34 controls. All participants completed MRI at two timepoints, baseline and a follow-up, approximately four and a half years later. This interval, determined by the design of the HD-YAS study, provides sufficient sensitivity to detect meaningful longitudinal changes and supports robust comparisons of atrophy rates between HDGE participants and controls, aligning with the objectives of this project.

Within the HDGE group, participants were stratified according to the HD-ISS staging (Tabrizi *et al.*, 2022) to classify disease progression. At visit 1, 44 of the HDGE participants were classified as stage 0, with 9 participants in stage 1 and only a single participant in stage 2. By visit 2, ten participants had progressed from stage 0 (34 participants) to stage 1(19 participants).

2.2. Recruitment, Inclusion, Exclusion criteria & Ethics

2.2.1. Recruitment

HDGE participants and controls were recruited through the HD clinic at the National Hospital for Neurology and Neurosurgery, Queen Square, UK, the Huntington's Disease Youth Organisation (HDYO) and broader community outreach. Baseline assessments occurred between 2nd August 2017 and 25th April 2019; follow-up visits took place between 6th April 2022 and 21st March 2024.

2.2.2. Inclusion criteria

At baseline, participants were aged 18-40 provided informed consent and were able to comply with study procedures. HDGE individuals had a confirmed HD gene expansion (CAG repeat length ≥ 40), a Disease Burden Score (DBS = age x [CAG – 35.5]) of ≤240, and no clinical signs of manifest disease, corresponding to ≥18 years from clinical motor diagnosis. Controls were aged 18-40 with no family history of HD or a confirmed negative genetic test (CAG<36). Controls were stratified into genotype negative (first/second-degree relatives with CAG<36), family controls (non-biological relatives, e.g., partners), and community controls (no HD family history. At follow-up, participants provided renewed consent and continued to meet the baseline inclusion criteria (Scahill *et al.*, 2020b).

2.2.3. Exclusion criteria

Exclusions included significant comorbidities, genetic risk without manifest disease, or reduced penetrance CAG lengths (36-39). MRI contraindications (e.g., metal implants, pacemakers, pregnancy, severe claustrophobia) led to exclusion. Additional criteria included current drug trial participation, investigational drug use within 30 days, substance/alcohol abuse, unstable medications/nutraceuticals use, and medical, neurological, or psychiatric conditions inferring with study procedures or compliance (Scahill *et al.*, 2020b).

2.2.4. Ethics

HD-YAS was approved by the Bloomsbury Research Ethics Committee and conducted in accordance with the Declaration of Helsinki. All participants provided informed consent after receiving a full explanation of study procedures. Confidentiality and participant safety were maintained throughout (Scahill *et al.*, 2020b).

2.3. Imaging

2.3.1. Properties of scans acquired.

MRI scans were acquired at the Welcome Centre for Human Neuroimaging, using a Siemens PRISMA 3-Tesla MRI scanner (*MAGNETOM Prisma - MRI Scanner - Siemens Healthineers*), equipped with a radiofrequency body coil for transmission and a 64-channel head coil for reception. The protocol was optimised for this study and remained identical at baseline and follow-up, with no hardware or major software upgrades over the 4.5-year interval. Participants were scanned supine with foam pads placed laterally to minimise head motion. Follow-up scans were acquired on the same scanner as baseline, ensuring consistency across timepoints and enabling reliable assessment of longitudinal brain changes.

Only structural T1-weighted whole-brain scans were used for analysis, as this modality provides high-resolution volumetric data suitable for sensitive and widely validated techniques (Scahill *et al.*, 2025). Images were acquired using a 3D Magnetisation Prepared Rapid Gradient Echo sequence (TR = 2530 ms; TE = 3.34 ms; TI = 1100 ms; flip angle = 7°; voxel size/FOV = 256 x 256 x 176 mm^3). Baseline and follow-up scans were reprocessed in parallel to ensure consistent software pipelines and eliminate potential bias across timepoints.

2.3.2. MRI pre-processing

Prior to project initiation, all T1-weighted structural MRI scans underwent standardized pre-processing on a Linux-based system, including bias field correction using N3 and Intracranial volume (ICV) estimation via semi-automated protocols within Medical Imaging Display and Analysis System (MIDAS).

2.3.3. Brain segmentation using MALP-EM

Brain segmentation was performed using the Multi-Atlas Label Propagation with Expectation-Maximisation (MALP-EM v1.2)(*GitHub - ledigchr/MALPEM: MALPEM whole-brain segmentation framework*), a whole-brain tool developed by Ledig et al. at Imperial College London to delineate cortical and subcortical structures. T1-weighted images were linearly registered to MNI standard space using the ICBM152 template (http://www.bic.mni.mcgill.ca/ServicesAtlases/ICBM152NLin2009).

MALP-EM integrates several preprocessing and segmentation steps within a unified pipeline. It first corrects for intensity non-uniformities across the scan (bias correction), removes non-brain tissue such as skull and scalp (brain extraction), and then applies multi-atlas label fusion, where anatomical labels from a library of reference brains are aligned and combined to accurately identify regions in the target image. This enables precise volumetric estimation of 138 ROIs and is optimised for reproducibility across heterogeneous datasets and neurodegenerative cohorts (Ledig *et al.*, 2015). Its robustness in segmenting atrophied subcortical structures makes it well-suited for studies of neurodegenerative conditions such as HD, where anatomical variability and progressive volume loss present significant challenges. All segmentations were visually inspected using Freeview (FreeSurfer suite v7.2.0; https://surfer.nmr.mgh.harvard.edu/).

2.4. Standard Operating Procedure (SOP) for Visual Quality Control (QC)

For this project, I worked with MALP-EM-processed outputs (segmentations and volumes) and developed a structured SOP for visual QC to ensure data integrity. All datasets underwent systematic inspection.

2.4.1. QC workflow

T1-weighted MRI scans with MALP-EM ROI overlays were evaluated for image artefacts and regional delineation accuracy. Visual inspections were performed blinded to participants' genetic status to minimise bias. To reduce rater drift and batch effects, a within-subject, across-timepoint review model was adopted: each participant's visit 1 and visit 2 scans were reviewed consecutively, enabling direct comparison and identification of longitudinal segmentations inconsistencies. This structured QC approach ensured consistent visual assessments across all 88 participants, enhancing confidence in the longitudinal integrity of extracted regional volumes.

2.4.2. ROIs for Volumetric Analysis

Each scan yielded 138 ROIs via MALP-EM, with separate left/right hemisphere labels inspected individually. Seven ROIs were excluded, background (code 0), CSF, left/right vessels, left/right cerebral exterior, and optic chiasm, as they represent non-parenchymal or extracortical structures with limited relevance to volumetric analysis. Although the optic chiasm is a brain structure, it was excluded due to its small size and segmentation variability. This exclusion strategy aligns with prior MALP-EM applications (Ledig *et al.*, 2015), which omit such labels to focus on anatomically meaningful regions. The final set comprised 132 ROIs per scan, including 98 cortical and 34 subcortical regions, analysed across both visit 1 and visit 2 scans for all 88 participants. To streamline QC, all non-cortical ROIs (n = 34) were grouped in Excel by anatomical region: ventricular system, basal ganglia, limbic system, thalamus, brainstem, cerebellum, and cerebral white matter. Cortical ROIs (n = 98) were similarly organized by lobe frontal, parietal, temporal, occipital, cingulate, and opercular. Bilateral structures were reported separately for the right (R) and left (L) hemispheres. Each ROI was visually assessed across all scans to ensure consistent delineation quality (Table 2).

Table 2. Regions of interest (ROIs) included in volumetric analysis.

NON-CORTICAL ROIs – 34 in Total	
Ventricular System ROIs (6)	Third ventricle
	Fourth Ventricle
	Inferior Lateral Ventricle (Right, Left)
	Lateral ventricle (Right, Left)
Basal Ganglia ROIs (8)	Accumbens Area (Right, Left)
	Caudate (Right, Left)
	Putamen (Right, Left)
	Pallidum (Right, Left)
Limbic System ROIs (6)	Amygdala (Right, Left)
	Hippocampus (Right, Left)
	Basal Forebrain (Right, Left)
Thalamic structure ROIs (4)	Thalamus Proper (Right, Left)
	Ventral Diencephalon (Right, Left)
Brainstem ROI (1)	Brainstem
Cerebellum (7)	Cerebellum Exterior (Right, Left)
	Cerebellum White Matter (Right, Left)
	Cerebellar Vermal Lobules I - V
	Cerebellar Vermal Lobules VI - VII
	Cerebellar Vermal Lobules VIII - X
White Matter ROIs (2)	Cerebral White Matter (Right, Left)
CORTICAL ROIs BY LOBE – 98 in Total	
Frontal lobe ROIs (36)	Frontal Pole (Right, Left)
	Superior Frontal Gyrus (Right, Left)
	Superior Frontal Gyrus Medial Segment (Right, Left)
	Middle Frontal Gyrus (Right, Left)
	Precentral Gyrus (Right, Left)
	Precentral Gyrus Medial Segment (Right, Left)
	Supplementary Motor Cortex (Right, Left)
	Opercular part of the Inferior Frontal Gyrus (Right, Left)
	Orbital part of the Inferior Frontal Gyrus (Right, Left)
	Triangular part of the Inferior Frontal Gyrus (Right, Left)
	Frontal Operculum (Right, Left)
	Anterior Orbital Gyrus (Right, Left)
	Lateral Orbital Gyrus (Right, Left)
	Medial Orbital Gyrus (Right, Left)
	Posterior Orbital Gyrus (Right, Left)
	Medial Frontal Cortex (Right, Left)
	Subcallosal Area (Right, Left)
	Gyrus Rectus (Right, Left)

Parietal Lobe ROIs (14)	Post Central Gyrus (Right, Left)
	Post Central Gyrus Medial Segment (Right, Left)
	Superior Parietal Lobule (Right, Left)
	Supramarginal Gyrus (Right, Left)
	Angular Gyrus (Right, Left)
	Precuneus (Right, Left)
	Parietal Operculum (Right, Left)
Temporal Lobe ROIs (22)	Superior Temporal Gyrus (Right, Left)
	Middle Temporal Gyrus (Right, Left)
	Inferior Temporal Gyrus (Right, Left)
	Temporal Pole (Right, Left)
	Transverse Temporal Gyrus (Right, Left)
	Planum Temporale (Right, Left)
	Planum Polare (Right, Left)
	Para hippocampal Gyrus (Right, Left)
	Ententorhinal Area (Right, Left)
	Anterior Insula (Right, Left)
	Posterior Insula (Right, Left)
Occipital Lobe (18)	Calcarine cortex (Right, Left)
	Cuneus (Right, Left)
	Lingual Gyrus (Right, Left)
	Fusiform Gyrus (Right, Left)
	Occipital Pole (Right, Left)
	Superior Occipital Gyrus (Right, Left)
	Middle Occipital Gyrus (Right, Left)
	Inferior Occipital Gyrus (Right, Left)
	Occipital Fusiform Gyrus (Right, Left)
Cingulate Lobe ROIs (6)	Anterior Cingulate Gyrus (Right, Left)
	Middle Cingulate Gyrus (Right, Left)
	Posterior Cingulate Gyrus (Right, Left)
Opercular region ROIs (2)	Central Operculum (Right, Left)

Non-cortical ROIs (n = 34) include structures from the ventricular system, basal ganglia, limbic system, thalamus, brainstem, cerebellum, and cerebral white matter. Cortical ROIs (n = 98) are grouped by anatomical lobes: frontal, parietal, temporal, occipital, cingulate, and opercular regions. All bilateral structures are reported separately for the right (R) and left (L) hemispheres.

Abbreviations: ROI = region of interest; R = right hemisphere; L = left hemisphere.

2.4.3. Multi-Plane Inspection of all ROIs

Each ROI was inspected across axial, coronal, and sagittal planes to support quality control and minimize view-dependent misinterpretation. Horizontal slices were optimal for visualizing subcortical structures and enabled bilateral inspection for comparative assessment (Figure 4a). Coronal views best captured cortical folding and lateral regions (Figure 4b), while sagittal slices provided clear visualization of midline structures (Figure 4c). Cerebral white matter ROIs were loaded individually to ensure precise delineation and reliable evaluation. These orthogonal views offered complementary anatomical context, allowing cross-validation of ROI boundaries and enhancing detection of segmentation errors, partial volume effects, or tissue misclassifications. This multi-plane approach was especially critical for cortical regions, where convoluted morphology can obscure subtle inaccuracies when viewed in isolation.

2.4.4. Grading of ROIs

To standardize the evaluation of ROI quality, a structured 0 to 3 grading scheme was implemented (Table 3, Figure 5). Scores were assigned through visual inspection of binary masks generated by MALP-EM, where each voxel is labelled 1 if it belongs to the ROI and 0 if it does not. Whenever image quality or registration was ambiguous, experienced analysts jointly re-examined the scan and agreed on the final score.

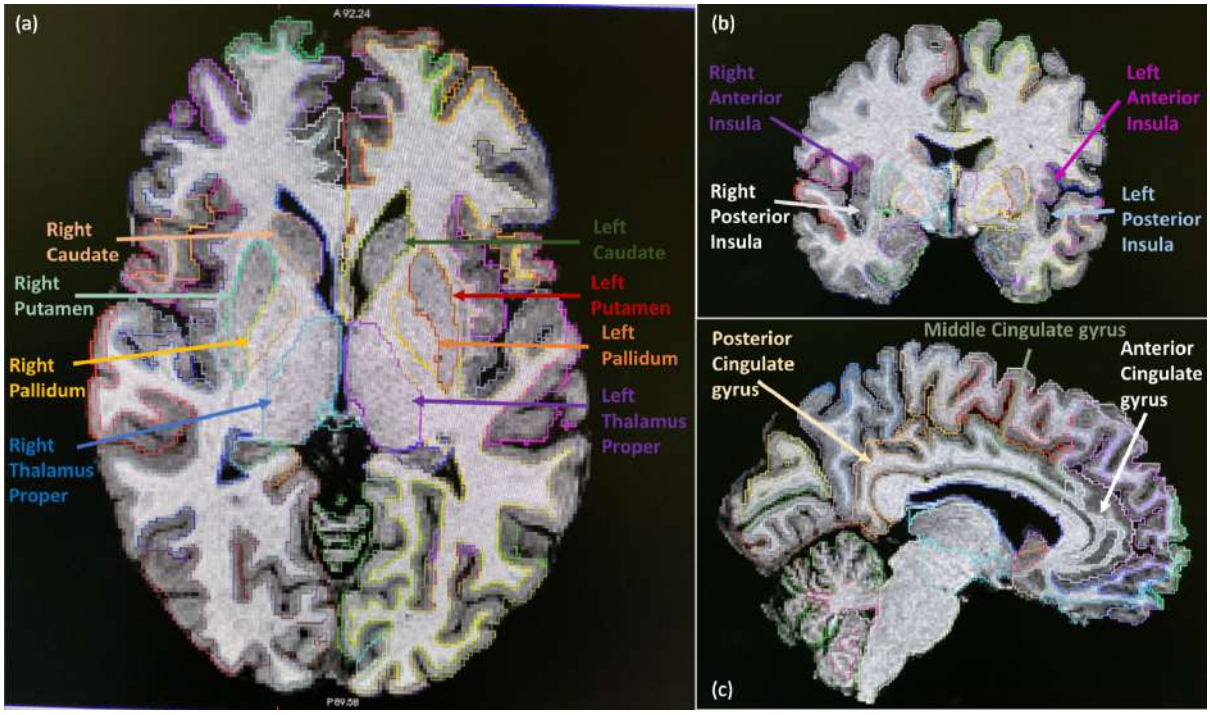

Figure 4. Multi-Plane ROI Inspection Across Orthogonal Views Using Freeview. Panel (a) displays horizontal slices optimized for subcortical regions, including the striatum, pallidum, and thalamus, with bilateral visualization. Panel (b) presents coronal slices, emphasizing cortical folding and lateral structures such as the insula. Panel (c) shows sagittal slices, best suited for midline cortical ROIs including the anterior, middle, and posterior cingulate gyri. ROIs were delineated using MALPEM and visualized in Freeview with color-coded overlays.

Table 3. ROI quality control grading criteria, associated scores, colour codes, and inclusion in statistical analysis.

Grade	Score	Colour code	Use in statistical analysis	Description
Fail	0		No	Gross errors in ROI delineation
Borderline Fail	1		No	The ROI shows consistent but minor issues with the 'tightness' of the segmentation fit, often due to partial volume effects
Borderline Pass	2		Yes	Minor issues with ROI boundaries
Pass	3		Yes	The ROI segmentation is accurate with no significant issues.

Grades range from Fail (score 0) to Pass (score 3) and are visually colour-coded in Excel to facilitate quick identification during data quality checks, red for Fail, pink for Borderline Fail, yellow for Borderline Pass, and green for Pass. These reflect the accuracy of region of interest (ROI) delineation. Grades 0 and 1 (Fail and Borderline Fail) are excluded from statistical analysis due to significant or consistent boundary errors. Grades 2 and 3 (Borderline Pass and Pass) are included, indicating acceptable or high-quality segmentations suitable for analysis.

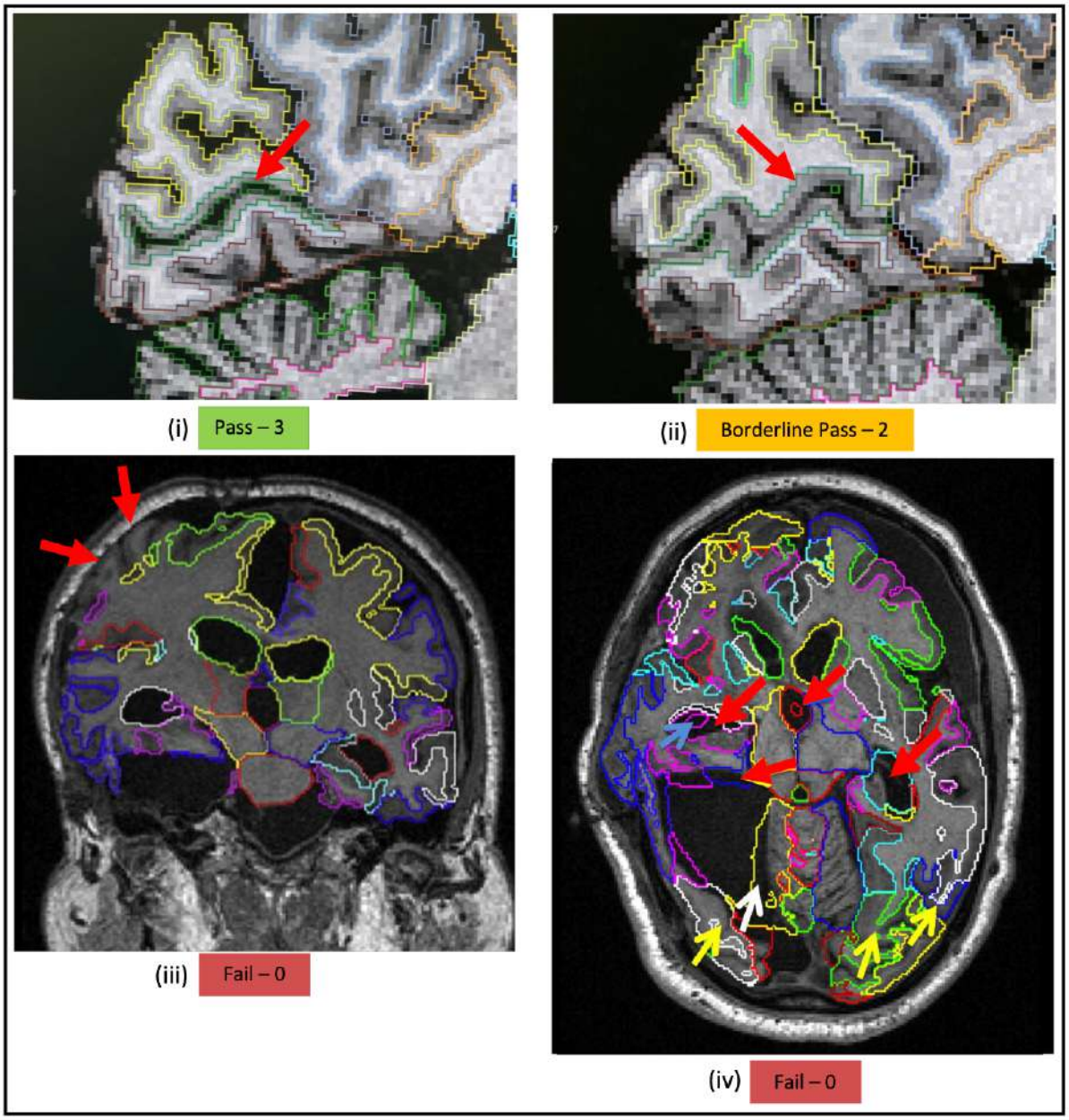

Figure 5. Example of ROI Grading Scheme. This figure illustrates the grading criteria used for assessing regions of interest (ROIs). Panel (i) shows an example of a well-defined ROI that meets all quality standards, receiving a grade of "pass" with a score of 3. Panel (ii) presents a borderline pass ROI with minor imperfections, scored as 2, which is still considered acceptable for analysis. In contrast, panels (iii) and (iv) display ROIs with significant errors, making them unsuitable for inclusion in analysis as these are graded as "fail" (score of 0) (adapted from Ledig *et al.*, 2015). It is important to note that these two examples were obtained from an external source, as no such poor-quality ROIs were present in our dataset, and no borderline fail cases could be found externally.

2.5. Statistical analysis

2.5.1. AIM 1: Between group differences in Atrophy Rates

2.5.1.1. Volume extraction

Post-QC, ROI volumes were extracted using MALP-EM's probabilistic segmentations, which assign each voxel a likelihood (0-1) of belonging to a specific anatomical region. Each scan yielded 132 ROI volumes at both visit 1 and visit 2. For analysis, left and right hemisphere ROIs were combined, resulting in 69 bilateral volumes per scan.

2.5.1.2. Calculations

For plotting cross-sectional data, in order to account for individual differences in head size, ROI volumes (in mls) were normalised using intracranial volume (ICV) - equation a. For longitudinal analysis, annualised atrophy rates were calculated in absolute terms (mls per year) as the primary readout (c). To enable comparison with some previous studies, proportional atrophy rates were also calculated (volume change as a percentage of baseline volume for the region; d).

a. $$TIV\ adjusted\ ROI\ volume\ at\ visit\ 1 = \left(\frac{ROI\ volume\ at\ visit\ 1}{TIV\ of\ the\ participnat}\right) x\ 100$$

b. $$TIV\ adjusted\ ROI\ volume\ at\ visit\ 2 = \left(\frac{ROI\ volume\ at\ visit\ 2}{TIV\ of\ the\ participnat}\right) x\ 100$$

c. $$Annulaised\ Atrophy\ Rate\ (ml/year) = \frac{Total\ volume\ at\ visit\ 1 - Total\ volume\ at\ visit\ 2}{Scan\ interval\ (years)}$$

d. $$Annualised\ Atrophy\ Rate\ (\%peryear) =$$

$$\left(\frac{\frac{Total\ volume\ at\ visit\ 1\ -\ Total\ volume\ at\ visit\ 2}{Total\ volume\ at\ visit\ 1}}{Scan\ Interval\ (years)}\right) x\ 100$$

2.5.1.3. Regression Analysis

Between-group differences in annualised atrophy rates (ml/year and %/year scale) across 69 ROIs were assessed using linear regression in STATA v18.5 (StataCorp, TX, USA), adjusting for age and sex. To account for multiple comparisons, False Discovery Rate (FDR) correction was applied using the Benjamini-Hochberg

procedure. FDR-adjusted p-value (q-values) was reported for each ROI, with statistical significance defined as q<0.15 consistent with Scahill et al. (2025*)*.

2.5.2. AIM 2: Longitudinal Effect Size

2.5.2.1. Effect size estimation

To complement significance testing across 69 ROIs, Cohen's d effect sizes (ES) to quantify between-group differences in standardized units. ES were classified as small (d≤0.2), medium (0.2<d<0.8), or large (d≥0.8) (Cohen, 2013).

Longitudinal ES with 95% CIs were calculated as the covariate-adjusted mean difference in annual change between HDGE and controls, divided by the residual SD of change in HDGE. Unit free ES enables cross-metric comparisons across brain regions. ES^2 is inversely proportional to sample size requirements, assuming a fully effective treatment reduces the mean rate of change in HD cases to that observed in controls, without altering variability (Hobbs *et al.*, 2015). CIs were calculated using bias corrected and accelerated (BCa) bootstrapping (2000 replications). An ES of 2 implies that the mean change in HD is two SD away from that in controls.

2.5.3. AIM 3: Association of regional Atrophy Rates with HD-ISS Stage and CAP100

Linear regression assessed the relationship between regional atrophy rates and both HD-ISS staging (0 vs 1) and CAP100 scores across 16 significant ROIs, identified from the initial 69-region analysis. All models were run in STATA v18.5, adjusting for age and sex. One Stage 2 participant was excluded due to insufficient group size. For each ROI, regression coefficients were reported alongside 95% CIs, p-values ($p < 0.05$), and FDR-adjusted q-values ($q < 0.15$).

Chapter 3. Results

3.1. Visual quality control

Visual QC and grading of ROIs extracted from MALP-EM (Table 2) were most reliable in regions with clear GM-CSF and relatively distinct GM-WM boundaries. The putamen showed a continuous GM-WM boundary, though slightly less distinct than the caudate but still providing sufficient contrast for accurate segmentations. Other subcortical structures like the pallidum and thalamus exhibited less well-defined GM-WM contrast on T1-weighted MRI, making QC more challenging than for other deep GM structures. All subcortical GM ROIs passed QC (Grade 3) at both timepoints, with no gross segmentation's errors. Few of these subcortical ROIs like the putamen, pallidum, and thalamus, even though graded a pass, were still peer-reviewed as they displayed perivascular spaces (PVS) in inferior slices (Figure 6), which did not affect segmentation quality.

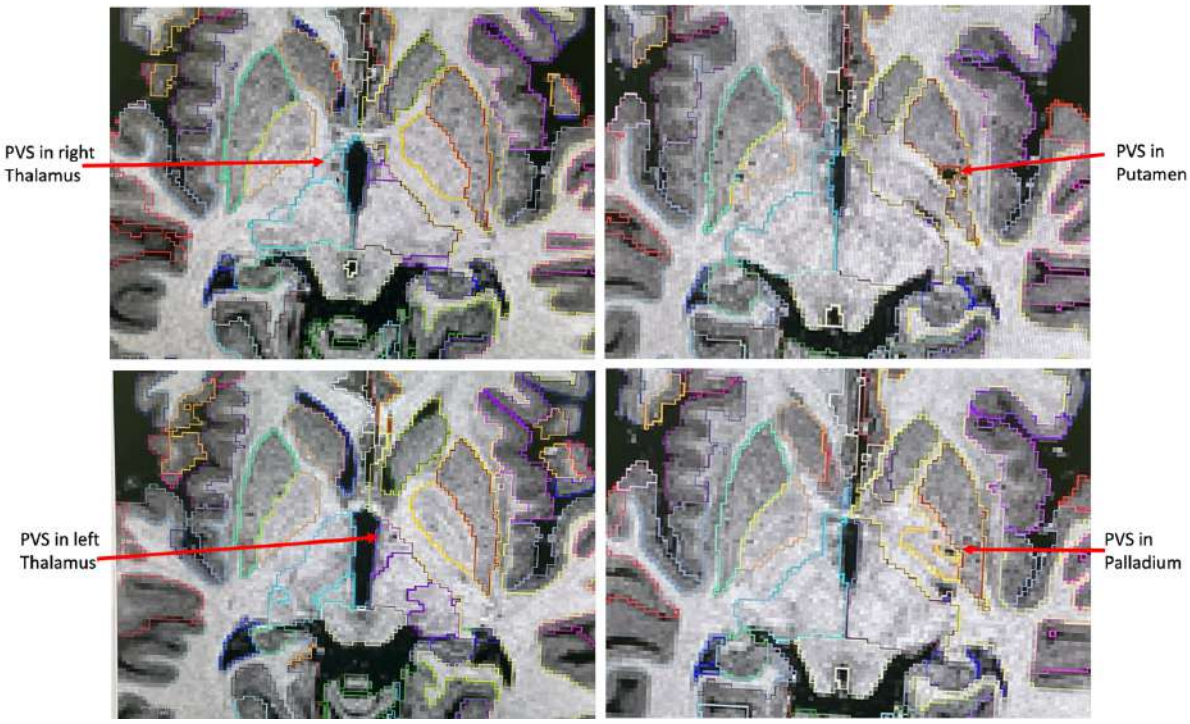

Figure 6. Perivascular Spaces (PVS) in Deep Grey Matter Regions. MRI slices showing examples of PVS in the putamen, thalamus, and pallidum. PVS were more noticeable in inferior slices, especially within the basal ganglia. Their presence did not appear to grossly impact segmentation quality.

Overall, visual QC of cortical ROIs (Table 3) was more challenging than for subcortical regions due to the cortex's highly convoluted anatomy and poor boundary definition between adjacent areas. As a result, cortical ROIs were reviewed with extra caution across all planes to be ensure thorough and comprehensive grading (Table 4).

Of the 69 bilateral ROIs assessed, 60 were graded as "Pass" (Score 3) and 9 as "Borderline Pass" (Score 2). ROIs for the brainstem (n = 1), cerebellum (n = 5), and cerebral white matter (n = 1) were consistently graded as borderline pass across participants. Within the occipital lobe, the calcarine cortex passed QC in 30% of participants, with 70% graded borderline pass, while the lingual gyrus showed the opposite pattern, 70% pass and 30% borderline pass (Table 4).

Table 4. Summary of ROI-level quality control (QC) grading across 69 bilateral regions of interest (ROIs) in 88 participants.

ROI domain	No. of ROIs	No of ROIs per QC grade			
		Pass Score 3	Borderline Pass Score 2	Borderline Fail Score 1	Fail Score 0
Non-cortical ROIs					
Ventricular System	4	4	0	0	0
Basal Ganglia	4	4	0	0	0
Limbic System	3	3	0	0	0
Thalamic structure	2	2	0	0	0
Brainstem	1	0	1(88/88)	0	0
Cerebellum	5	0	5(88/88)	0	0
White Matter	1	0	1(88/88)	0	0
Cortical ROIs					
Frontal lobe	18	18	0	0	0
Parietal Lobe	7	7	0	0	0
Temporal Lobe	11	11	0	0	0
Occipital Lobe	9	7	2	0	0
	Calcarine cortex	✓ (26/88)	✓ (62/88)	X	X
	Cuneus	✓	X	X	X
	Lingual Gyrus	✓ (61/88)	✓ (27/88)	X	X
	Fusiform Gyrus	✓	X	X	X
	Occipital Pole	✓	X	X	X
	Superior Occipital Gyrus	✓	X	X	X
	Middle Occipital Gyrus	✓	X	X	X
	Inferior Occipital Gyrus	✓	X	X	X
	Occipital Fusiform gyrus	✓	X	X	X

Cingulate Lobe	3	3	0	0	0
Opercular region	1	1	0	0	0

ROIs are grouped by anatomical region and classified by QC grade: Grade 3 (Pass), Grade 2 (Borderline Pass), Grade 1 (Borderline Fail), and Grade 0 (Fail). Non-cortical ROIs include ventricular, subcortical, brainstem, cerebellar, and white matter structures. Cortical ROIs are organized by lobe. For ROIs with variable QC outcomes across participants (e.g., brainstem, cerebellum, white matter, and occipital subregions), participant-level grading counts are indicated in parentheses (e.g., 88/88). Occipital lobe subregions are further detailed to show the distribution of QC grades across participants. Symbols denote QC outcomes: ✓ = assigned to the QC grade, X = not assigned to this QC grade.

Abbreviations: ROI, region of interest; L+R, left and right hemispheres combined; QC, quality control; n = number.

The calcarine cortex (Figure 7) and lingual gyrus were assigned a borderline pass (Score 2) due to segmentation inconsistencies across participants. In some cases, the ROIs included adjacent CSF; in others, they captured only the cortical boundaries. Minor boundary issues, such as a 'loose' fit, suggesting volume over-estimation, were observed but the segmentation generally identified the correct ROI. Following peer review, both ROIs were deemed acceptable for analysis.

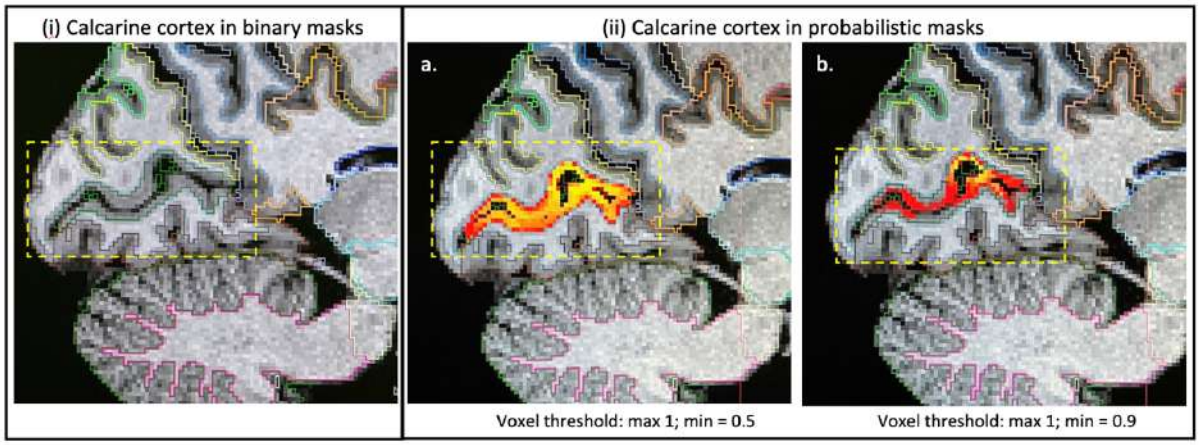

Figure 7. Visualisation of the Calcarine Cortex ROI Using Binary and Probabilistic masks. Two visualisation approaches were applied to the calcarine cortex ROI, Panel (i) Binary masks (threshold: max = 1; min = 0) with ROI boundaries as outline only; Panel (ii) Probabilistic masks (a. threshold: max = 1; min = 0.5; b. threshold: max = 1; min = 0.9) as heat map overlayed on the binary mask, outline only. The calcarine cortex ROI was graded as a borderline pass since these visualizations revealed some segmentation limitations. However, after peer review, it was accepted for statistical analysis because it correctly identified the intended ROI.

The cerebellar grey matter ROI was graded as borderline pass due to its ‘loose’ fit and dense foliation, which complicated segmentation and led to inclusion of adjacent CSF. They were also viewed and checked using probabilistic maps, with adjusted thresholds to provide additional information to guide QC and were concluded to be acceptable for analysis (Figure 8).

Global structures including cerebral and cerebellar white matter (Figure 8), as well as brainstem were also graded as borderline pass, reflecting a slightly ‘tight’ fit, suggesting volume under-estimation. Nonetheless, each segmentation correctly identified the ROI and was approved for analysis.

No ROIs received a Borderline Fail or Fail grade, indicating the absence of gross or consistent boundary errors. Consequently, all ROIs from all participants in both visit 1 and 2 were included in the final statistical analysis.

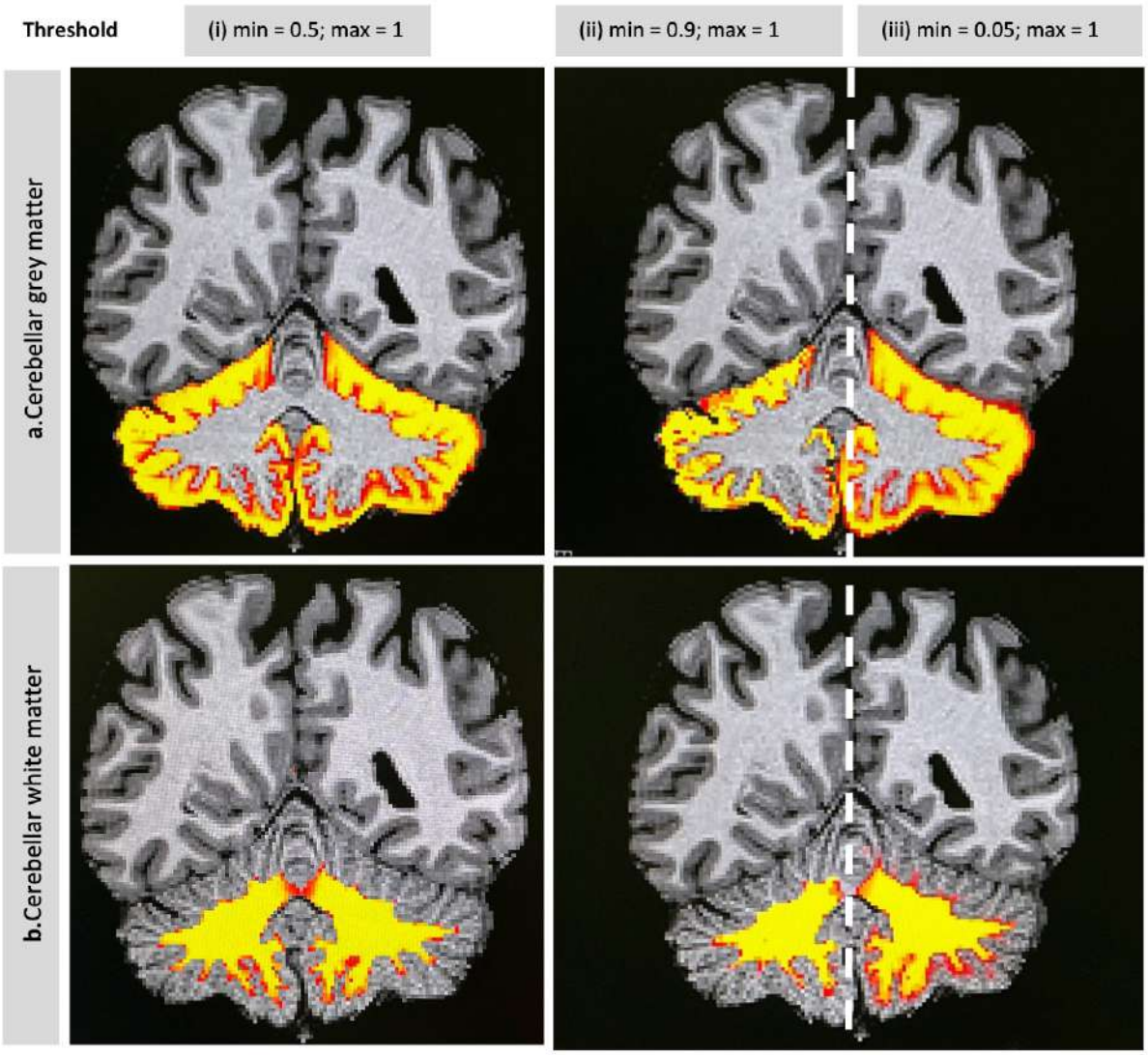

Figure 8. Segmentation of Cerebellar Grey and White Matter. MRI slices showing cerebellar grey matter (a.) and white matter (b.) segmentations using probabilistic masks as heat maps. Cerebellar grey matter appeared 'loose' and white matter appeared 'tight' fit, reflecting challenges from partial volume effects and dense foliation. For grading these regions ROIs, they were viewed using probabilistic masks adjusting for different thresholds, (i) similar to the binary masks threshold, (ii) a stricter and (iii) loose threshold to better visual and assess the segmentation for QC.

All ventricular ROIs were well segmented and graded as pass. QC was more challenging in three cases due to anatomical variation: (1) asymmetrical brain anatomy in one participant and (2) unusually enlarged ventricles in two participants (Figure 9). Despite these deviations, MALP-EM generated high-quality segmentations, that passed visual QC, demonstrating the technique's robustness to anatomical variability.

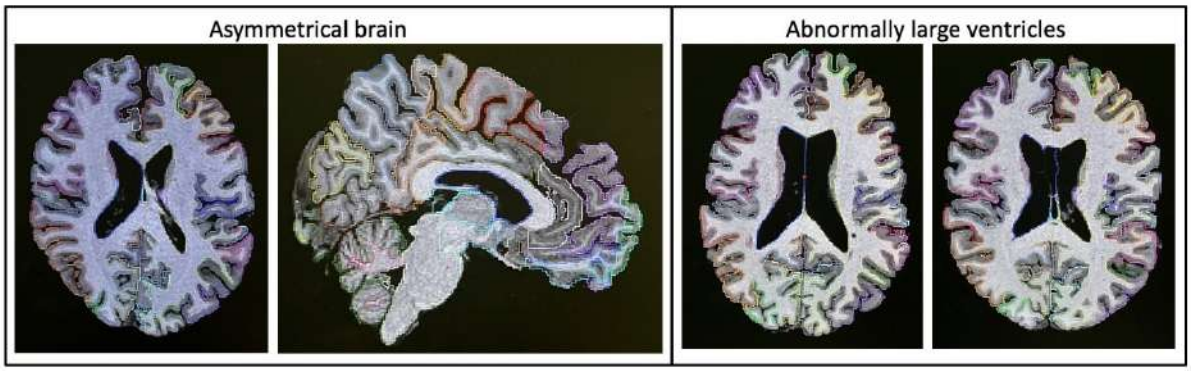

Figure 9. Anatomical Variability in T1-Weighted MRI: Asymmetry and Ventricular Enlargement. Examples of structural variation observed during visual QC, including asymmetrical brain anatomy and enlarged ventricles. These features, while potentially challenging for automated segmentation, did not compromise output quality. Segmentations remained consistently accurate across individuals with pronounced anatomical differences, demonstrating methodological robustness across different anatomies.

3.2. Group demographics

The final cohort included 54 participants in the HDGE group and 34 healthy controls (Table 5). Groups were well-matched for age and sex, with no significant differences in mean age at baseline (HDGE: 29.85±5.65 years; controls: 30.11±5.87 years; $p = 0.84$) or follow-up (HDGE: 34.55±5.95 years; controls: 34.88±5.95 years; $p = 0.80$). Sex distribution was balanced (HDGE: 27 males, 27 females; controls: 14 males, 20 females), with no significant group differences as indicated by a Pearson chi-square test (χ^2 (1, N = 88) = 0.653, $p = 0.419$) (Table 5).

In the HDGE group, the mean CAG repeat length was 42.31±1.53, consistent with the pathogenic range for HD. Mean DBS increased from 198.11±36.89 at visit 1 to 230.28±41.56 at visit 2, reflecting the progressive nature of disease pathology.

Overall, the HDGE and control groups were demographically comparable, minimizing the risk of confounding in subsequent analyses.

Table 5. Participant demographics, sex distribution, age, CAG repeat length, and DBS scores for HDGE and controls participants.

	HDGE (n = 54)	Control (n = 34)	p-value
Age at visit 1 (years)	29.85 (5.65)	30.11 (5.87)	0.84
Age at visit 2 (years)	34.55 (5.95	34.88 (5.95)	0.80
Sex			
Male (n = 41) **Female (n = 47)**	27 (50%) 27 (50%)	14 (41.18%) 20 (58.82%)	0.42
CAG repeat length	42.31 (1.53)	N/A	N/A
DBS at visit 1	198.11 (36.89)	N/A	N/A
DBS at visit 2	230.28 (41.56)	N/A	N/A

Summary of sex distribution, age at each visit, CAG repeat length, and Disease Burden Scores (DBS) for individuals with HD gene expansion (HDGE; *n* = 54) and controls (*n* = 34). Age, CAG repeat length, and DBS are reported as mean ± standard deviation (SD). Sex is presented as number and percentage of participants per group. p-values reflect between-group comparisons where applicable. CAG and DBS values are reported for HDGE only; N/A = not applicable.

Abbreviations: HDGE = Huntington's disease gene expansion; DBS = Disease Burden Score; CAG = Cytosine-Adenine-Guanine repeat; SD = standard deviation; N/A = not applicable; M/F = male/female.

3.3. AIM 1: Between group differences in Atrophy Rates

Significant group differences in annualised atrophy rates were observed in 16 of 69 ROIs between HDGE and controls, after adjusting for age, sex and correcting for multiple comparisons using FDR (Table 6; $p<0.05$, $q<0.15$,).

Table 6. Annualised Atrophy Rates (mls/year) in Controls and HDGE participants.

Annualised atrophy rates (mls per year)						
ROI	**Control (n = 34)**	**HDGE (n = 54)**	**Between group diff. in atrophy rate (adjusted for age & sex)**			
			Mean (HDGE – Controls)	**95% CI**	**p-value**	**q-value**
Caudate	0.021 (0.022), 0.013 to 0.028	0.106 (0.072), 0.086 to 0.125	0.082	0.062 to 0.103	4.74×10^{-12}	3.27×10^{-10}
Putamen	0.043 (0.036), 0.031 to 0.056	0.172 (0.113), 0.141 to 0.203	0.124	0.092 to 0.157	3.48×10^{-11}	2.37×10^{-9}
Pallidum	0.010 (0.018), 0.004 to 0.017	0.045 (0.037), 0.035 to 0.055	0.035	0.023 to 0.046	5.23×10^{-8}	3.51×10^{-6}
Third ventricle	0.0004 (0.0142), -0.0046 to 0.0054	-0.018 (0.023), -0.024 to -0.011	-0.017	-0.025 to -0.010	5.2×10^{-6}	0.0003
Lateral ventricle	-0.047 (0.213), -0.121 to 0.027	-0.368 (0.461), -0.494 to -0.242	-0.309	-0.442 to -0.175	0.00002	0.001
Cerebral white matter	-1.567 (1.202), -1.987 to -1.148	-0.643 (1.551), -1.067 to -0.220	0.983	0.507 to 1.459	0.0001	0.006
Angular gyrus	0.074 (0.078), 0.046 to 0.101	0.143 (0.073), 0.123 to 0.163	0.067	0.033 to 0.101	0.0002	0.012
Triangular part of the inferior frontal gyrus	0.042 (0.036), 0.030 to 0.055	0.075 (0.039), 0.064 to 0.085	0.031	0.014 to 0.047	0.0003	0.019
Ventral diencephalon	0.005 (0.024), -0.004 to 0.013	0.026 (0.031), 0.018 to 0.035	0.021	0.010 to 0.032	0.0004	0.025
Superior frontal gyrus medial segment	0.051 (0.051), 0.033 to 0.068	0.092 (0.051), 0.078 to 0.105	0.037	0.016 to 0.058	0.001	0.037
Calcarine cortex	-0.029 (0.038), -0.043 to -0.016	0.001 (0.035), -0.009 to 0.010	0.029	0.013 to 0.046	0.001	0.038
Parietal operculum	0.011 (0.016), 0.005 to 0.016	0.025 (0.024), 0.019 to 0.032	0.014	0.006 to 0.023	0.001	0.038
Accumbens area	-0.0001 (0.010), -0.0035 to 0.0033	0.008 (0.011), 0.005 to 0.011	0.008	0.004 to 0.013	0.001	0.038

Brainstem	-0.074 (0.053), -0.093 to -0.055	-0.033 (0.068), -0.052 to -0.014	0.041	0.016 to 0.067	0.002	0.104
Lingual gyrus	-0.037 (0.058), -0.058 to -0.017	0.006 (0.083), -0.017 to 0.029	0.046	0.017 to 0.074	0.002	0.128
Precuneus	0.020 (0.065), -0.003 to 0.043	0.069 (0.075), 0.049 to 0.090	0.045	0.016 to 0.074	0.003	0.142
Occipital fusiform gyrus	0.075 (0.040), 0.061 to 0.089	0.107 (0.065), 0.089 to 0.124	0.030	0.008 to 0.051	0.007	0.379
Frontal operculum	0.018 (0.018), 0.012 to 0.024	0.028 (0.016), 0.024 to 0.033	0.010	0.002 to 0.018	0.011	0.550
Planumtemporale	0.006 (0.023), -0.002 to 0.015	0.018 (0.021), 0.013 to 0.024	0.011	0.001 to 0.021	0.028	0.942
Middle frontal gyrus	0.228 (0.124), 0.185 to 0.272	0.289 (0.110), 0.259 to 0.319	0.056	0.004 to 0.108	0.036	0.942
Post central gyrus	0.129 (0.115), 0.089 to 0.170	0.179 (0.087), 0.155 to 0.203	0.046	0.001 to 0.092	0.045	0.942
Anterior cingulate gyrus	0.015 (0.041), 0.000 to 0.029	0.035 (0.039), 0.024 to 0.045	0.020	0.003 to 0.038	0.026	0.942
Superior parietal lobule	0.119 (0.103), 0.083 to 0.155	0.160 (0.079), 0.139 to 0.181	0.037	-0.002 to 0.077	0.065	0.942
Supramarginal gyrus	0.082 (0.076), 0.056 to 0.109	0.122 (0.079), 0.101 to 0.144	0.036	0.003 to 0.070	0.032	0.942
Lateral orbital gyrus	0.033 (0.021), 0.025 to 0.040	0.043 (0.021), 0.037 to 0.049	0.010	0.0002 to 0.0194	0.044	0.942
Frontal pole	0.045 (0.055), 0.025 to 0.064	0.067 (0.052), 0.053 to 0.081	0.023	-0.001 to 0.047	0.060	0.942
Precentral gyrus medial segment	0.011 (0.029), 0.001 to 0.021	0.027 (0.035), 0.017 to 0.036	0.015	0.001 to 0.029	0.031	0.942
Middle cingulate gyrus	-0.002 (0.044), -0.017 to 0.013	0.013 (0.035), 0.004 to 0.023	0.015	-0.003 to 0.033	0.099	0.942
Precentral gyrus	0.195 (0.134), 0.148 to 0.242	0.244 (0.108), 0.215 to 0.274	0.045	-0.009 to 0.098	0.102	0.942
Superior temporal gyrus	0.026 (0.059), 0.006 to 0.047	0.051 (0.060), 0.035 to 0.067	0.023	-0.003 to 0.050	0.084	0.942
Parahippocampal gyrus	0.002 (0.032), -0.010 to 0.013	-0.008 (0.023), -0.014 to -0.002	-0.010	-0.022 to 0.003	0.135	0.942

Cerebellar vermal lobules VI-VII	-0.005 (0.012), -0.009 to -0.001	-0.001 (0.012), -0.004 to 0.003	0.004	-0.001 to 0.009	0.110	0.942
Medial frontal cortex	0.010 (0.022), 0.002 to 0.018	0.020 (0.027), 0.013 to 0.027	0.009	-0.001 to 0.020	0.087	0.942
Hippocampus	-0.001 (0.020), -0.008 to 0.006	-0.008 (0.022), -0.014 to -0.002	-0.007	-0.016 to 0.002	0.117	0.942
Fusiform gyrus	-0.005 (0.047), -0.021 to 0.011	0.011 (0.055), -0.004 to 0.026	0.016	-0.007 to 0.038	0.164	0.942
Opercular part of the inferior frontal gyrus	0.038 (0.029), 0.027 to 0.048	0.046 (0.031), 0.038 to 0.055	0.009	-0.005 to 0.022	0.217	0.942
Anterior insula	0.019 (0.025), 0.010 to 0.028	0.028 (0.031), 0.020 to 0.036	0.008	-0.004 to 0.020	0.174	0.942
Superior frontal gyrus	0.260 (0.110), 0.221 to 0.298	0.291 (0.106), 0.262 to 0.320	0.028	-0.022 to 0.078	0.264	0.942
Inferior occipital gyrus	0.140 (0.087), 0.109 to 0.170	0.159 (0.066), 0.141 to 0.177	0.014	-0.017 to 0.046	0.362	0.942
Inferior temporal gyrus	0.101 (0.101), 0.066 to 0.136	0.124 (0.085), 0.101 to 0.148	0.018	-0.022 to 0.059	0.376	0.942
Anterior orbital gyrus	0.015 (0.018), 0.009 to 0.021	0.020 (0.019), 0.015 to 0.025	0.004	-0.004 to 0.012	0.333	0.942
Inferior lateral ventricle	0.002 (0.012), -0.002 to 0.006	0.005 (0.013), 0.001 to 0.008	0.003	-0.003 to 0.008	0.342	0.942
Supplementary motor cortex	0.082 (0.046), 0.066 to 0.098	0.091 (0.045), 0.079 to 0.103	0.009	-0.011 to 0.029	0.386	0.942
Thalamus	0.053 (0.066), 0.030 to 0.076	0.064 (0.065), 0.046 to 0.082	0.012	-0.016 to 0.039	0.392	0.942
Fourth ventricle	0.001 (0.020), -0.006 to 0.008	-0.002 (0.014), -0.006 to 0.002	-0.003	-0.010 to 0.005	0.504	0.942
Cuneus	-0.048 (0.057), -0.068 to -0.028	-0.038 (0.048), -0.051 to -0.025	0.008	-0.015 to 0.032	0.473	0.942
Amygdala	-0.002 (0.017), -0.007 to 0.004	0.002 (0.018), -0.003 to 0.007	0.003	-0.004 to 0.010	0.406	0.942
Middle temporal gyrus	0.115 (0.107), 0.077 to 0.152	0.136 (0.103), 0.108 to 0.164	0.017	-0.030 to 0.064	0.483	0.942

Subcallosal area	-0.003 (0.014), -0.008 to 0.002	-0.006 (0.020), -0.011 to -0.0001	-0.003	-0.010 to 0.004	0.390	0.942
Basal forebrain	0.005 (0.009), 0.001 to 0.008	0.007 (0.016), 0.003 to 0.011	0.002	-0.003 to 0.007	0.432	0.942
Middle occipital gyrus	0.044 (0.057), 0.025 to 0.064	0.055 (0.063), 0.038 to 0.072	0.008	-0.018 to 0.034	0.524	0.942
Cerebellum white matter	-0.333 (0.374), -0.463 to -0.203	-0.276 (0.498), -0.412 to -0.140	0.062	-0.130 to 0.254	0.521	0.942
Cerebellum exterior	0.382 (0.424), 0.234 to 0.530	0.328 (0.475), 0.198 to 0.458	-0.062	-0.263 to 0.139	0.542	0.942
Entorhinal area	-0.001 (0.031), -0.011 to 0.010	0.004 (0.027), -0.004 to 0.011	0.003	-0.009 to 0.016	0.609	0.942
Posterior orbital gyrus	0.017 (0.027), 0.007 to 0.026	0.021 (0.032), 0.012 to 0.029	0.004	-0.009 to 0.017	0.554	0.942
Transverse temporal gyrus	0.007 (0.015), 0.002 to 0.012	0.009 (0.018), 0.004 to 0.014	0.002	-0.005 to 0.009	0.612	0.942
Cerebellar vermal lobules VIII-X	-0.004 (0.012), -0.008 to 0.0004	-0.003 (0.011), -0.006 to 0.000	0.001	-0.004 to 0.006	0.660	0.942
Central operculum	0.021 (0.027), 0.012 to 0.030	0.025 (0.029), 0.017 to 0.033	0.003	-0.009 to 0.015	0.663	0.942
Planumpolare	-0.003 (0.023), -0.011 to 0.005	-0.004 (0.020), -0.010 to 0.001	-0.001	-0.011 to 0.008	0.761	0.942
Occipital pole	-0.006 (0.074), -0.032 to 0.019	-0.010 (0.079), -0.031 to 0.011	-0.005	-0.038 to 0.028	0.755	0.942
Temporal pole	0.198 (0.097), 0.164 to 0.232	0.209 (0.093), 0.184 to 0.235	0.006	-0.034 to 0.045	0.776	0.942
Postcentral gyrus medial segment	0.018 (0.022), 0.011 to 0.026	0.020 (0.021), 0.014 to 0.026	0.001	-0.008 to 0.011	0.794	0.942
Cerebellar vermal lobules I-V	-0.018 (0.022), -0.025 to -0.010	-0.020 (0.024), -0.026 to -0.013	-0.001	-0.011 to 0.009	0.840	0.942
Orbital part of the inferior frontal gyrus	0.025 (0.020), 0.018 to 0.032	0.026 (0.018), 0.021 to 0.031	0.001	-0.008 to 0.009	0.853	0.942
Posterior cingulate gyrus	0.015 (0.035), 0.003 to 0.027	0.017 (0.033), 0.008 to 0.026	0.001	-0.014 to 0.016	0.889	0.942

Medial orbital gyrus	0.029 (0.039), 0.016 to 0.043	0.030 (0.042), 0.019 to 0.042	-0.001	-0.018 to 0.016	0.911	0.942
Superior occipital gyrus	0.004 (0.044), -0.012 to 0.019	0.004 (0.059), -0.012 to 0.020	-0.001	-0.024 to 0.021	0.894	0.942
Gyrus rectus	0.028 (0.026), 0.019 to 0.037	0.028 (0.029), 0.020 to 0.036	-0.001	-0.012 to 0.011	0.920	0.942
Posterior insula	-0.004 (0.016), -0.009 to 0.002	-0.003 (0.018), -0.008 to 0.002	0.0003	-0.007 to 0.007	0.942	0.942

This table reports mean annualised atrophy rates (mls/year) for each ROI, with SD in parentheses and 95% CI. Data are presented separately for control participants (n = 34) and HDGE participants (n = 54). Between-group differences in mean atrophy rates are shown alongside p-values adjusted for age and sex, and FDR-corrected q-values to account for multiple comparisons. ROIs are ranked by ascending statistical significance (FDR-corrected q-values, threshold at $q \leq 0.15$). 61/69 ROI names in bold indicate statistically significant group differences based on the FDR threshold.

Abbreviations: mls: millilitre; ROI: Region of Interest; n: number; HDGE: Huntington Disease Gene Expansion group; FDR: False Discovery Rate; diff.: difference; CI: Confidence interval; Ventral diencephalon: Ventral diencephalon.

3.3.1. Subcortical deep grey matter regions

As expected, subtle but highly significant between-group differences in atrophy rates were observed in the caudate (p = 4.74x10^{-12}, FDR = 3.27x10^{-10}) and putamen (p = 3.48x10^{-11}, FDR = 2.37x10^{-9}) compared with controls (Table 6, Figure 10. a-b: i). HDGE participants showed mean annualised volume loss rates of 1.48% in the caudate and 1.85% in the putamen, five to seven times higher than controls (0.26% and 0.45%, respectively).

Other subcortical grey matter regions also showed significant, though subtler increases in atrophy rates of HDGE, including the pallidum (p = 5.23x10^{-8}, FDR = 3.51x10^{-6}), nucleus accumbens (p = 0.001, FDR = 0.038) and ventral diencephalon (p = 0.0004, FDR = 0.025) (Figure 10. b-e: i). Although changes were less pronounced than in the striatum, annualised percent volume loss revealed consistent subcortical vulnerability in HDGE: pallidum (1.58% vs. 0.35%), accumbens area (1.05% vs. 0.03%), and ventral diencephalon (0.27% vs. 0.05%) showed subtle increases relative to controls. These percent-based metrics complement the absolute volumetric findings. Detailed statistics, including confidence intervals and effect sizes, are in supplementary Table 1.

No elevated atrophy rates were observed in other subcortical GM regions, such as the thalamus, in HDGE compared with controls (Table 6).

Spaghetti plots of regional volume (% of ICV) against age, illustrate individual-level longitudinal data for the six subcortical grey matter ROIs (Figure 10). Controls showed relatively flat trajectories with minimal volume change over time, consistent with expectations for this young adult cohort (Figure 10. a-e: ii). In contrast, HDGE participants exhibited clear downward trajectories in the caudate, putamen and pallidum (Figure 10. a-e: iii), suggestive of early atrophy. Despite significant group-level differences in mean trajectories (Table 6), cross-sectional volumes showed substantial overlap between HDGE and controls (Figure 10).

Interestingly, the single HD-ISS stage 2 participant (denoted in blue, Figure 10, iii) had the smallest volume for the caudate, putamen, and pallidum across all participants. This is expected for the caudate and putamen, as minimum volumes define the transition from HD-ISS stage 0 to stage 1 (Tabrizi *et al.*, 2022); however, it is

noteworthy that the pallidum, showing the third most significant group between differences, also followed this pattern, unlike other subcortical grey matter regions.

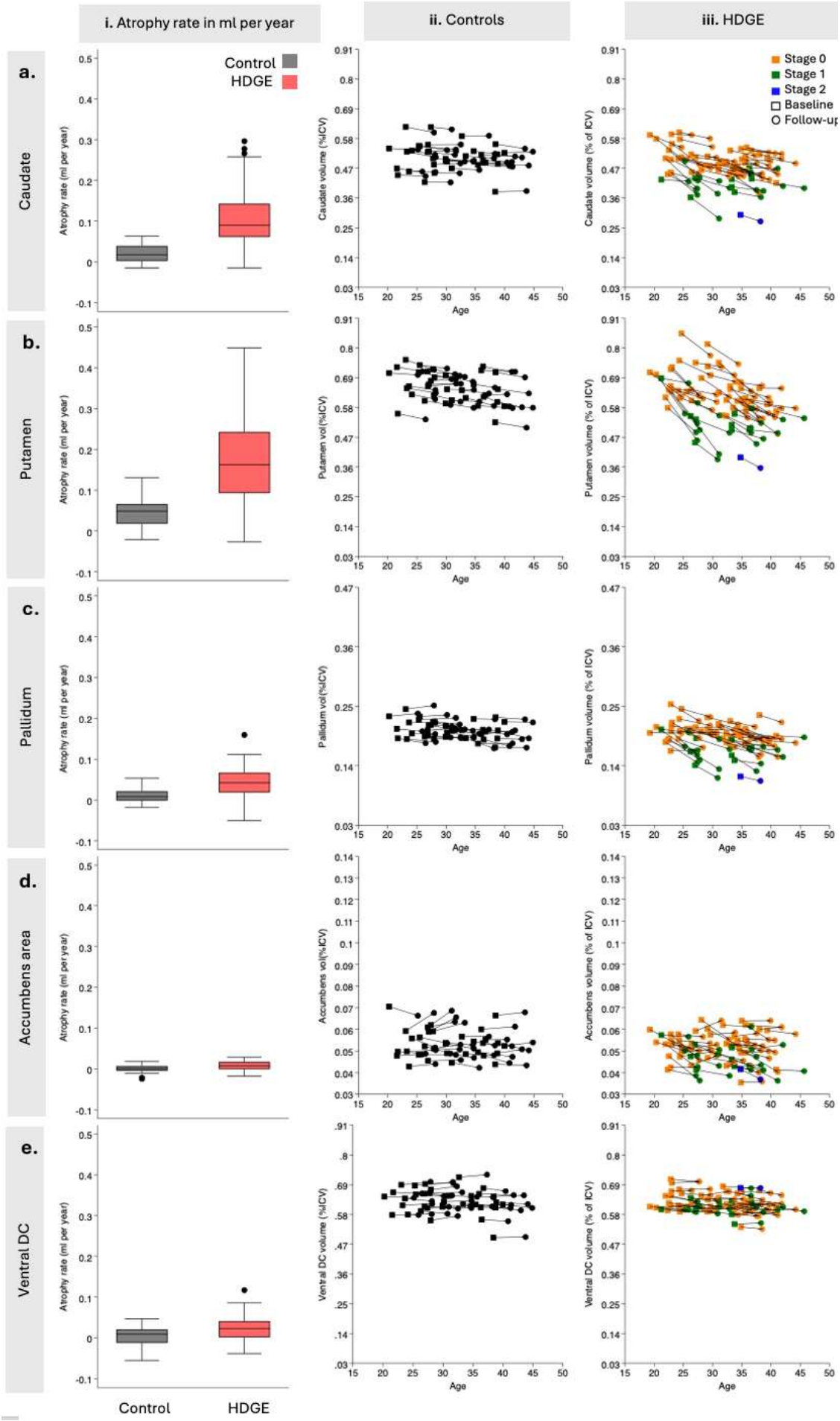
i. Atrophy rate in ml per year
ii. Controls
iii. HDGE
a.
Caudate
b.
Putamen
c.
Pallidum
d.
Accumbens area
e.
Ventral DC
Control
HDGE
Atrophy rate (ml per year)
Caudate volume (%ICV)
Putamen vol (%ICV)
Pallidum vol (%ICV)
Accumbens vol (%ICV)
Ventral DC volume (%ICV)
Caudate volume (% of ICV)
Putamen volume (% of ICV)
Pallidum volume (% of ICV)
Accumbens volume (% of ICV)
Ventral DC volume (% of ICV)
Age
Stage 0
Stage 1
Stage 2
Baseline
Follow-up

Figure 10. Annualized Atrophy Rates and Age-Related Volume Trajectories in HDGE and Control Participants in Subcortical ROIs. Annualized atrophy rates (age- and sex-adjusted, FDR-corrected; Table 3) are shown for six subcortical regions: (a) caudate, (b) putamen, (c) pallidum, (d) accumbens, and (e) ventral diencephalon, comparing HDGE (n = 54; red) and controls (n = 34; grey). Box plot displays group means (horizontal line), interquartile range (box), and whiskers (1.5xIQR), and outliers (filled circles). Scatterplots show regional volumes (%ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green) and 2 (blue) (iii). Longitudinal baseline and follow-up data are connected by black lines; squares and circles indicate baseline and follow-up, respectively. The pallidum (c, ii–iii) y-axis is zoomed to one tick interval at the same scale as other regions, while the accumbens (d, ii–iii) axes are truncated to highlight age-related changes.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, percentage of intracranial volume; FDR, false discovery rate; ml/year, millilitres per year.

3.3.2. Cortical regions

Subtle but significant between-group differences in atrophy rates were observed in 7 out of 34 cortical regions across the frontal, parietal, and occipital lobes in HDGE participants compared to controls (Table 6). Notably, the p and q values for these cortical regions were closer to the significance thresholds (e.g., angular gyrus: p=0.0002, q=0.012) than those observed in deep GM regions such as the caudate and putamen, which showed much stronger statistical significance (e.g., caudate: $p=4.7\times10^{-12}$, $q=3.27\times10^{-10}$; Table 6).

3.3.2.1. Frontal lobe

Of the 18 frontal lobe ROIs examined (Table 2), the triangular part of the inferior frontal gyrus and the superior frontal gyrus medial segment, showed significant between-group differences in atrophy rates (p=0.003, q=0.019; p=0.001, q=0.037, respectively; Table 11, Figure 11 a-b: i).

Controls showed relatively flat volume trajectories with minimal age-related decline (Figure 11 a-b: ii), whereas HDGE participants showed slightly downward trajectories in both regions (Figure 11 a-b: iii), consistent with early progressive neurodegeneration.

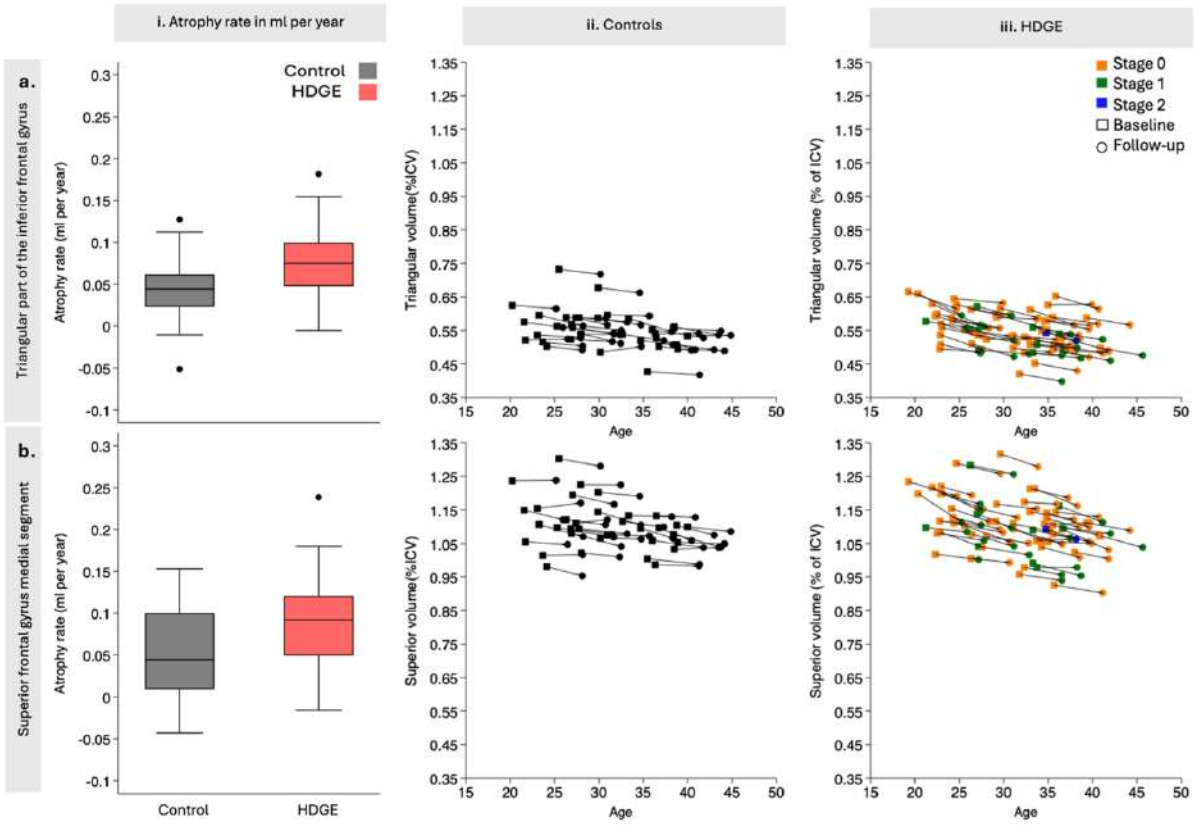

Figure 11. Annualized Atrophy Rates and Age-Related Volume Trajectories in Control and HD Gene Expansion Participants in frontal lobe ROIs. Annualized atrophy rates (age- and sex-adjusted, FDR-corrected; Table 3) are shown for two frontal lobe regions with significant group differences: the triangular part of the inferior frontal gyrus (a) and the superior frontal gyrus medial segment (b), comparing HDGE (n=54, red) and controls (n=34, grey). Box plots display group means (horizontal line), interquartile ranges (box), whiskers (1.5× IQR), and outliers (filled circles). Scatterplots show regional volumes (% ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green), and 2 (blue) (iii). Longitudinal baseline and follow-up data are connected by black lines; squares and circles indicate baseline and follow-up, respectively

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, FDR, false discovery rate; ml/year, percentage of intracranial volume; ml/year, millilitres per year; triangular, triangular part of the inferior frontal gyrus; superior, superior frontal gyrus medial segment.

3.3.2.2. Parietal lobe

Among the 7 parietal lobe ROIs (Table 2), 3 regions demonstrated significantly elevated atrophy rates in HDGE participants compared to controls: angular gyrus, parietal operculum, and precuneus (q=0.012, 0.038, 0.142 respectively; Figure 12).

Spaghetti plots of individual volume trajectories revealed a downward trend in volume of the angular gyrus for the HDGE participants, whereas the control data appear more variable compared with other structures, with some control participants losing volume, some gaining volume and some remaining stable (Figure 12. a: ii-iii), resulting in an overall atrophy rate of 0.074 ml/year in controls and 0.143 ml/yr in HDGE (Table 6).

Unlike the angular gyrus, volume trajectories for controls in the parietal operculum and precuneus remained largely stable (Figure 12. b-c: ii), whereas HDGE participants showed a mix of stable and slightly declining trajectories (Figure 12. b-c: iii). There was a substantial overlap between groups in these two regions (Figure 12. b-c: ii-iii). However, the slight downward trajectories observed in some HDGE participants is indicative of early atrophy due to disease (Figure 12. b-c: iii). This also supports the idea that these regions can be identified early in disease progression, as evident by their significant between group differences.

The overlap in individual atrophy rates trajectories also highlights variability within each group (Figure 12. b-c: ii-iii).

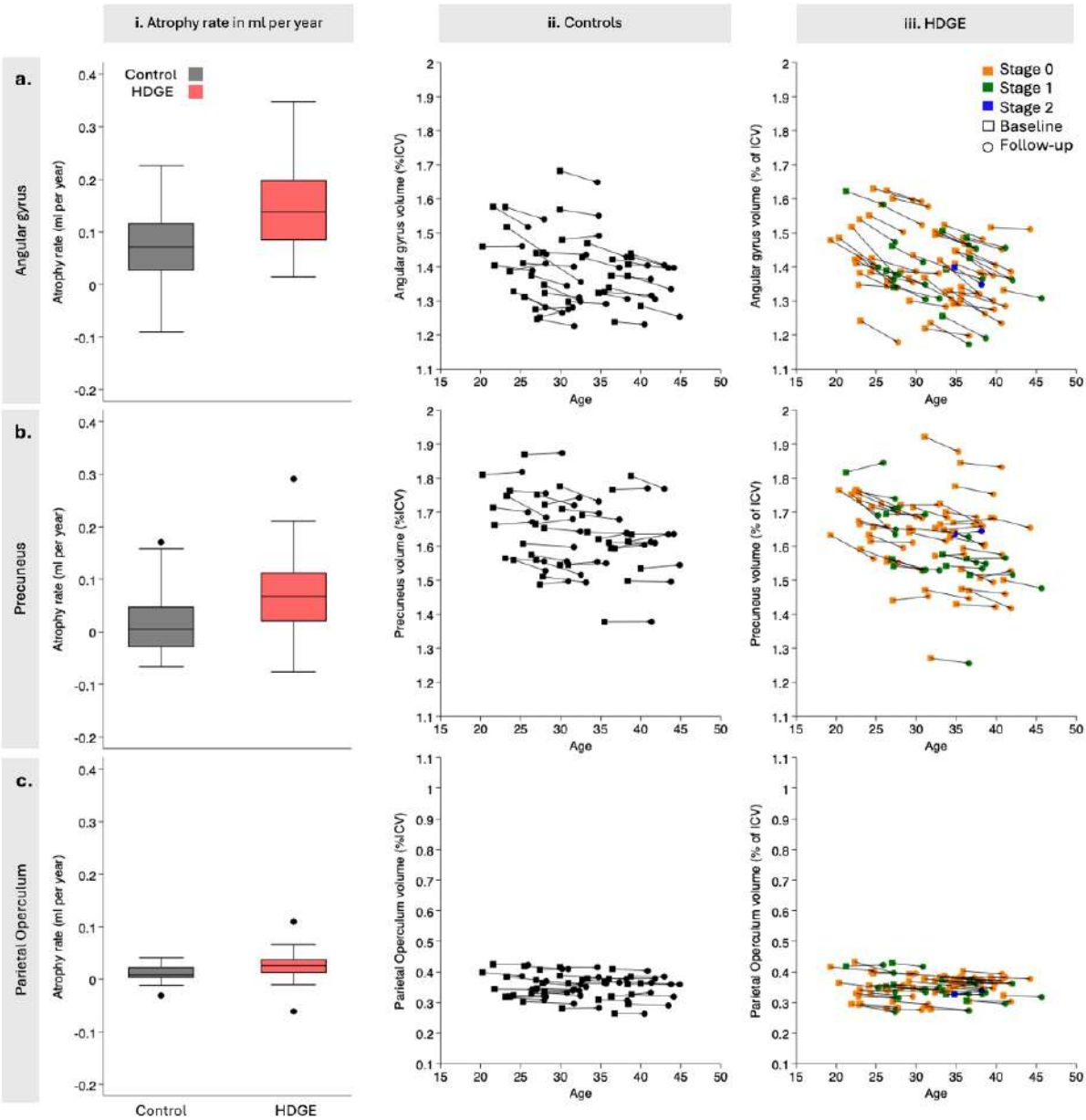

Figure 12. Annualized Atrophy Rates and Age-Related Volume Trajectories in Control and HD Gene Expansion Participants in Parietal Lobe ROIs. Annualized atrophy rates (age- and sex-adjusted, FDR-corrected; Table 3) are shown for three parietal regions with significant group differences: angular gyrus (a), precuneus (b), and parietal operculum (c) comparing HDGE (n=54, red) and controls (n=34, grey). Box plots display group means (horizontal line), interquartile ranges (box), whiskers (1.5× IQR), and outliers (filled circles). Spaghetti plots illustrate flat volume trajectories in controls and consistent declines in HDGE participants across all three regions (iii–iv). Scatterplots show regional volumes (% ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green), and 2 (blue) (iii). Longitudinal baseline and follow-up data are connected by black lines; squares and circles indicate baseline and follow-up, respectively. Y-axes are scaled consistently across regions except for the parietal operculum (c), to better highlight age-related volume changes.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, percentage of intracranial volume; FDR, false discovery rate; ml/year, millilitres per year.

3.3.2.3. Occipital lobe

Among the 9 occipital lobe ROIs (Table 2), the calcarine cortex and the lingual gyrus exhibited significant between group differences in atrophy rates, p=0.001, q=0.038; p=0.002, q=0.128, respectively (Table 6).

Notably, mean atrophy rates in the HDGE group were close to zero for both of these structures, with 95% CIs spanning zero, suggesting the volume of these regions is stable over time in this group (Table 6, Figure 13). However, within the control group, there is evidence of slight increases in volume over time for both regions (i.e. negative atrophy rates (mean(SD) CI: calcarine cortex: -0.029(0.038) -0.043 to -0.016; lingual gyrus: -0.037(0.058), -0.058 to -0.017). It is the subtle increases in volume in the control group, in the absence of change in the HDGE group, that is driving the significant between-group differences for these regions. (Figure 13).

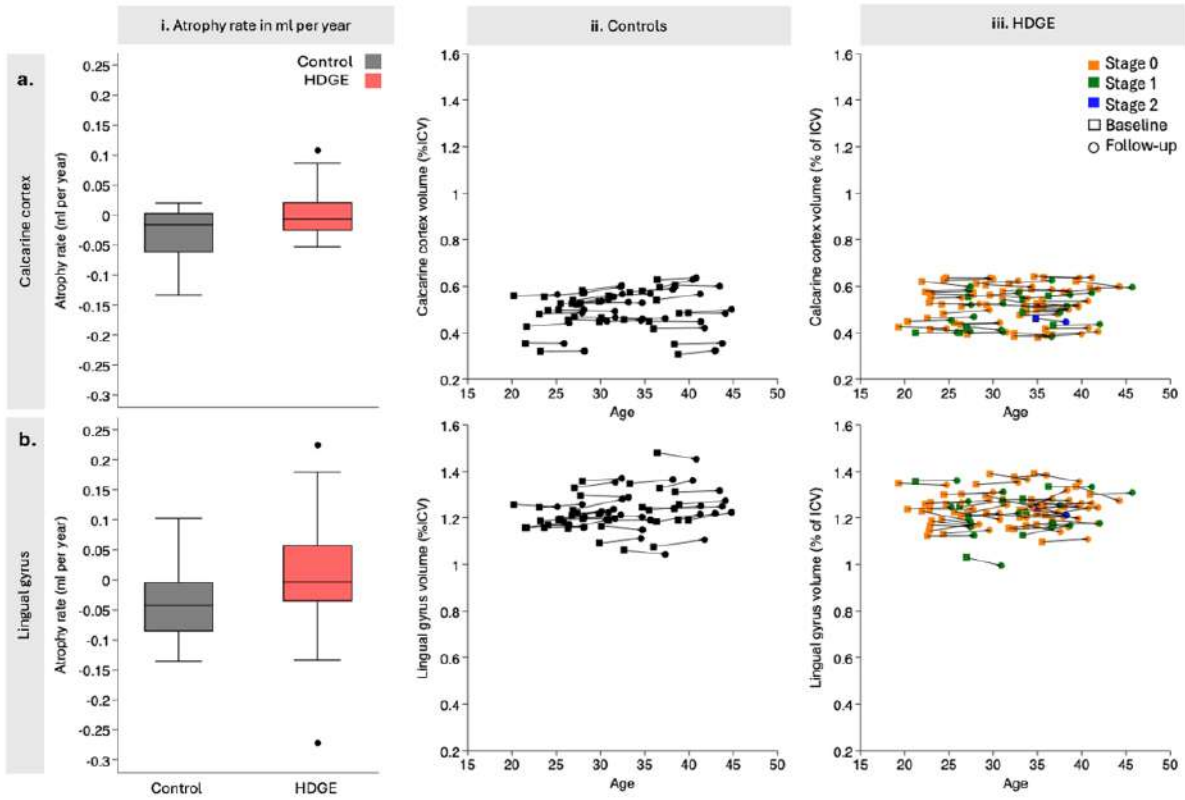

Figure 13. Annualized Atrophy Rates and Age-Related Volume Trajectories in Control and HD Gene Expansion Participants in Occipital Lobe ROIs. Annualized atrophy rates (age- and sex-adjusted, FDR-corrected; Table 3) are shown for two occipital regions with significant group differences: calcarine cortex (a) and lingual gyrus (b), comparing HDGE (n=54, red) and controls (n=34, grey). Box plots display group means (horizontal line), interquartile ranges (box), whiskers (1.5× IQR), and outliers (filled circles). Scatterplots show regional volumes (% ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green), and 2 (blue) (iii). Longitudinal baseline and follow-up data are connected by black lines; squares and circles indicate baseline and follow-up, respectively.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, percentage of intracranial volume; FDR, false discovery rate; ml/year, millilitres per

3.3.3. Cerebral white matter and brainstem

Significant differences in atrophy rates were observed in the cerebral white matter (p=0.001, q-value=0.006) and brainstem (p=0.002, q=0.104) of HDGE participants compared with controls. No significant difference was found in the only other white matter region assessed, the cerebellar white matter (Table 6).

As expected in the control group of this young adult study, cerebral white matter volume increased with age, with the increase appearing to plateau around the mid-30s (visual inspection of data in Figure 14, a ii), with a mean increase of WM volume of 0.983 mls per year across the group. In contrast, white-matter volume change in the HDGE group was more variable with smaller increases of on average -0.643 ml per year). A similar pattern of change was observed for the brainstem. With controls generally maintaining stable or slightly increasing volumes over time (Figure 14, b: ii), whereas HDGE trajectories were mostly flat, with some showing gradual decline (Figure 14, b: iii).

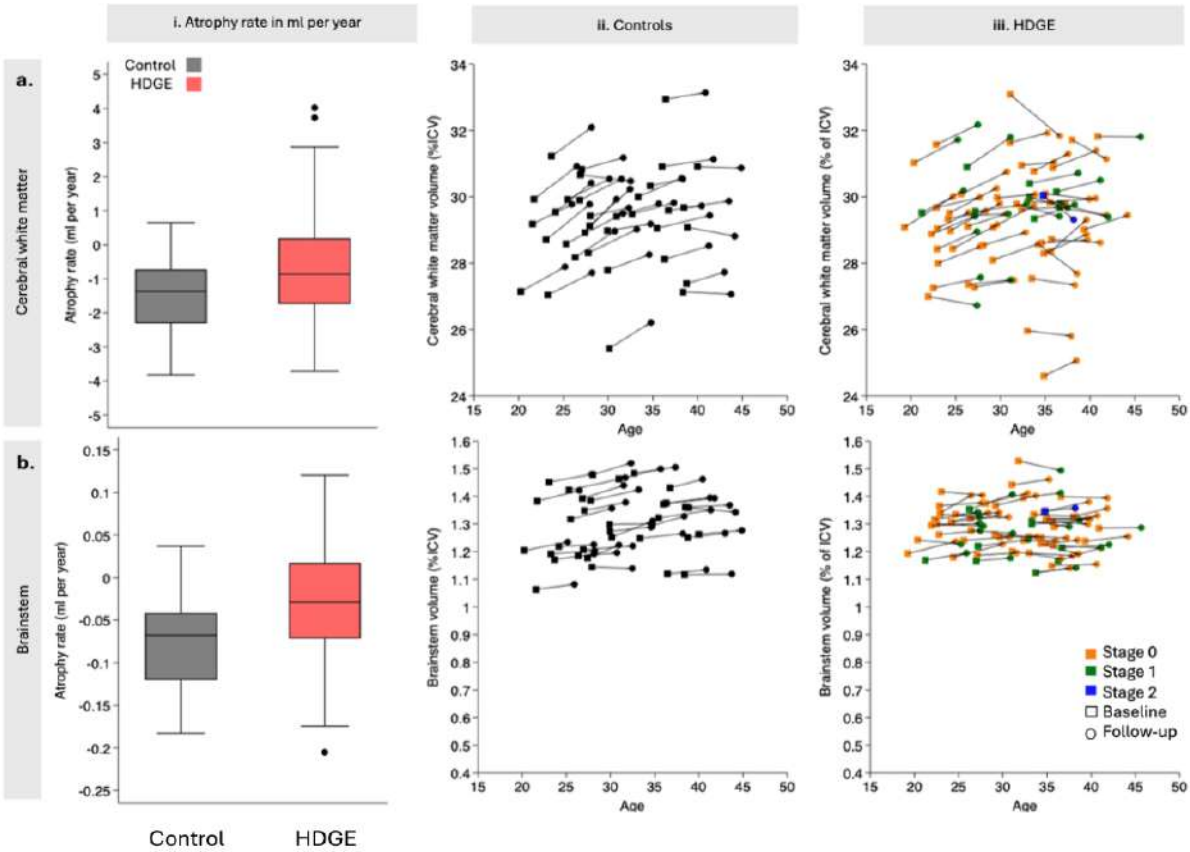

Figure 14. Annualized Atrophy Rates and Age-Related Volume Trajectories in Control and HD Gene Expansion Participants in whole brain ROIs. Annualized atrophy rates (adjusted for age and sex; FDR-corrected, Table 3) are shown for whole-brain regions with significant group-level differences, including cerebral white matter (a), brainstem (b), and ventral diencephalon (c), comparing HDGE (n=54, red) and controls (n=34, grey). Box plots display group means (horizontal line), interquartile ranges (box), whiskers (1.5× IQR), and outliers (filled circles). Scatterplots illustrate regional volumes (% ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green), and 2 (blue) (iii). Longitudinal trajectories are visualized by connecting baseline (squares) and follow-up (circles) data points with black lines.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, percentage of intracranial volume; FDR, false discovery rate; ml/year, millilitres per year.

3.3.4. Ventricular regions

HDGE participants showed increased enlargement of both the lateral ($p=2x10^5$, q=0.001) and third ventricles ($p=5.2x10^-6$, q-value = 0.0003) over time, compared to controls (Figure 15, a-b: i), suggesting changes linked to disease progression. There was no evidence of increased expansion of the fourth ventricle in HDGE compared with controls (p=0.504, q=0.942).

Spaghetti plots showed that ventricular volumes were variable between control participants but relatively stable over time, reflected by relatively flat trajectories (Figure 15, a-b: ii). In contrast, HDGE participants displayed increasing ventricular volumes over time, shown by upward volume trajectories in both the lateral and third ventricles, where several individuals exhibited pronounced volume increases during follow-up, evident as outliers (Figure 15, a: iii).

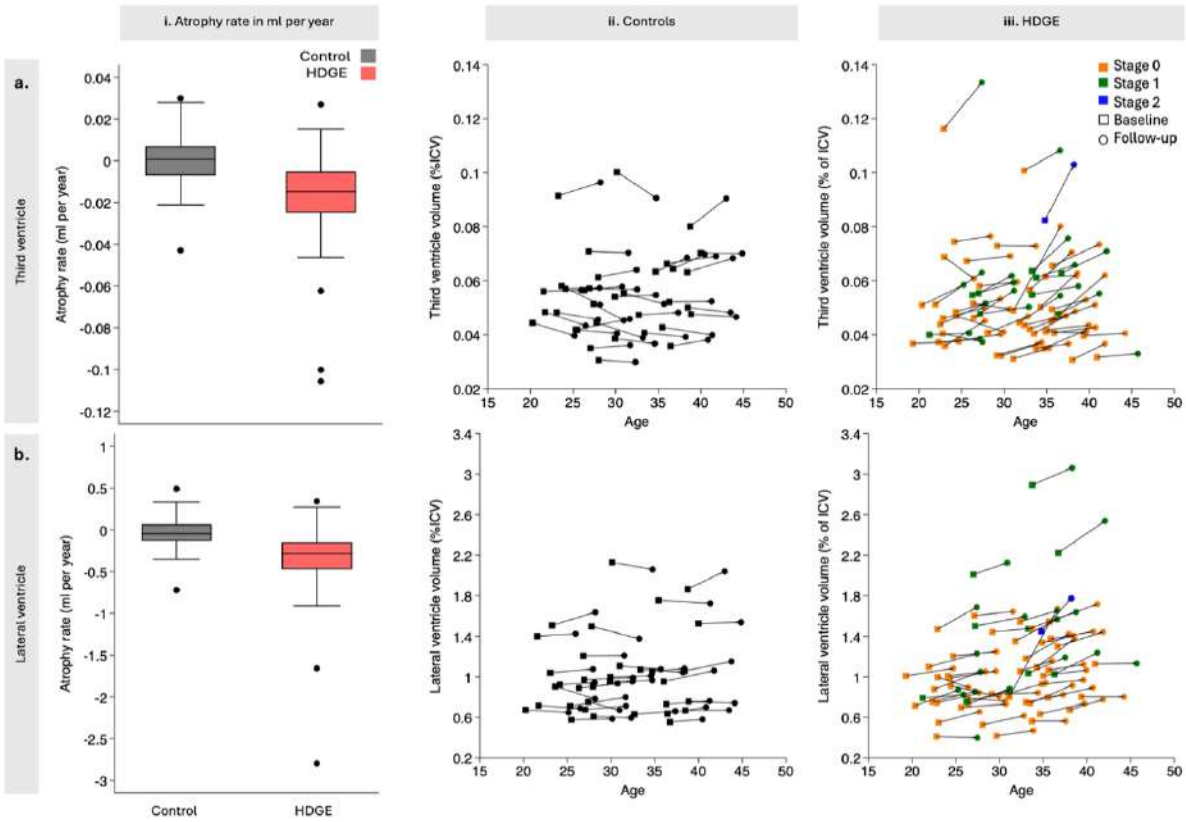

Figure 15. Annualized Atrophy Rates and Age-Related Volume Trajectories in Control and HD Gene Expansion Participants in ventricular ROIs. Annualized volumetric change rates (adjusted for age and sex; FDR-corrected, Table 3) are shown for ventricular regions with significant group-level differences, including the lateral ventricles (a) and third ventricle (b), comparing HDGE (n=54, red) and controls (n=34, grey). Box plots display group means (horizontal line), interquartile ranges (box), whiskers (1.5× IQR), and outliers (filled circles). Scatterplots illustrate ventricular volumes (% ICV) versus age for controls (ii) and HDGE participants stratified by HD-ISS stage 0 (orange), 1 (green), and 2 (blue) (iii). Longitudinal trajectories are visualized by connecting baseline (squares) and follow-up (circles) data points with black lines.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; %ICV, percentage of intracranial volume; FDR, false discovery rate; ml/year, millilitres per year.

3.4. Aim 2: Longitudinal effect size.

Out of the 69 ROIs assessed, 16 showed significant between-group differences in annualised atrophy rates ($q < 0.15$) after adjusting for age and sex and applying FDR correction (Table 6, Figure 16).

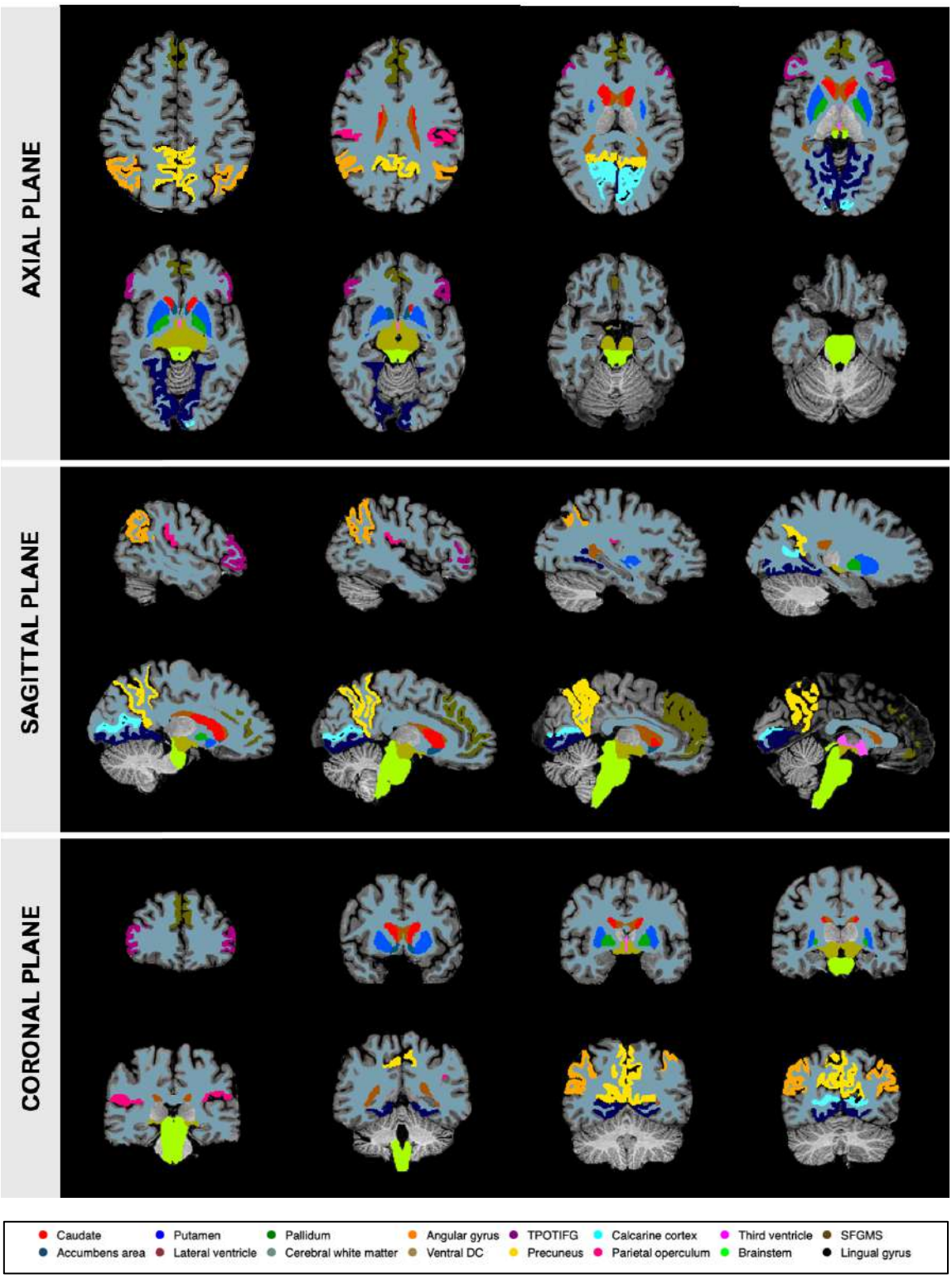

Figure 16. Anatomical distribution of the 16 ROIs showing significant longitudinal volume change. Whole-brain segmentation was performed to extract regional volumes across cortical and subcortical structures. Figure 1 shows longitudinal effect sizes across the 16 ROIs that demonstrated significant between-group differences in annualised volume change; these same ROIs are visualised here. Colour codes match those in Figure 1, ordered by descending effect size (ml/year), enabling direct

comparison between anatomical location and magnitude of atrophy. ROIs are displayed on axial, sagittal, and coronal slices to illustrate their spatial distribution across the brain.

Abbreviations: TPOTIFG = triangular part of the inferior frontal gyrus; SFGMS = superior frontal gyrus medial segment; ROI = region of interest; ml = millilitres.

Longitudinal effect sizes (ES) for the 16 regions are listed in Table 7, ranked by descending magnitude: large (Cohen's $d \geq 0.8$), medium ($0.2 < d < 0.8$), and small ($d \leq 0.2$). To aid interpretation, Table 7 also includes segmentation quality assessments. Cerebral white matter and brainstem received consistent borderline QC grades across participants, indicating uniform but limited segmentation confidence. The calcarine cortex and lingual gyrus showed mixed QC grading (pass and borderline pass), reflecting greater variability. All other regions received a pass grade, supporting higher confidence in their volumetric estimates. ROIs with borderline or mixed QC should be interpreted cautiously due to potential segmentation uncertainty.

Table 7. Ranked effect sizes for the 16/69 ROIs showing significant group differences.

ROI	Effect size (d)	CI
Caudate	1.305	0.994 to 1.568
Putamen	1.269	0.943 to 1.581
Pallidum	0.991	0.571 to 1.302
Angular gyrus	0.958	0.444 to 1.419
Triangular part of the inferior frontal gyrus	0.884	0.378 to 1.390
Calcarine cortex	0.860	0.426 to 1.360
Third ventricle	0.830	1.145 to 0.494
Superior frontal gyrus medial segment	0.818	0.298 to 1.281
Accumbens area	0.745	0.324 to 1.181
Lateral ventricle	0.722	1.031 to 0.364
Cerebral white matter	0.704	0.323 to 1.014
Ventral diencephalon	0.695	0.313 to 1.071
Precuneus	0.672	0.166 to 1.088
Parietal operculum	0.635	0.228 to 1.034
Brainstem	0.627	0.211 to 1.034
Lingual gyrus	0.561	0.112 to 0.899

Cohen's *d* values quantify the magnitude of between-group differences in annualised volume change, with ROIs ordered by descending effect size. Corresponding 95% CIs were derived via bootstrapping. All regions listed met the threshold for statistical significance ($q<0.15$).

Note: ROIs shaded in light grey received consistent borderline QC grades across participants, while dark grey regions showed mixed grading. Both should be interpreted cautiously due to segmentation uncertainty. In light grey regions, group differences were primarily driven by subtle volume increases in controls rather than decline in HDGE, making effect size interpretation more complex and less directly indicative of disease-related atrophy.

Abbreviations: ROI = region of interest; CI = confidence interval.

As hypothesised, the caudate (d = 1.305) and putamen (d = 1.269) showed the largest effect sizes, with the pallidum (d = 0.991) also demonstrating a large effect. These findings underscore early and pronounced basal ganglia vulnerability in HDGE, consistent with its central role in HD pathology.

Medium effect sizes in the accumbens area and ventral diencephalon suggest deep grey matter loss beyond the striatum, even in early disease (Table 7).

The caudate and putamen has narrow 95% CIs (0.994-1.568 and 0.943-1.581, respectivley) that did not cross zero, indicating robust group-level differences. The pallidum's broader CI (0.571-1.302) suggest greater variability, likely due to segmentation uncertainty, highlighting the importance to ROI quality when interpreting ES estimates.

Large effect sizes were observed in two frontal cortical ROIs, the triangular part of the inferior frontal gyrus and the superior frontal gyrus medial segment, as well as the angular gyrus (parietal lobe) and calcarine cortex (occipital lobe) (Table 7). Despite greater anatomical variability and segmentation complexity in cortical regions, these ROIs showed similar CIs (Figure 17), indicating consistent atrophy patterns and uniform group-level changes. All other cortical ROIs fell within the medium effect size range (Table 7; Figures 16 & 17).

Global ROIs, including cerebral white matter and brainstem, showed medium effect sizes, while ventricular ROIs demonstrated marked expansion, large for the third ventricle and medium for the lateral ventricle (Table 7, Figure 17). These patterns reflect widespread structural changes, with tissue loss accompanied by compensatory ventricular enlargement.

Of the 16 regions with significant group differences, 12 are recommended for further validation based on strong effect sizes and consistently high segmentation quality (Table 7, Figure 17), supporting their anatomical specificity and reliability.

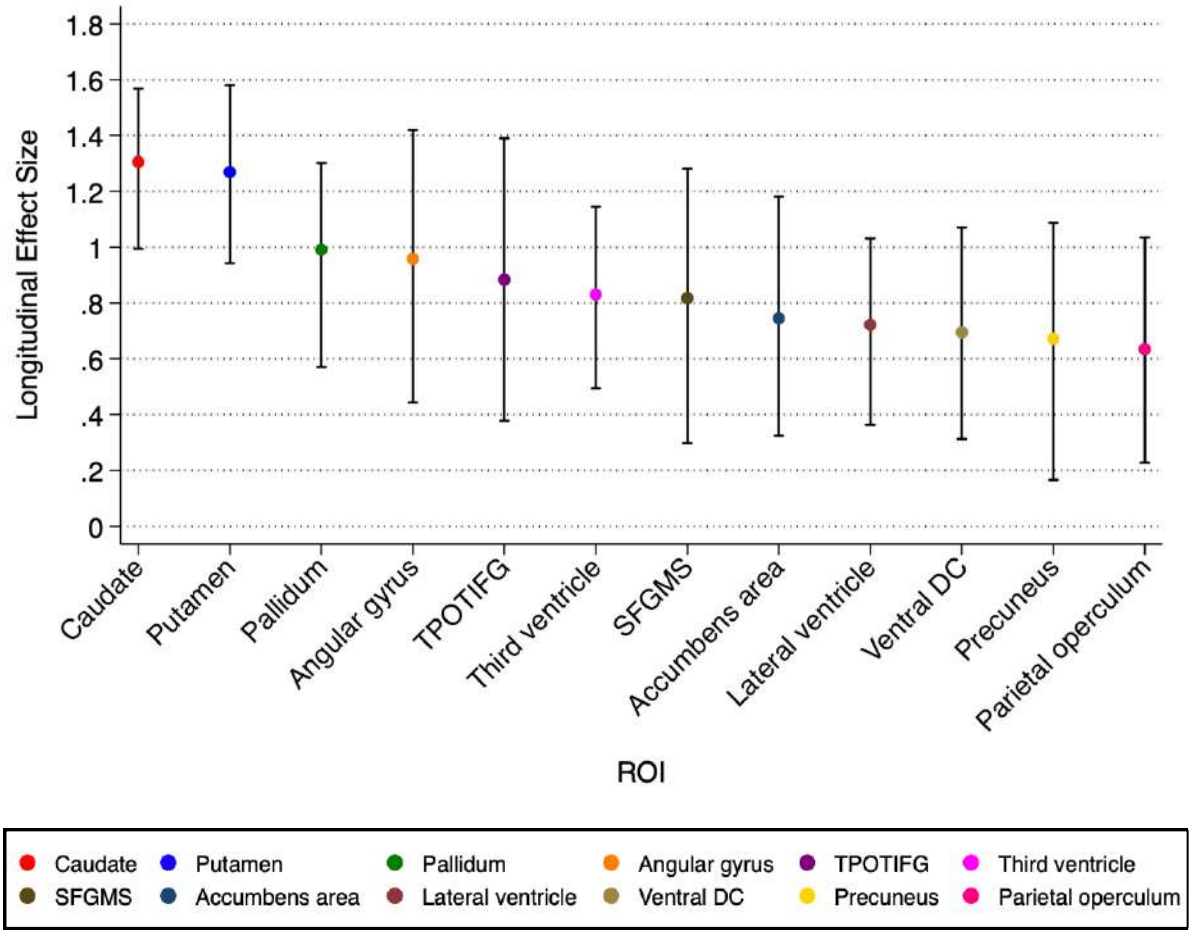

Figure 17. Effect sizes for longitudinal volume change across 12 ROIs with significant between-group differences and consistently high segmentation quality.

ROIs are ordered by descending effect size. Each dot represents the group-level mean Cohen's *d* for annualised volume change, with error bars indicating bootstrapped 95% confidence intervals. The 12 regions shown were selected based on statistical significance and consistently good QC grading, excluding ROIs with borderline or mixed segmentation outcomes.

Abbreviations: ml = millilitres; ROI = region of interest; TPOTIFG = triangular part of the inferior frontal gyrus; SFGMS = superior frontal gyrus medial segment.

3.5. Aim 3: Association of Regional Atrophy Rates with HD-ISS Stage & CAP100

3.5.1. HD-ISS Stage

Due to limited representation in HD-ISS Stage 2 (n = 1), statistical comparisons were restricted to participants in Stages 0 and 1. Whole-brain analysis identified 16 regions with significant longitudinal atrophy rates across the HDGE cohort (Table 6), however, none of these regions showed statistically significant differences in atrophy rate between HD-ISS Stages 0 and 1 (Figure 18. a-d: i). Detailed results for all remaining subcortical regions, as well as cortical, ventricular, and global measures, are provided in the appendix.

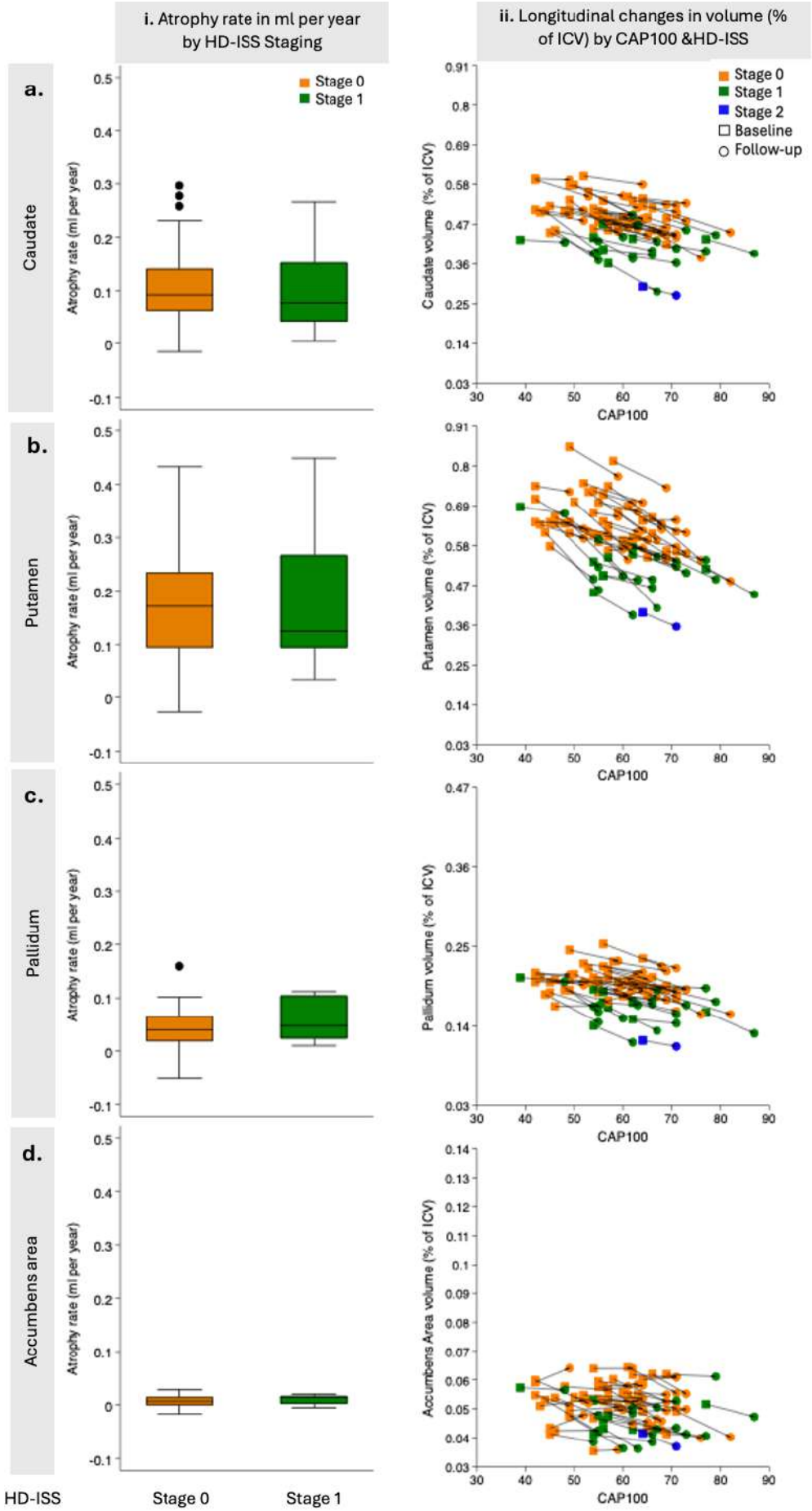
i. Atrophy rate in ml per year by HD-ISS Staging
ii. Longitudinal changes in volume (% of ICV) by CAP100 &HD-ISS
a.
Caudate
b.
Putamen
c.
Pallidum
d.
Accumbens area
Stage 0
Stage 1
Stage 2
Baseline
Follow-up
Atrophy rate (ml per year)
Caudate volume (% of ICV)
Putamen volume (% of ICV)
Pallidum volume (% of ICV)
Accumbens Area volume (% of ICV)
CAP100
HD-ISS
Stage 0
Stage 1

Figure 18. ***Annualized Atrophy Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in subcortical ROIs.*** This figure illustrates regional atrophy rates and volumetric decline across the (a) caudate, (b) putamen, (c) pallidum, and (d) accumbens area. For each subregion boxplots compare annualized atrophy rates in millilitres per year across HD-ISS stages 0 (orange), 1 (green), and 2 (blue). Each boxplot displays the group median (horizontal line), interquartile range (box), and whiskers extending to 1.5× IQR, with individual outliers shown as filled circles. Scatterplots (a-d) show regional volume, expressed as percentage of intracranial volume (%ICV), plotted against CAP100 score for HD-ISS stages 0 to 2. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes. The pallidum (c: i) y-axis is zoomed to one tick interval at the same scale as other regions, while the accumbens (d: i) axes are truncated to highlight age-related changes CAP100-related trends. Although between-group differences in atrophy rates were not statistically significant, CAP100-related trajectories within the HD group illustrate volume change across individuals.

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest.

3.5.2. CAP100

Of the 69 whole-brain ROIs, 16 showed significant between-group differences in annualized atrophy rates (Table 6). Among these, all subcortical grey matter regions, except the ventral diencephalon, were significantly correlated with CAP100 ($q < 0.15$, age- and sex-adjusted, FDR-corrected), a cumulative disease burden metric (Figure 18a. d:ii). While cortical ROIs showed group-level differences in earlier analyses, none were significantly associated with CAP100 (Table 8). Notably, the brainstem and third ventricle also showed significant CAP100 correlations, reflecting ongoing neurodegeneration.

Table 8. Association between CAP100 and annualized atrophy rates (%ICV) in regions with significant longitudinal atrophy.

ROI	Coefficient	95% CI	p-value	q-value
Putamen	0.009	0.005 to 0.014	0.0001	0.002
Pallidum	0.003	0.002 to 0.005	0.0001	0.002
Caudate	0.005	0.002 to 0.008	0.001	0.013
Accumbens area	0.001	0.0003 to 0.001	0.002	0.028
Brainstem	0.005	0.001 to 0.008	0.008	0.092
Third ventricle	-0.001	-0.002 to -0.0003	0.012	0.135
Cerebral white matter	0.068	-0.005 to 0.140	0.066	0.643
Ventral diencephalon	0.001	-0.0001 to 0.003	0.071	0.643
Lateral ventricles	-0.018	-0.041 to 0.004	0.110	0.880
Angular gyrus	0.003	-0.001 to 0.006	0.143	0.898
TPOTIFG	0.001	-0.001 to 0.003	0.361	0.898
Calcarine cortex	-0.0003	-0.002 to 0.002	0.744	0.898
Precuneus	0.0005	-0.003 to 0.004	0.782	0.898
SFGMS	0.0003	-0.002 to 0.003	0.784	0.898
Lingual gyrus	-0.001	-0.005 to 0.004	0.788	0.898
Parietal operculum	0.00008	-0.001 to 0.001	0.898	0.898

Linear regression results examining the association between CAP100 and annualized regional atrophy rates, calculated as change in volume normalized to intracranial volume (%ICV), across 16 ROIs previously identified as showing significant longitudinal atrophy. For each ROI, the regression coefficient (mean), 95% CI, uncorrected p-value, and FDR-corrected q-value are reported CAP100 was calculated as: CAP100 = Age × (CAG – 33.66) where Age is the participant's age at scan and CAG is the number of CAG repeats in the HTT gene. This metric reflects cumulative exposure to mutant huntingtin and is scaled to approximate the onset threshold. A negative CAP100 coefficient indicates faster expansion with higher CAP100, consistent with advancing disease.

Abbreviations: ROI, region of interest; CI, confidence interval; FDR, false discovery rate; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66).

Chapter 5. Discussion

This whole-brain analysis revealed significant between-group differences in longitudinal volume change across 16 brain regions in the HD-YAS cohort, nearly 20 years before predicted clinical motor diagnosis. While many reflect progressive atrophy in HDGE participants, others, such as cerebral white matter, brainstem, and two occipital lobe regions, were driven by volume increases in controls and should cautiously interpreted with due to known segmentation limitations in these regions. These findings reinforce the striatum's role as a standard MRI-derived volumetric biomarker in HD, while also highlighting additional subcortical deep grey matter and cortical regions that merit further investigation as alternative outcome measure in clinical trials, particularly where striatal biomarkers maybe suboptimal, such as in gene therapy trials (Hobbs *et al.*, 2024).

Previous MRI studies have focused primarily on striatal changes (Tabrizi *et al.*, 2009), with subcortical and cortical alterations typically reported only near or after motor onset, when interventions may be less effective (Tabrizi *et al.*, 2012). In contrast, our findings identify both known and novel regions affected much earlier, revealing initial structural vulnerabilities that could inform preventative trial design. These regions may serve as efficacy endpoints in future disease-modifying studies and support the development of targeted early interventions.

5.1. Aim 1: Between group differences in Atrophy Rates

5.1.1. Subcortical deep grey matter regions

5.1.1.1. Caudate and Putamen

In this study of the HD-YAS cohort, where most HDGE participants were classified within HD-ISS stages 0-1, significantly elevated striatal atrophy rates were detected relative to matched controls. Annualized atrophy rates were, on average, 1.48% in the caudate and 1.85% in the putamen, compared to 0.26% and 0.45%, respectively, in controls. These early-stage findings are consistent with prior longitudinal imaging

studies, including PREDICT-HD (HD-ISS stages 1-2) (Paulsen et al., 2010) and TRACK-HD (stages 2-3) (Tabrizi et al., 2013), which demonstrated stepwise increases in striatal atrophy, as a proportion of baseline volume, with advancing disease stage.

Importantly, while TRACK-HD employed a legacy staging framework based on estimated years to onset and diagnostic confidence to define premanifest and manifest stages (Tabrizi et al., 2013), HD-YAS uses the newer HD-ISS system, which anchors staging to biological and clinical milestones. Despite these methodological differences, striatal atrophy remains consistent, reinforcing its reliability as a marker of disease progression, detectable well before clinical motor onset in individuals who remain otherwise asymptomatic.

5.1.2. Other subcortical regions

Beyond the striatum, HDGE participants in the HD-YAS cohort showed significant atrophy in subcortical regions including the pallidum, nucleus accumbens, and ventral diencephalon. These findings align with van den Bogaard et al. (2011), who reported early pallidum and thalamic atrophy near motor onset in premanifest HD. The involvement of limbic-associated regions such as the accumbens and ventral diencephalon suggests that neurodegeneration may extend beyond motor circuits even in early disease stages (van den Bogaard *et al.*, 2011). Thalamic atrophy was not elevated in our cohort, consistent with prior reports of relative sparing in early HD (Douaud *et al.*, 2006). While van den Bogaard et al. (2011) observed thalamic volume loss, it mirrored whole-brain atrophy rates rather than showing disproportionate subcortical vulnerability. This discrepancy may reflect cohort differences in proximity to clinical motor diagnosis, with thalamic involvement emerging later and, when present in premanifest individuals, likely reflects global rather than region-specific atrophy. Thalamic segmentation on T1-weighted MRI is also challenging due to poor boundary contrast, especially relative to structures like the caudate and putamen. Similarly, the hippocampus is preserved in early stages, as confirmed by my findings, but shows measurable atrophy near clinical diagnosis (van den Bogaard *et al.*, 2011).

The ventral diencephalon comprises deep structures, including the hypothalamus, mammillary bodies, subthalamic nucleus, substantia nigra, red nucleus, geniculate

nuclei, and basal forebrain, linked to autonomic, motor, and limbic functions (Mai, Majtanik and Paxinos, 2015). These are difficult to resolve individually on 3T MRI and are segmented collectively, whereas 7T imaging offers improved anatomical definition (Björkman-Burtscher, Bloch and Maly Sundgren, 2018). The statistically significant atrophy observed in this region suggests early involvement in HDGE, requiring high-resolution studies to clarify substructure-specific contributions to disease mechanisms. Neuropathological studies have identified mutant huntingtin aggregates in ventral forebrain regions, including the accumbens and extended amygdala, adjacent to the ventral diencephalon and acting as interfaces between motor, limbic, and olfactory systems. These findings suggest early pathological involvement of diencephalic circuits in non-striatal symptomatology (Petrasch-Parwez *et al.*, 2022). Complementary evidence from fused dorsal and ventral brain organoids, modelling cortical and subcortical regions, respectively, reveals disrupted development and abnormal gene activity in HD, including changes in the choroid plexus near the ventral diencephalon. Together, these data support the role of ventral DC in early HD pathogenesis (Świtońska-Kurkowska *et al.*, 2025).

5.1.3. Cortical regions

5.1.3.1. Frontal lobe

My findings show early frontal lobe atrophy in HDGE participants, with significantly elevated atrophy rates in the triangular part of the inferior frontal gyrus and the medial segment of the superior frontal gyrus. These regions exhibited clear group differences, with HDGE participants showing declining volumes compared to stable
profiles in controls. Earlier studies linked cognitive impairment and symptom severity to frontal lobe atrophy, predominantly in later HD stages (Aylward *et al.*, 1998), while more recent evidence shows modestly elevated atrophy rates even before clinical motor diagnosis (Johnson, Ziegler, *et al.*, 2021). This aligns with findings that cognitive and psychiatric symptoms often linked to fronto-striatal dysfunction emerges years before motor decline (Jellinger, 2024). Atrophy in regions like the inferior frontal gyrus was associated with impaired psychomotor speed and executive function, reinforcing the role of cortical degeneration in predicting cognitive outcomes (Johnson, Ziegler, *et*

al., 2021). Together, these studies and my findings underscore that frontal atrophy begins earlier in HD than traditionally thought.

5.1.3.2. Parietal lobe

The angular gyrus showed significantly elevated atrophy rates in HDGE participants, with a consistent downward volume trajectory and an annual atrophy rate nearly twice than that of controls (0.143 vs. 0.074 ml/year), who exhibited more variable trajectories. Anatomically, the angular gyrus projects to the caudate, a key site of HD pathology and post-mortem studies have revealed marked pyramidal neuron loss in this region (Macdonald *et al.*, 1997). Lesions here disrupt motion perception and prolong reflexive saccade latency, the time delay in automatic eye movements toward visual stimuli. HD patients often present with early saccadic slowing, impaired voluntary movements execution, falls, staggering gait, and limited awareness of motor deficits (Grabska *et al.*, 2024). Despite these features, no in vivo studies have examined the angular gyrus in HD-ISS stages 0-1. However, resting-state fMRI findings show altered intrinsic activity in this region (Liu *et al.*, 2016). Given its role in diverse cognitive functions, these results suggest that HD may affect broader cortical networks beyond the striatum.

In this study, the precuneus showed significant atrophy rate in HDGE participants as compared to controls. Prior HD-YAS findings reported altered iron and myelin content in this region (Johnson, Parker, *et al.*, 2021). Precuneus atrophy has been linked to visuospatial deficits (Labuschagne *et al.*, 2016) and impaired emotional recognition in early preHD (Scahill *et al.*, 2013), highlighting its structural and functional vulnerability in early HD.

Although the parietal operculum has not been directly examined in HD, its potential involvement warrants further investigation.

5.1.3.3. Occipital lobe

The occipital lobe has emerged as a key site of cortical vulnerability in HD, with converging evidence across HD-ISS stages linking it to symptom progression (Johnson *et al.*, 2014). In my HD-YAS study (HD-ISS Stage 0-1), I observed significant between-group differences in calcarine cortex atrophy rates. While volumes in HDGE participants remained stable over time, controls showed subtle increases in both regions. In controls, small volume increases in the calcarine cortex and lingual gyrus, likely reflect segmentation inaccuracies, absent in HDGE participants, suggesting early disruption of occipital lobes structural stability prior to motor clinical diagnosis. These findings extend TRACK-HD results (HD-ISS Stages 2-3), which reported progressive occipital thinning, particularly in the lingual gyrus, linked to worsening cognitive and motor function (Johnson, Ziegler, *et al.*, 2021). My study reveals even earlier occipital involvement, with significant atrophy in the calcarine cortex and lingual gyrus in HD-ISS Stages 0-1. The calcarine cortex, a primary visual region, and the lingual gyrus, implicated in visual memory and psychomotor processing, appear to contribute distinctly to symptom progression (Coppen *et al.*, 2018). Together, these results underscore the occipital lobe's structural changes across HD-ISS stages and highlight its early role in disease evolution.

5.1.3.4. Cerebral white matter and brainstem

In the HD-YAS cohort, cross-sectional analyses showed no significant group differences in white matter volume, suggesting preserved tissue integrity at this early stage (Scahill *et al.*, 2020). In contrast, TRACK-HD cross-sectional data (Tabrizi *et al.*, 2009) demonstrated pronounced white matter loss prior to clinical motor diagnosis, indicating earlier structural changes in some individuals.

White matter trajectories observed in our control group align with normative lifespan patterns, reported in a large-scale study of over 120,000 MRI scans (Bethlehem *et al.*, 2022), showing volume increases through early adulthood, peaking near age 30, and declining after 50. The control data in our study reflect this expected developmental arc, lending confidence to the reliability of our measurements. In contrast, HDGE

participants showed more variable and attenuated white matter changes, possibly reflecting early deviation from normative maturation. Given ongoing white matter development into the third decade, interpretation in young adults with neurodegeneration is complex due to overlapping developmental and pathological processes. These findings underscore the need for age-specific baselines when evaluating white matter change and suggest it may offer complementary insight alongside grey matter in early-stage HD. Compared to cerebral white matter, HDGE participants showed a subtle volumetric increase in the brainstem relative to controls, potentially reflecting early involvement. This may relate to brainstem's mixed white and grey matter composition. Post-mortem evidence supports this, showing selective neuronal loss and widespread axonal inclusions across multiple brainstem nuclei, implicating the region in underexplored HD symptoms (Rüb *et al.*, 2014a).

5.1.3.5. Ventricular expansion

In this HDGE group, annualised atrophy rates were 2.33% in the lateral ventricles and 2.26% in the third ventricle, approximately 7-fold and 14-fold higher, respectively, than in controls. Unlike prior findings reporting a 9.27 mL larger total ventricular volume before clinical motor diagnosis and a 4-fold higher overall ventricular atrophy rate in HD ($p<0.05$), the HDGE group did not show increased lateral ventricular expansion relative to controls. Importantly, that study did not assess third or fourth ventricles.

Ventricular expansion reflects progressive tissue loss in adjacent subcortical and cortical regions and is detectable from early disease stages. As a global marker, it captures diffuse atrophy in a single measure, though sensitivity and specificity may vary with disease stage and regional involvement. Although the third ventricle is rarely emphasised in HD literature, the event-based model (EBM), a data-driven framework inferring biomarker progression, predicts its early involvement (Wijeratne *et al.*, 2018).

5.2. Aim 2: Longitudinal Effect sizes

As hypothesised, the caudate ($d = 1.305$) and putamen ($d = 1.269$) showed the largest longitudinal effect sizes, confirming early structural vulnerability in HD. These findings

align with TRACK-HD, which reported similarly high effect sizes at a later disease stage (Tabrizi *et al.*, 2013). The pallidum (d = 0.991) ranked third in effect size, suggesting early involvement in HD. Its wider confidence interval (95% CI: 0.571-1.302), compared to the caudate and putamen, likely reflects segmentation variability due to less distinct WM-GM boundaries on T1- weighted scans, as well as the possibility that structural changes emerge slightly later than striatal atrophy. Although segmentations passed visual QC, perivascular spaces were noted; their impact on volume remains unquantified. Therefore, it is important to consider the quality of ROI delineation when interpreting effect size estimates and the 95 % CI. Medium effect sizes in the accumbens area and ventral diencephalon support the hypothesis that non-striatal deep grey matter regions also exhibit measurable atrophy early in the disease process.

All subcortical ROIs, including the caudate, putamen, pallidum, accumbens area and ventral diencephalon, showed significant between-group differences and large effect sizes, and are recommended for further validation as MRI-derived biomarkers for clinical trial outcomes.

Occipital ROIs, the calcarine cortex and lingual gyrus, showed high and medium effect sizes, respectively, though smaller than striatal regions. However, both exhibited segmentation uncertainty (borderline QC grades, 'loose fit' patterns), likely due to partial volume effects, where voxels contained mixed tissue types (e.g., GM and CSF), leading to misclassification and volume overestimation. This issue was only observed in these two cortical ROIs and directly impacted segmentation reliability. Additionally, both regions showed a subtle volume increase over time in controls, driven by variability rather than disease-related change. Together, these sources of uncertainty complicate interpretation and, despite significant between-group differences, limit the suitability of these ROIs as outcome measures.

Additionally, several novel regions, including frontal ROIs (triangular part of the inferior frontal gyrus, superior frontal gyrus medial segment) and parietal ROIs (angular gyrus, precuneus, parietal operculum) showed significant atrophy with medium to high effect sizes. Clear between-group differences were observed in the HD-YAS cohort, approximately 20 years before predicted onset, highlighting early vulnerability outside

the striatum. Importantly, the gradient in effect sizes reflects the expected hierarchy of structural vulnerability, most pronounced in the striatum and extending to select cortical regions, highlighting a spatial pattern of early change. These findings support the utility of these regions as early biomarkers for stratification and outcome measures in preventative clinical trials.

ROIs such as the cerebral white matter and brainstem showed medium effect sizes, with cerebral white matter serving as a global structural marker, potentially indicative of diffuse structural decline throughout the brain. However, both regions received borderline QC grades across participants, suggesting segmentation uncertainty and warranting cautious interpretation. White matter trajectories in controls followed age-related increases, while HDGE participants showed greater variability and a mean decline, suggesting disrupted maturation. These deviations underscore the importance of age-specific baselines when evaluating early-stage HD-related white matter change.

Of the 16 ROIs with significant between-group differences, 12 are recommended for further validation based on robust effect sizes, HDGE-drive atrophy, and consistent segmentation quality (Table 4, Figure 12).

Together, these findings demonstrate that longitudinal effect sizes provide a quantitative framework for comparing regional atrophy trajectories in HD. A multi-region imaging approach that integrates both localized and global metrics enhances our ability to track disease progression and refine biomarker selection, particularly for early-stage clinical trials where anatomical relevance and statistical efficiency are critical.

From a clinical trial perspective, regions with larger longitudinal effect sizes, such as the caudate, offer greater statistical efficiency, requiring fewer participants to detect treatment effects, as demonstrated by (Hobbs *et al.*, 2015), making them making them strong candidates for imaging-based outcome measures in early-phase trials. (Figure 19).

These preliminary estimates suggest that regions with large effect sizes (e.g., striatum, pallidum) may require fewer than 50 participants per arm to detect a 25% treatment effect with 80% power, assuming standard variance. Moderate-effect regions (e.g., thalamus, insula) fall within the 70-120 range, while smaller-effect regions (e.g., cortical areas) may exceed 150 per arm. These estimates support initial ROI prioritization for trial design, though further research is needed to validate each region's reliability, sensitivity, and longitudinal stability. Definitive trials will require more advanced modelling and cohort-specific validation to confirm the utility of these imaging endpoints.

In this study, both annualised atrophy rates in ml/year (Table 6) and annualised atrophy rates in %/year (Supplementary Table 1) were assessed to capture distinct aspects of regional change. Atrophy rates in ml/year quantify absolute tissue loss, whereas atrophy rates in %/year reflect proportional change relative to baseline volume. Together, these metrics offer complementary insights for biomarker evaluation and staging.

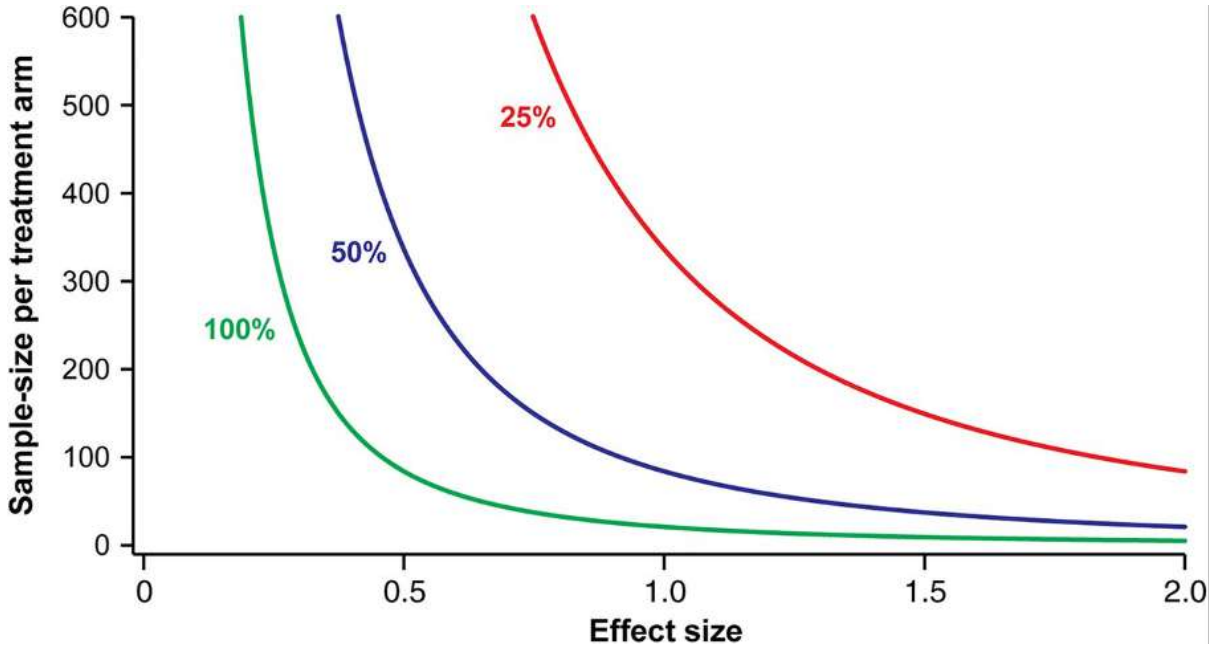

Figure 19. ***Relationship between effect size and sample-size requirements for randomized controlled trials.*** This graph illustrates how increasing effect size reduces the number of participants required per treatment arm to detect a change between two time points. Curves represent treatments with efficacy levels of 25% (red), 50% (blue), and 100% (green), assuming 90% power and a 5% significance level. The x-axis shows effect size (0–2.0); the y-axis shows sample size per arm (0–600). This figure is adapted from (Hobbs *et al.*, 2015).

Table 9. Estimated sample size requirements for imaging-based clinical trials across selected brain regions.

ROI	Effect Size (d)	Estimated Sample Size per Arm
Large Effect Size (Cohen's d ≥ 0.8)		
Caudate	1.305	60 participants
Putamen	1.269	65 participants
Pallidum	0.991	100 participants
Angular gyrus	0.958	105 participants
IFG (triangular part)	0.884	120 participants
Third ventricle	0.830	135 participants
SFG medial segment	0.818	140 participants
Medium Effect Size (Cohen's d 0.6–0.8)		
Accumbens area	0.745	160 participants
Lateral ventricle	0.722	170 participants
Ventral diencephalon	0.695	180 participants
Precuneus	0.672	190 participants
Parietal operculum	0.635	210 participants

Sample size estimates are based on observed effect sizes (Cohen's d) for each ROI, assuming a 50% treatment effect, 90% power, and a two-tailed $\alpha = 0.05$, following the modelling framework of Hobbs et al. (2015). ROIs are grouped by effect size to illustrate relative sensitivity to neurodegenerative change. Regions with $d \geq 0.8$ typically require <50-140 participants per arm, while those with $d = 0.6$-0.8 fall in the 160-210 range. ROIs with $d < 0.5$ are not shown due to limited feasibility as primary endpoints in early-phase trials.

5.3. Aim 3: Association of regional atrophy rates with HD-ISS stage and CAP100

5.3.1. HD-ISS stage

The absence of significant differences in atrophy rates between HD-ISS Stages 0-1 suggests that while structural decline is detectable, its progression may be relatively uniform across these early disease stages. This aligns with prior evidenced that neurodegeneration can precede clinical milestones but may not accelerate until later stages (Scahill *et al.*, 2025).

Limited representation of Stage 2 (n = 1) precluded statistical comparisons beyond Stage 1, highlighting challenges in capturing transitional dynamics within early-stage cohorts. Future studies with enriched Stage 2 sampling will be essential for identifying inflection points and validating stage-specific imaging biomarkers. Detailed subcortical, cortical, ventricular, and global results are provided in the appendix to support transparency and reproducibility, offering a foundation for longitudinal modelling and biomarker refinement as HD frameworks evolve.

Although regions identified in whole-brain analysis were statistically robust, they did not differentiate between HD-ISS stages, suggesting that volumetric decline alone may lack sensitivity for early-stage stratification. This highlights the need for multimodal approaches, including microstructural imaging, functional connectivity, and biofluid markers, to complement volumetric metrics and improve early-stage resolution.

To further investigate disease progression, regional atrophy rates were examined in relation to HD-ISS stage and CAP100.

5.3.2. CAP100

Among the 16 ROIs with significant group differences, all subcortical grey matter regions, except the ventral diencephalon, showed significant correlations between

atrophy rate and CAP100, indicating sensitivity to cumulative disease burden. This aligns with Scahill *et al.*, (2025), who reported strong CAP100 associations int he caudate and putamen decades before clinical motor diagnosis.

In contrast, cortical regions with significant group-level atrophy did not correlate with CAP100, suggesting a more variable or delayed trajectory in early HD. Estevez-Fraga *et al.*, (2023) found that cortical regions most affected in HD also show high expression of developmental genes, indicating that early brain development may shape regional vulnerability. They also proposed that compensatory mechanisms in cortical circuits could mask the impact of structural decline in early stages.

Significant CAP100 associations with third ventricular enlargement and subtle brainstem volume increases, further support diffuse structural involvement in early HD. While ventricular expansion is a known marker of global tissue loss that accelerates prior to clinical motor diagnosis, previous studies focused on lateral ventricles and did not assess third or fourth ventricular volumes (Hobbs *et al.*, 2010). Our findings specifically highlight third ventricular enlargement in HDGE participants, with a robust CAP100 correlation, indicating early midline regions neurodegeneration. Additionally, brainstem involvement further suggests early disruption of motor and autonomic circuits beyond striatal pathways, consistent with prior neuropathological evidence (Rüb *et al.*, 2014b).

This study's strong CAP100 associations in subcortical grey matter but not cortical ROIs align with circuit-level insights from Blumenstock and Dudanova (2020), who emphasize that striatal dysfunction precedes cortical degeneration in HD. Their review highlights early cortico-striatal disruptions, particularly in the caudate and putamen, which are highly vulnerable to mutant huntingtin toxicity. While I observed significant atrophy rates in specific cortical regions, the lack of consistent CAP100 associations may reflect greater variability in cortical trajectories or involvement driven by mechanisms less tightly coupled to cumulative CAG burden.

Significant CAP100 associations in the brainstem and third ventricle further support HD's diffuse impact, consistent with early involvement of motor and autonomic circuits. Blumenstock and Dudanova (2020) also stress the need to examine non-striatal

pathways, including brainstem projections and glial contributions, to fully capture the diffuse nature of HD-related neurodegeneration.

5.4. Strengths, Limitations, and Future Directions

This study offers several strengths that enhance its scientific and translational value. It leverages a unique, well-characterised cohort of young adult gene expansion carriers (YAS) in HD-ISS stages 0 and 1, those most likely to be eligible for future prevention trials. Its longitudinal design enables detection of subtle neurodegenerative changes, while a spatially unbiased, whole-brain approach allows comprehensive mapping of regional vulnerability. Imaging was acquired using a standardised protocol and underwent rigorous visual QC, with all segmentations reviewed for anatomical accuracy and reliability.

Several limitations should be acknowledged. The 4.5-year imaging interval, although annualised, may reduce sensitivity to short-term changes relevant for trial design. While the single-site nature of YAS supports consistency, it may limit generalisability to multi-site trials. Clinical and cognitive performance remained near ceiling, precluding imaging-clinical correlation analysis. Although HD-ISS stages 0-1 are ideal for prevention-focused studies, current gene therapy trials often target stages 2-3, underscoring the need to extend analyses to more advanced cohorts such as TRACK-HD.

PVS, fluid-filled cavities surrounding cerebral vessels, may influence gene therapy distribution and efficacy (Coleman *et al.*, 2024). Although visual QC suggested PVS did not affect segmentation quality, their potential contribution to regional volume measures warrants consideration.

MALPEM-derived ROIs were generally high quality; however, cerebellar grey matter regions showed a ‘loose fit,’ likely due to partial volume effects, while cerebral and cerebellar white matter and brainstem ROIs were tightly delineated. These discrepancies highlight segmentation limitations and the need for more anatomically precise modelling.

Future work should incorporate three timepoints to better model individual trajectories and explore links between imaging and biofluid markers such as NfL and PENK. Longer follow-up would clarify how imaging metrics relate to HD-ISS progression and early symptom emergence. Extending analyses to larger, multi-site cohorts with broader HD-ISS representation, such as TRACK-HD, will be essential for validating findings and enhancing translational relevance.

5.5. Conclusion

This study demonstrates that longitudinal atrophy is detectable in HDGE participants nearly two decades before motor onset, validating previous reports that neurodegeneration begins early and is strongly evident in the striatum. Effect size analyses suggest additional regions may also serve as potential trial endpoints, though further validation is needed. While atrophy rates did not differ between the earliest HD-ISS stages, significant CAP100 associations in subcortical regions indicate sensitivity to cumulative disease burden, whereas cortical changes appear more variable or delayed. Overall, these findings confirm established striatal vulnerability while highlighting that other regions are also implicated, providing preliminary evidence to inform future biomarker development and preventative trial design in Huntington's disease.

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Appendix

Supplementary Table 1. Annualised Atrophy Rates (%/year) in Controls and HDGE participants

Annualised atrophy rates (% per year)						
ROI	**Control (n = 34)**	**HD-GEG (n = 54)**	**Between group diff. in atrophy rate (adjusted for age & sex)**			
			Mean (HDGE - Controls)	**95% CI**	**p-value**	**q-value**
Caudate	0.261 (0.272), 0.166 to 0.356	1.478 (1.055), 1.190 to 1.766	1.189	0.895 to 1.484	5.33×10^{-12}	3.68×10^{-10}
Putamen	0.446 (0.367), 0.318 to 0.574	1.854 (1.248), 1.513 to 2.194	1.374	1.014 to 1.735	4.17×10^{-11}	2.84×10^{-9}
Third ventricle	0.160 (1.461), -0.350 to 0.670	-2.263 (2.413), -2.922 to -1.605	-2.392	-3.160 to -1.624	2.06×10^{-08}	1.38×10^{-6}
Pallidum	0.353 (0.608), 0.141 to 0.565	1.577 (1.317), 1.217 to 1.936	1.219	0.808 to 1.630	7.56×10^{-8}	4.99×10^{-6}
Lateral ventricle	-0.320 (1.234), -0.750 to 0.111	-2.327 (2.968), -3.137 to -1.517	-1.956	-2.813 to -1.100	0.00002	0.001
Cerebral white matter	-0.358 (0.270), -0.452 to -0.264	-0.153 (0.336), -0.245 to -0.062	0.215	0.110 to 0.319	0.0001	0.007
Angular gyrus	0.348 (0.360), 0.223 to 0.474	0.669 (0.342), 0.575 to 0.762	0.318	0.160 to 0.475	0.0001	0.008
Triangular part of the inferior frontal gyrus	0.503 (0.443), 0.348 to 0.657	0.883 (0.423), 0.767 to 0.998	0.368	0.178 to 0.559	0.0002	0.015

Ventral DC	0.045 (0.258), -0.045 to 0.135	0.272 (0.324), 0.184 to 0.361	0.221	0.102 to 0.340	0.0004	0.024
Parietal operculum	0.204 (0.327), 0.090 to 0.318	0.478 (0.439), 0.358 to 0.598	0.278	0.124 to 0.433	0.001	0.034
Superior frontal gyrus medial segment	0.297 (0.301), 0.192 to 0.402	0.536 (0.278), 0.460 to 0.612	0.222	0.099 to 0.346	0.001	0.034
Accumbens area	0.025 (1.136), -0.372 to 0.421	1.048 (1.441), 0.655 to 1.442	1.010	0.448 to 1.573	0.001	0.034
Calcarine cortex	-0.375 (0.480), -0.543 to -0.208	-0.007 (0.433), -0.126 to 0.111	0.357	0.152 to 0.561	0.001	0.048
Precuneus	0.072 (0.260), -0.019 to 0.162	0.275 (0.302), 0.193 to 0.358	0.190	0.074 to 0.305	0.002	0.091
Lingual gyrus	-0.209 (0.318), -0.320 to -0.098	0.024 (0.433), -0.094 to 0.143	0.244	0.090 to 0.399	0.002	0.123
Brainstem	-0.376 (0.269), -0.470 to -0.282	-0.172 (0.351), -0.268 to -0.076	0.201	0.071 to 0.330	0.003	0.149
Occipital fusiform gyrus	0.831 (0.423), 0.683 to 0.979	1.164 (0.664), 0.983 to 1.345	0.320	0.094 to 0.546	0.006	0.318
Frontal operculum	0.380 (0.366), 0.252 to 0.508	0.611 (0.363), 0.512 to 0.711	0.229	0.064 to 0.393	0.007	0.360
Planumtemporale	0.105 (0.497), -0.068 to 0.279	0.392 (0.425), 0.275 to 0.508	0.273	0.067 to 0.480	0.010	0.515

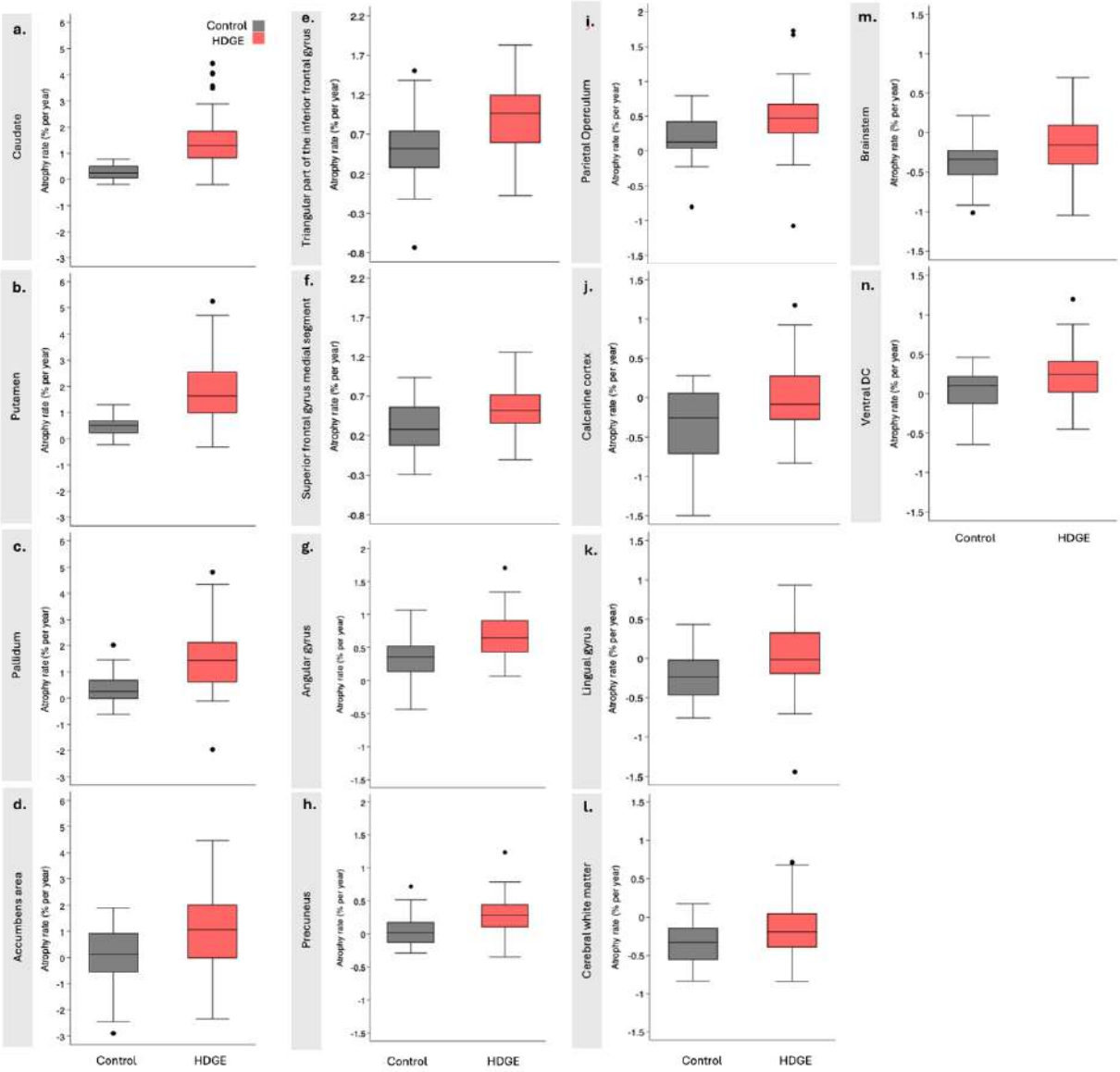
Fourth ventricle	0.151 (1.020), -0.205 to 0.506	-0.172 (0.868), -0.409 to 0.065	-0.295	-0.698 to 0.107	0.148	0.955
Amygdala	-0.089 (0.734), -0.345 to 0.167	0.066 (0.763), -0.142 to 0.274	0.143	-0.170 to 0.456	0.367	0.955
Cerebellum exterior	0.373 (0.400), 0.234 to 0.513	0.316 (0.477), 0.186 to 0.446	-0.061	-0.258 to 0.136	0.539	0.955
Cerebellum white matter	-0.962 (1.042), -1.326 to -0.599	-0.795 (1.392), -1.175 to -0.415	0.168	-0.374 to 0.710	0.540	0.955
Hippocampus	-0.014 (0.263), -0.106 to 0.078	-0.112 (0.305), -0.196 to -0.029	-0.094	-0.214 to 0.025	0.121	0.955
Inferior lateral ventricle	0.152 (0.891), -0.159 to 0.463	0.301 (0.849), 0.070 to 0.533	0.137	-0.241 to 0.514	0.473	0.955
Thalamus	0.328 (0.407), 0.186 to 0.470	0.386 (0.376), 0.283 to 0.488	0.068	-0.096 to 0.231	0.413	0.955
Cerebellar vermal lobules I-V	-0.316 (0.368), -0.444 to -0.187	-0.367 (0.438), -0.487 to -0.248	-0.033	-0.205 to 0.138	0.700	0.955
Cerebellar vermal lobules VI-VII	-0.239 (0.538), -0.426 to -0.051	-0.038 (0.548), -0.188 to 0.111	0.200	-0.033 to 0.432	0.091	0.955
Cerebellar vermal lobules VIII-X	-0.120 (0.381), -0.253 to 0.013	-0.093 (0.371), -0.194 to 0.009	0.029	-0.137 to 0.195	0.725	0.955
Basal forebrain	0.437 (0.894), 0.125 to 0.749	0.649 (1.464), 0.250 to 1.049	0.191	-0.319 to 0.700	0.459	0.955

Anterior cingulate gyrus	0.145 (0.392), 0.008 to 0.282	0.320 (0.373), 0.219 to 0.422	0.179	0.011 to 0.347	0.037	0.955
Anterior insula	0.207 (0.270), 0.112 to 0.301	0.282 (0.304), 0.199 to 0.365	0.072	-0.052 to 0.196	0.249	0.955
Anterior orbital gyrus	0.358 (0.423), 0.210 to 0.505	0.470 (0.444), 0.349 to 0.592	0.099	-0.090 to 0.288	0.300	0.955
Central operculum	0.216 (0.264), 0.124 to 0.308	0.267 (0.317), 0.181 to 0.354	0.043	-0.085 to 0.170	0.506	0.955
Cuneus	-0.462 (0.545), -0.652 to -0.272	-0.371 (0.432), -0.489 to -0.253	0.072	-0.143 to 0.288	0.505	0.955
Entorhinal area	-0.030 (0.689), -0.270 to 0.211	0.073 (0.586), -0.087 to 0.233	0.079	-0.200 to 0.358	0.576	0.955
Frontal pole	0.559 (0.678), 0.322 to 0.795	0.845 (0.647), 0.668 to 1.022	0.298	0.002 to 0.595	0.049	0.955
Fusiform gyrus	-0.027 (0.282), -0.125 to 0.071	0.062 (0.321), -0.026 to 0.150	0.086	-0.047 to 0.219	0.201	0.955
Gyrus rectus	0.636 (0.581), 0.433 to 0.839	0.647 (0.670), 0.464 to 0.830	0.018	-0.247 to 0.284	0.891	0.955
Inferior occipital gyrus	1.033 (0.582), 0.830 to 1.236	1.216 (0.509), 1.077 to 1.355	0.154	-0.072 to 0.380	0.178	0.955
Inferior temporal gyrus	0.396 (0.373), 0.266 to 0.526	0.483 (0.311), 0.398 to 0.567	0.074	-0.077 to 0.224	0.333	0.955
Lateral orbital gyrus	0.656 (0.404), 0.515 to 0.797	0.823 (0.395), 0.715 to 0.930	0.171	-0.010 to 0.351	0.064	0.955

Middle cingulate gyrus	-0.027 (0.435), -0.179 to 0.124	0.131 (0.355), 0.034 to 0.228	0.153	-0.024 to 0.329	0.090	0.955
Medial frontal cortex	0.236 (0.503), 0.060 to 0.411	0.479 (0.661), 0.298 to 0.659	0.229	-0.023 to 0.482	0.074	0.955
Middle frontalgyrus	0.557 (0.276), 0.461 to 0.653	0.698 (0.260), 0.627 to 0.769	0.138	0.018 to 0.258	0.024	0.955
Middle occipital gyrus	0.345 (0.446), 0.190 to 0.501	0.459 (0.523), 0.316 to 0.602	0.102	-0.109 to 0.312	0.341	0.955
Medial orbital gyrus	0.333 (0.422), 0.185 to 0.480	0.324 (0.460), 0.199 to 0.450	-0.025	-0.212 to 0.163	0.794	0.955
Postcentral gyrus medial segment	0.812 (0.960), 0.477 to 1.147	0.845 (0.911), 0.596 to 1.093	0.032	-0.382 to 0.447	0.877	0.955
Precentral gyrus medial segment	0.188 (0.478), 0.021 to 0.355	0.458 (0.616), 0.290 to 0.626	0.268	0.032 to 0.504	0.026	0.955
Middle temporal gyrus	0.365 (0.319), 0.253 to 0.476	0.444 (0.338), 0.352 to 0.536	0.069	-0.078 to 0.216	0.355	0.955
Occipital pole	-0.071 (1.440), -0.573 to 0.432	-0.147 (1.614), -0.588 to 0.293	-0.111	-0.766 to 0.544	0.737	0.955
Opercular part of the inferior frontal gyrus	0.494 (0.382), 0.360 to 0.627	0.598 (0.400), 0.489 to 0.707	0.106	-0.071 to 0.283	0.238	0.955
Orbital part of the inferior frontal gyrus	0.676 (0.490), 0.505 to 0.847	0.699 (0.488), 0.566 to 0.832	0.016	-0.201 to 0.233	0.884	0.955
Posterior cingulate gyrus	0.154 (0.350), 0.032 to 0.276	0.161 (0.316), 0.075 to 0.247	0.004	-0.143 to 0.151	0.955	0.955

Parahippocampal gyrus	0.025 (0.462), -0.136 to 0.186	-0.120 (0.335), -0.211 to -0.028	-0.146	-0.330 to 0.037	0.116	0.955
Posterior insula	-0.079 (0.330), -0.194 to 0.037	-0.062 (0.343), -0.155 to 0.032	0.018	-0.125 to 0.161	0.802	0.955
Postcentral gyrus	0.550 (0.485), 0.381 to 0.719	0.754 (0.355), 0.657 to 0.851	0.196	0.003 to 0.389	0.046	0.955
Posterior orbital gyrus	0.274 (0.429), 0.124 to 0.423	0.343 (0.546), 0.194 to 0.492	0.067	-0.142 to 0.277	0.525	0.955
Planumpolare	-0.077 (0.488), -0.247 to 0.093	-0.103 (0.440), -0.223 to 0.017	-0.030	-0.234 to 0.173	0.767	0.955
Precentral gyrus	0.693 (0.463), 0.532 to 0.855	0.866 (0.380), 0.762 to 0.970	0.165	-0.023 to 0.354	0.084	0.955
Subcallosal area	-0.146 (0.677), -0.382 to 0.090	-0.260 (0.872), -0.498 to -0.022	-0.134	-0.462 to 0.195	0.420	0.955
Superior frontal gyrus	0.789 (0.308), 0.681 to 0.896	0.863 (0.290), 0.784 to 0.943	0.075	-0.062 to 0.212	0.277	0.955
Supplementary motor cortex	0.672 (0.371), 0.542 to 0.802	0.749 (0.367), 0.649 to 0.849	0.081	-0.080 to 0.242	0.317	0.955
Supramarginal gyrus	0.432 (0.395), 0.294 to 0.570	0.624 (0.393), 0.517 to 0.731	0.182	0.009 to 0.354	0.040	0.955
Superior occipital gyrus	0.034 (0.532), -0.152 to 0.220	0.045 (0.692), -0.144 to 0.234	-0.008	-0.272 to 0.256	0.953	0.955
Superior parietal lobule	0.516 (0.418), 0.370 to 0.662	0.688 (0.313), 0.603 to 0.774	0.166	0.004 to 0.329	0.045	0.955

Superior temporal gyrus	0.161 (0.369), 0.032 to 0.290	0.339 (0.408), 0.227 to 0.450	0.175	0.002 to 0.347	0.047	0.955
Temporal pole	0.959 (0.458), 0.799 to 1.119	1.027 (0.444), 0.906 to 1.148	0.056	-0.138 to 0.249	0.569	0.955
Transverse temporal gyrus	0.209 (0.453), 0.051 to 0.367	0.261 (0.515), 0.120 to 0.401	0.050	-0.156 to 0.256	0.630	0.955



Supplementary Figure 1. Annualized Atrophy Rates in HDGE and Control Participants Across Multiple ROIs. Annualized atrophy rates (%/year; age- and sex-adjusted, FDR-corrected) are shown for 14 representative regions of interest (ROIs), comparing HDGE participants (n = 54; red) and controls (n = 34; grey). Box plots display group medians (horizontal line), interquartile range (box), whiskers (1.5×IQR), and outliers (filled circles). ROIs include subcortical structures (e.g., caudate, putamen, pallidum, accumbens, ventral diencephalon) and selected cortical regions. Significant between-group differences were observed in several regions, with the largest effects in the caudate and putamen. This figure highlights regional variability in atrophy rates and supports the identification of non-striatal biomarkers for future HD trials.

Abbreviations: HDGE, Huntington's disease gene expansion; IQR, interquartile range; FDR, false discovery rate; ROI, region of interest; %/year, percentage volume loss per year.

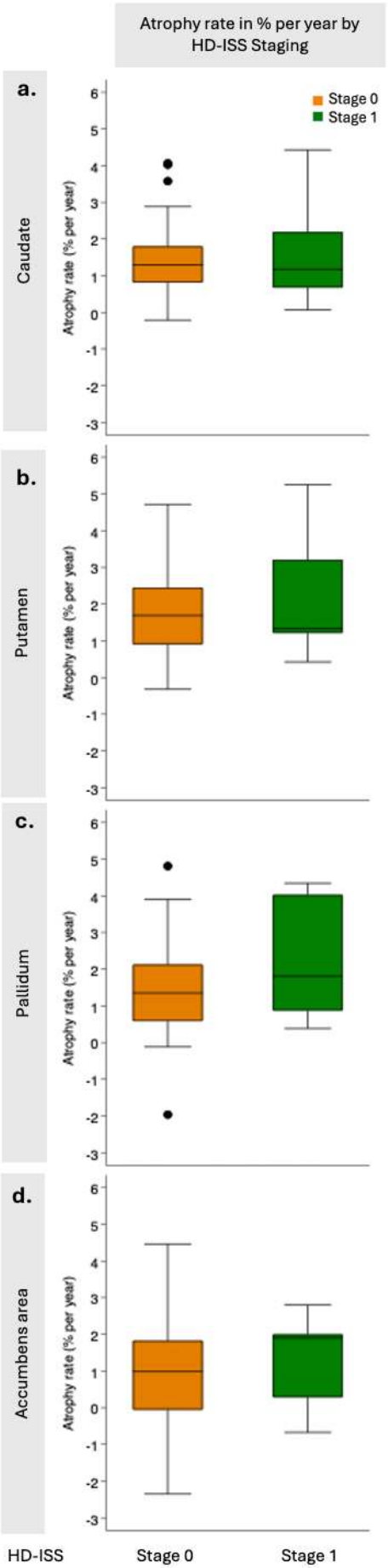
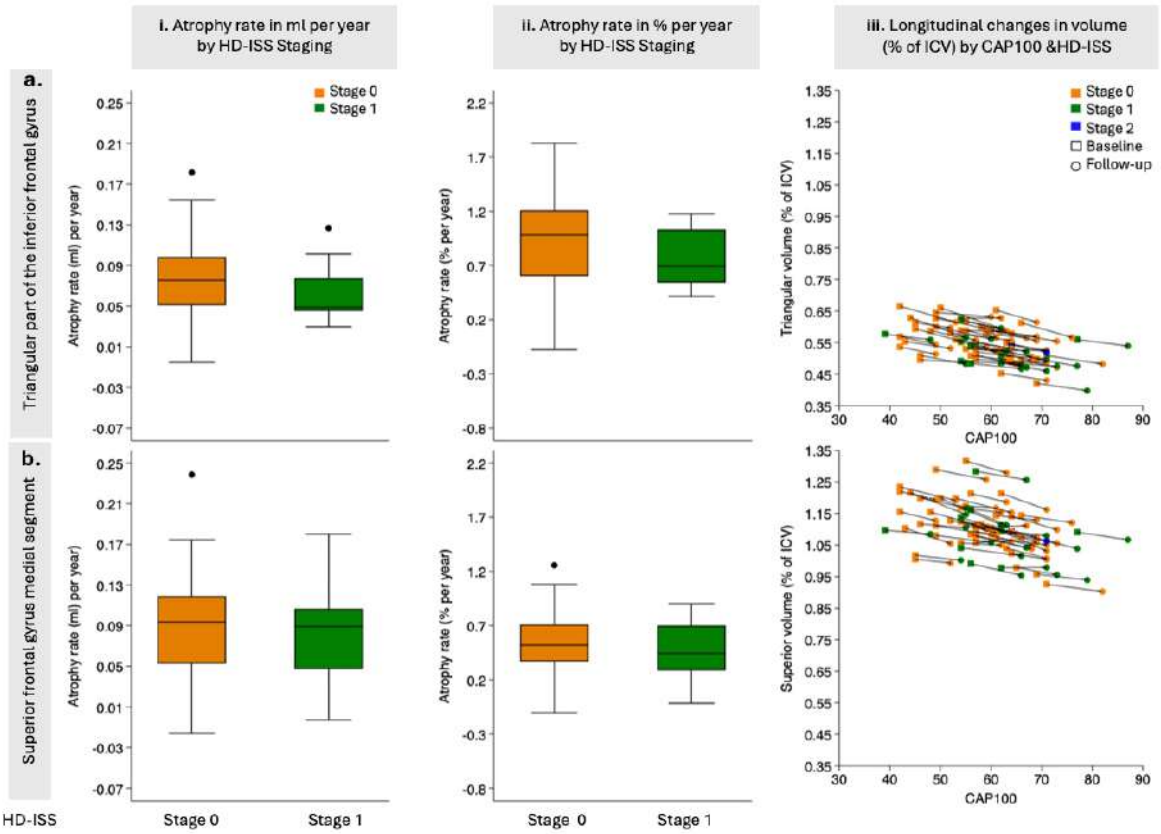

Atrophy rate in % per year by HD-ISS Staging
a.
Caudate
b.
Putamen
c.
Pallidum
d.
Accumbens area
Stage 0
Stage 1
Atrophy rate (% per year)
6
5
4
3
2
1
0
-1
-2
-3
HD-ISS
Stage 0
Stage 1

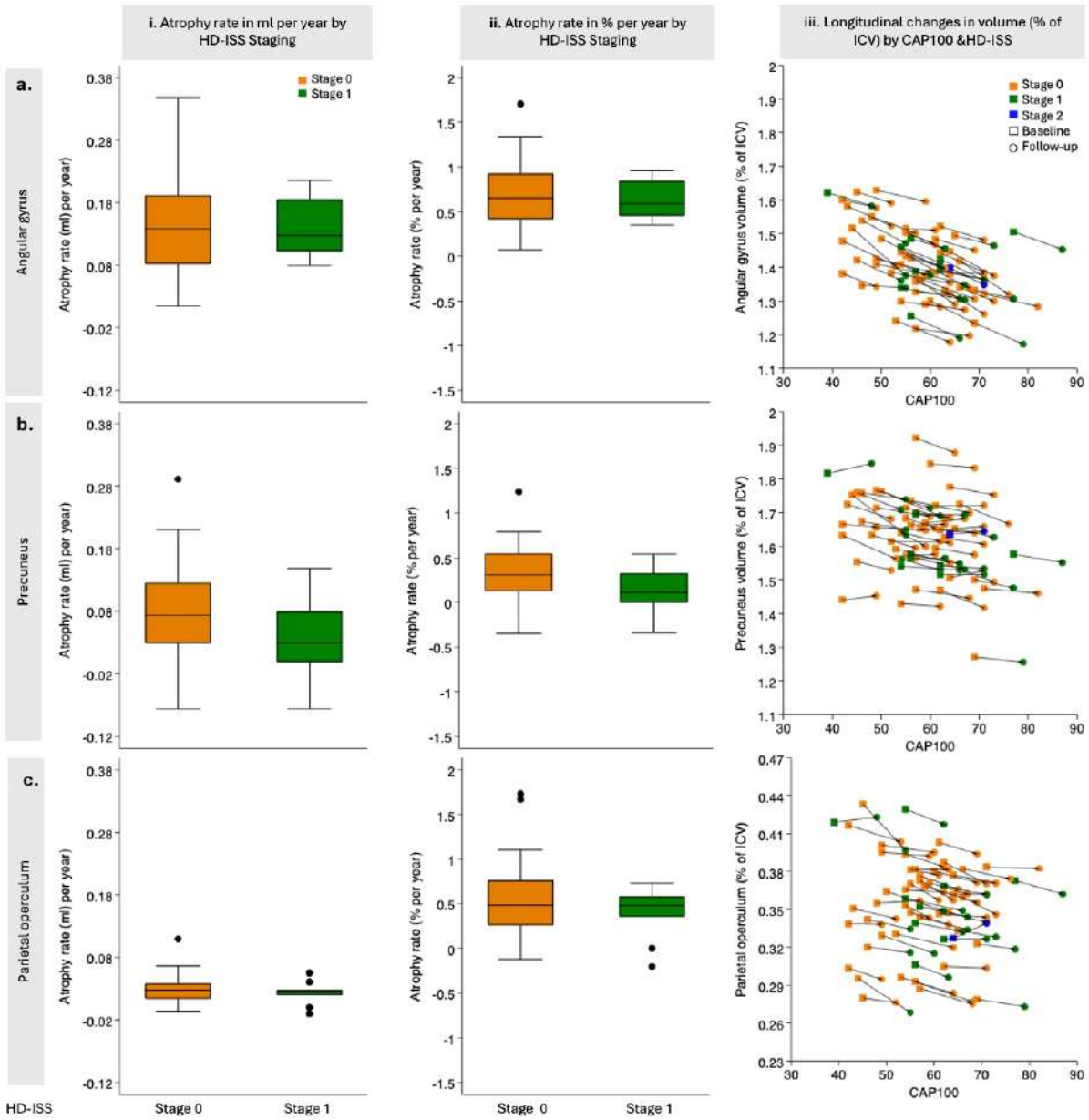
Figure 2. Annualized Atrophy Rates (%/year) by HD-ISS Stage in Subcortical ROIs. Box plots compare annualized atrophy rates, expressed as percentage volume loss per year (%/year), across HD-ISS stages 0 (orange) and 1 (green) in four subcortical regions: (a) caudate, (b) putamen, (c) pallidum, and (d) accumbens area. Each plot displays the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles).

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ROI, region of interest; %/year, percentage volume loss per year.



Supplementary Figure 3. Annualized Atrophy Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in Frontal Lobe ROIs. This figure illustrates regional atrophy rates and longitudinal volume decline across frontal lobe regions: (a) triangular part of the inferior frontal gyrus and (b) superior frontal gyrus (medial segment). For each region, boxplots compare annualized atrophy rates in millilitres per year across HD-ISS stages 0 (orange) and 1 (green). Boxplots display the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles). Scatterplots (c) show regional volume, expressed as percentage of intracranial volume (%ICV), plotted against CAP100 score for HD-ISS stages 0 and 1. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes.

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest; ml/year, millilitres per year; %/year, percentage volume loss per year.



Supplementary Figure 4. Annualized Atrophy Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in Parietal Lobe ROIs. This figure illustrates regional atrophy rates and longitudinal volume decline across parietal lobe regions: (a) superior parietal cortex, (b) inferior parietal cortex, and (c) precuneus. For each region, boxplots compare annualized atrophy rates in millilitres per year across HD-ISS stages 0 (orange) and 1 (green). Boxplots display the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles). Scatterplots show regional volume, expressed as percentage of intracranial volume (%ICV), plotted against CAP100 score for HD-ISS stages 0 and 1. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes.

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest; ml/year, millilitres per year; %/year, percentage volume loss per year.

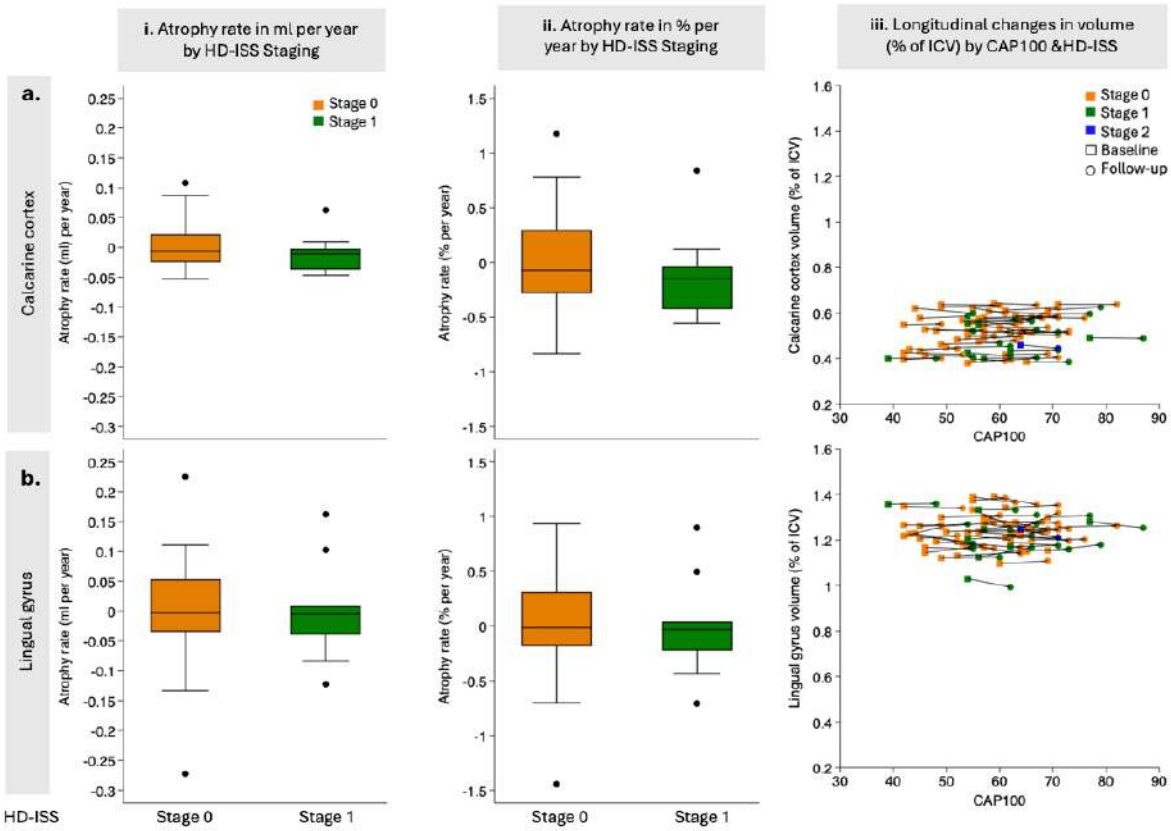


Figure 5. Annualized Atrophy Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in Occipital Lobe ROIs. This figure illustrates regional atrophy rates and longitudinal volume decline across occipital lobe regions: (a) caudate cortex and (b) lingual gyrus. For each region, boxplots compare annualized atrophy rates in millilitres per year across HD-ISS stages 0 (orange) and 1 (green). Boxplots display the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles). Scatterplots (c) show regional volume, expressed as percentage of intracranial volume (%ICV), plotted against CAP100 score for HD-ISS stages 0 and 1. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes.

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest; ml/year, millilitres per year; %/year, percentage volume loss per year.

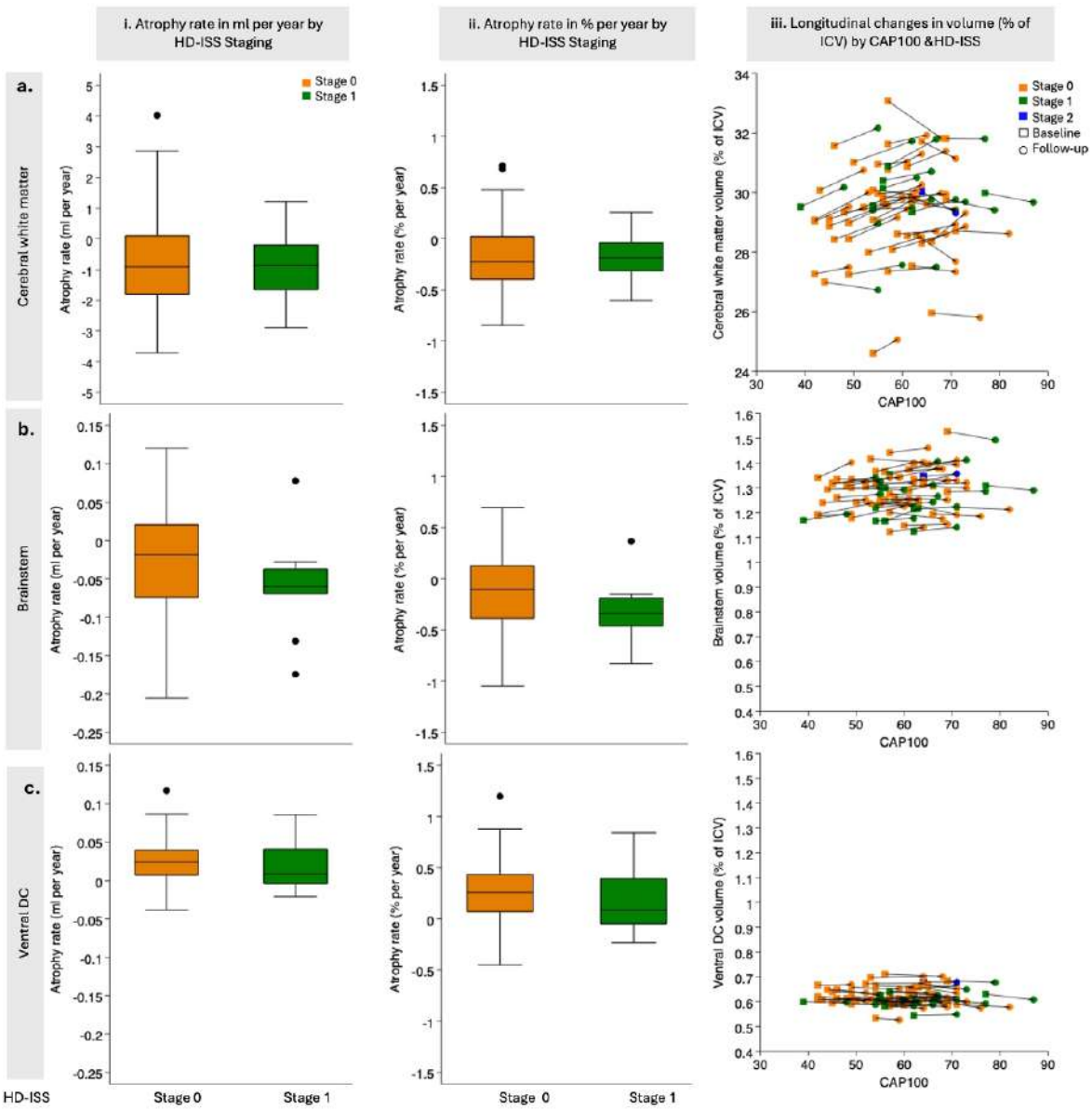


Figure 6. Annualized Atrophy Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in Cerebral white matter, Brainstem and Ventral DC ROIs. This figure illustrates regional atrophy rates and longitudinal volume decline across (a) brainstem, (b) cerebral white matter, and (c) ventral diencephalon. For each region, boxplots compare annualized atrophy rates in millilitres per year across HD-ISS stages 0 (orange) and 1 (green). Boxplots display the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles). Scatterplots show regional volume, expressed as percentage of intracranial volume (%ICV), plotted against CAP100 score for HD-ISS stages 0 and 1. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes.

Abbreviations**:** HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; ICV, intracranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest; ml/year, millilitres per year; %/year, percentage volume loss per year.

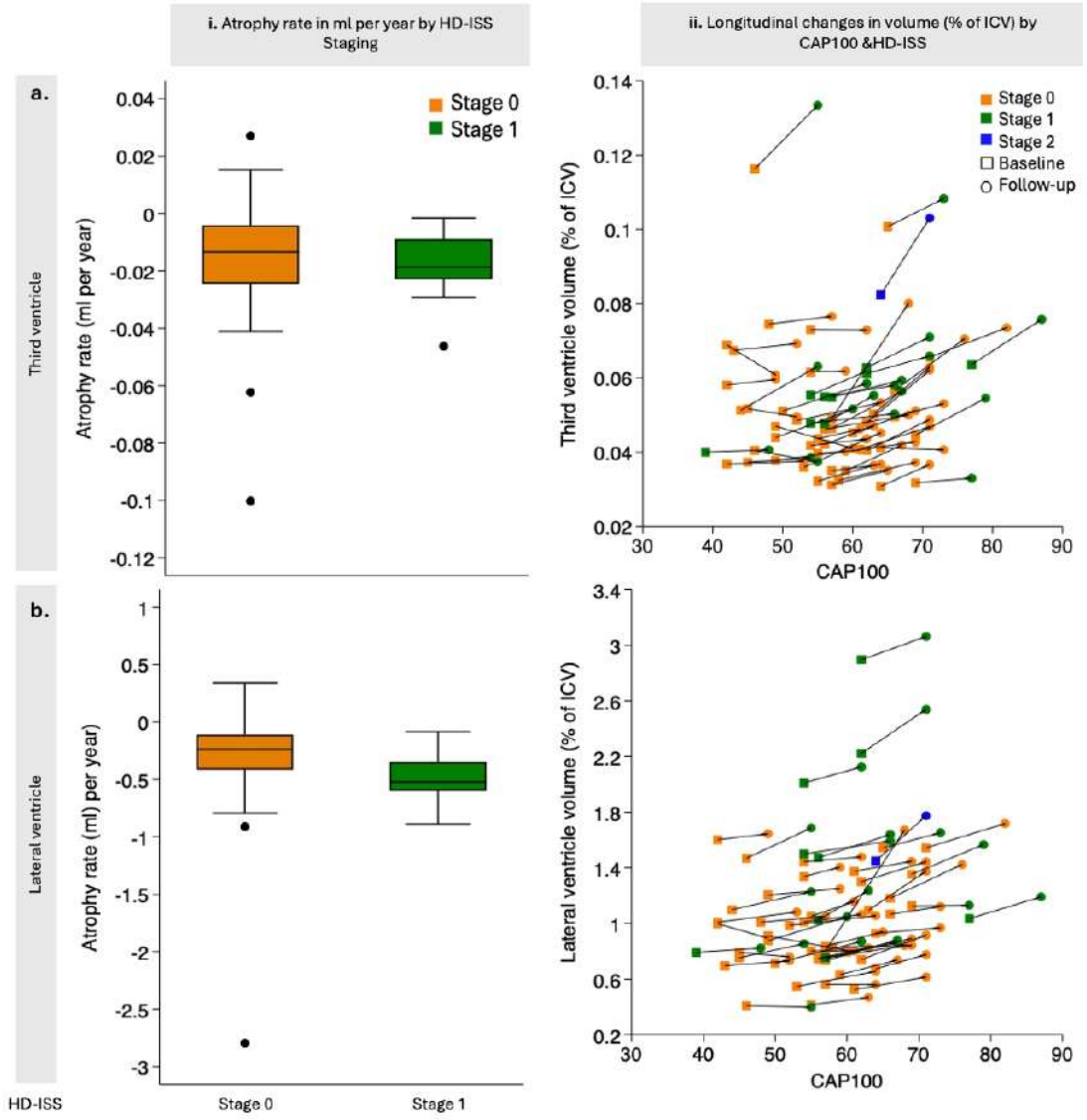


Figure 7. Annualized Expansion Rates and CAP100-Related Volume Trajectories Across HD-ISS Stages in Ventricular ROIs. This figure illustrates expansion rates and longitudinal volume changes in ventricular regions: (a) third ventricle and (b) lateral ventricles. For each region, boxplots compare annualized volume change in cubic millimetres per year (mm^3/year) across HD-ISS stages 0 (orange) and 1 (green). Boxplots display the group median (horizontal line), interquartile range (box), whiskers extending to 1.5× IQR, and individual outliers (filled circles). Scatterplots show regional volume, expressed as percentage of cranial volume (%CV), plotted against CAP100 score for HD-ISS stages 0, 1, and 2. Longitudinal measurements are connected by black lines per individual, with squares indicating baseline and circles indicating follow-up volumes.

Abbreviations: HD-ISS, Huntington's Disease Integrated Staging System; IQR, interquartile range; CV, cranial volume; CAP100, cumulative exposure metric calculated as Age × (CAG – 33.66); ROI, region of interest; mm^3/year, cubic millimetres per year.